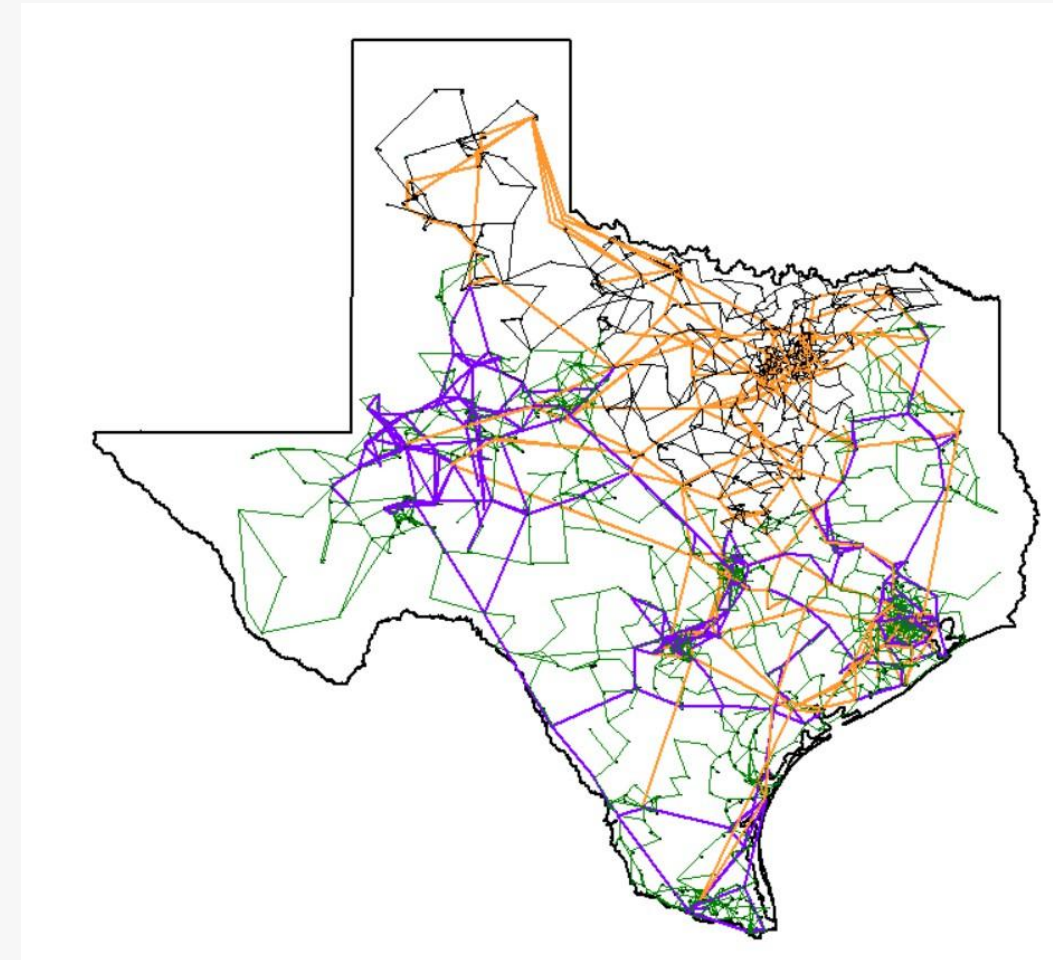


Research Proposal: A Weather-driven Integrated Approach to Assess Power Grid Resilience Under Extreme Winter Events

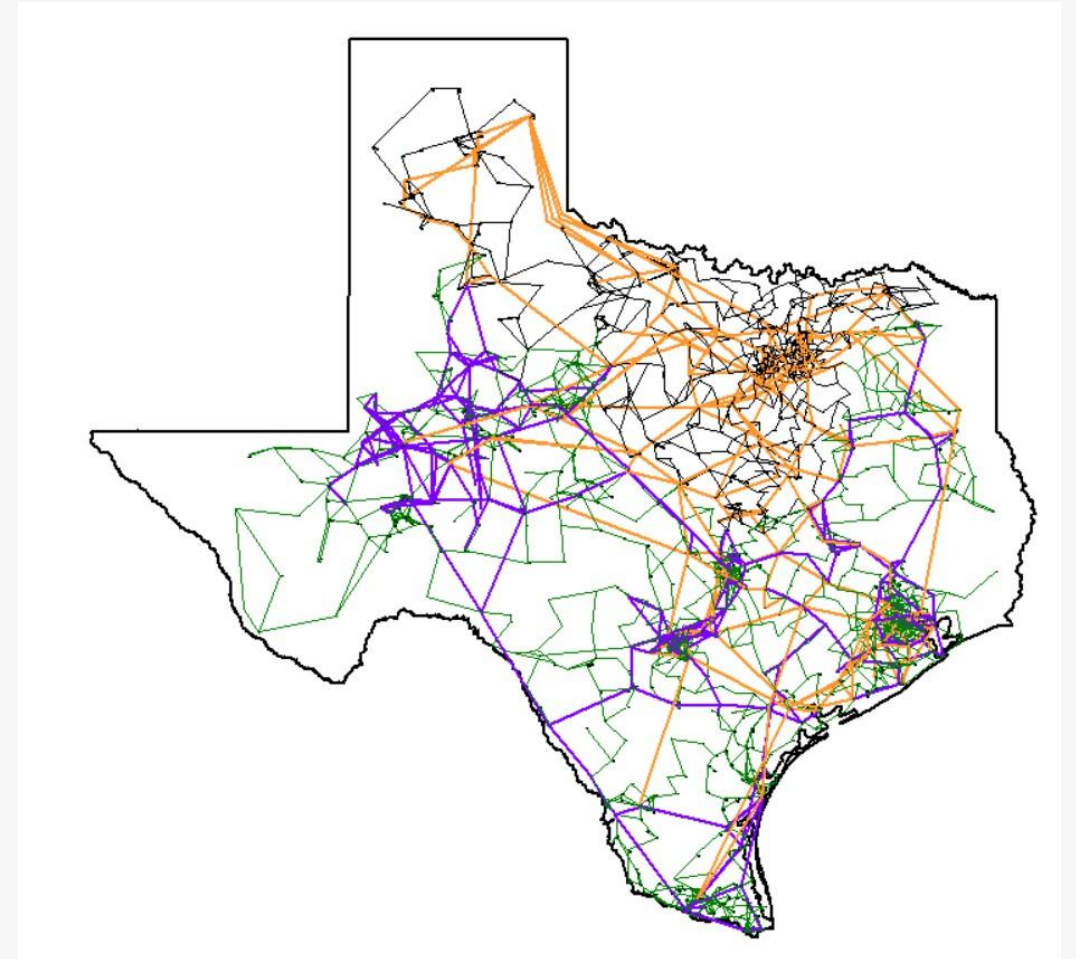


Chonseng Sou

Research Proposal: A Weather-driven Integrated Approach to Assess Power Grid Resilience Under Extreme Winter Events

Overview:

1. Research Objectives
2. Literature Review
3. Project Overview
4. Technical Details/Milestones



Research Objectives

Winter Risk Hasn't Gone Away — It Has Moved

1. The grid is now weather-constrained, not capacity-constrained.

The energy transition to high renewable penetration and heating electrification means grid performance is now constrained by **weather conditions**, not just installed capacity.

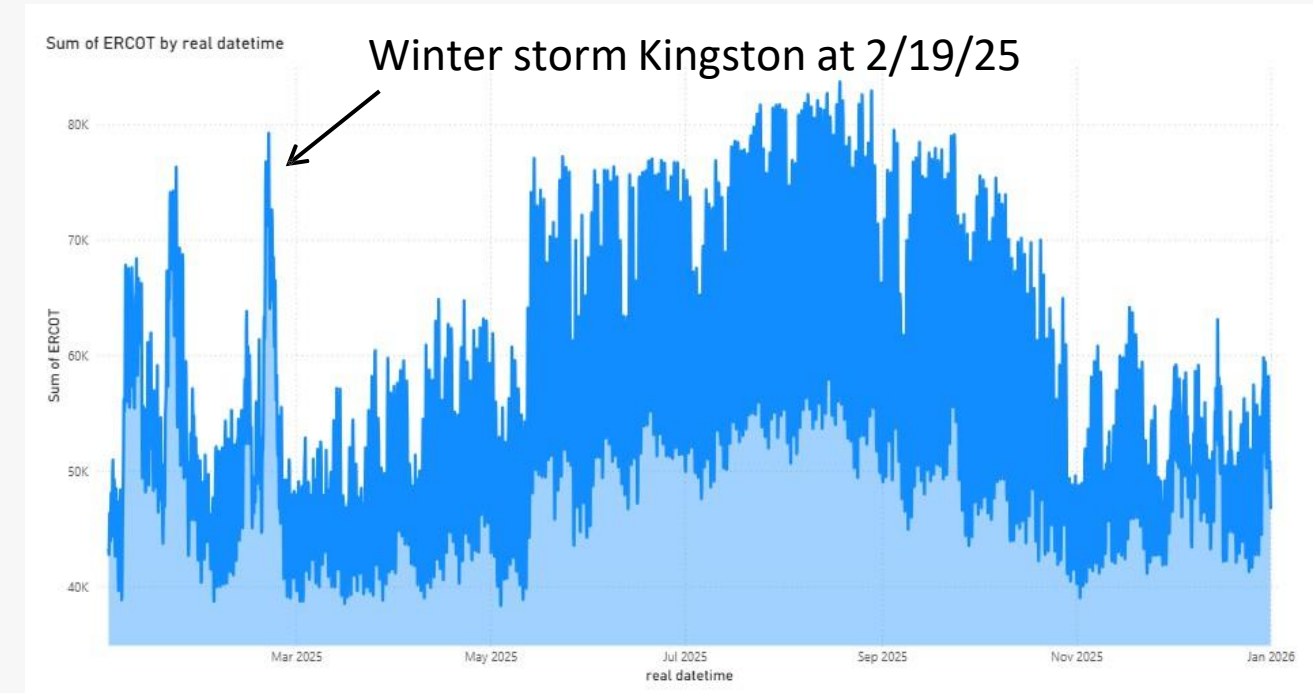
2. Device-level winterization has worked — the correlated collapse is fading.

Since EOP-012-3, the Uri-era pattern of correlated, common-mode generator failures has been largely mitigated. Winter Storm Cora (Jan 2025) and Kingston (Feb 2025) each set demand records yet never entered emergency operations.

3. So the residual risk moved to what winterization can't reach.

Under a post-EOP-012 winter, we observe stress in:

- **Demand side:** load consumption surges up.
- **Fuel-logistics side:** gas supply to generation gets constrained (not covered by EOP-012).
- **Generation side:** residual, largely *independent* forced outages remain.



ERCOT 2025 Load Profile

The ERCOT grid operated reliably during Winter Storm (W.S.) Kingston with adequate supply to meet demand. ERCOT issued a TXANS Weather Watch but did not call for conservation and did not enter emergency operations. **ERCOT set a new all-time winter and two new February peak demand records.**

- On February 19, ERCOT set a new unofficial February peak demand of 76,725* MW, surpassing the previous February record of 69,812 MW set on February 14, 2021, in the 7-8 p.m. hour, during W.S. Uri. On February 20, ERCOT set a new unofficial all-time winter and February peak demand of 80,1545* MW. The previous all-time winter record was 78,349 MW set January 16, 2024, in the 7-8 a.m. hour, during W.S. Heather. *A record is unofficial until final settlements occur, which can take several months.

February Winter Storm Kingston Recap

With NERC EOP-012-3, Grid Resilience is improved, but..

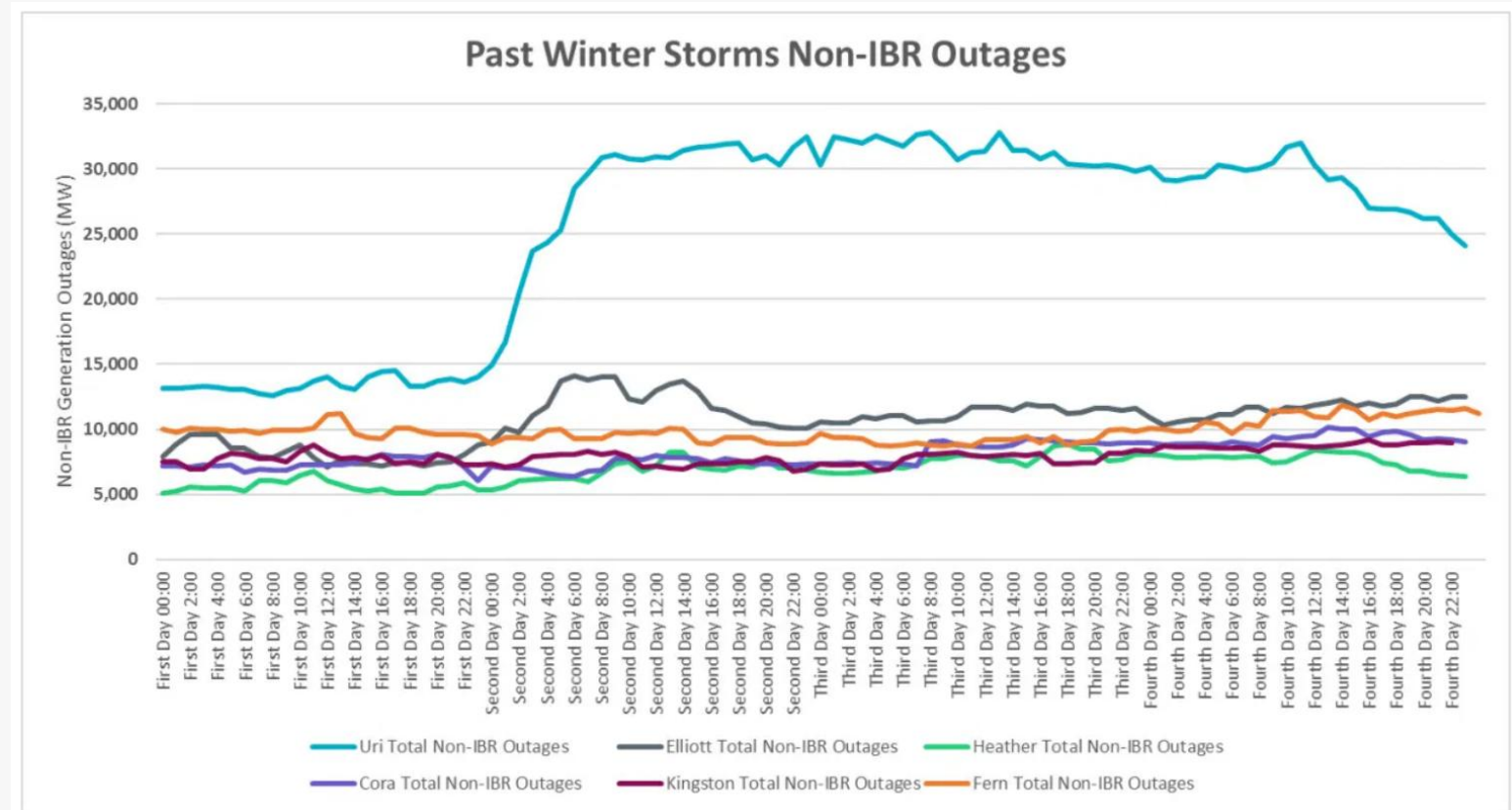
3. Device-level winterization has worked — the correlated collapse is fading.

Since EOP-012-3, non-IBR outages across five post-Uri winter storms have fallen to roughly one-third of Uri's peak and, more tellingly, changed *shape*: from Uri's correlated common-mode cliff to a flat, independent-looking band. Winter Storm Cora (Jan 2025) and Kingston (Feb 2025) each set demand records yet never entered emergency operations.

EOP-012-3

Extreme Cold Weather Preparedness and Operations

Mandatory Subject to Enforcement 10/01/2025



Source: figure - ERCOT Trending Topics,
TOPIC: WEATHERIZATION

The Problem — Adequacy Under a Post-EOP-012 Winter

Existing winter studies still struggle to represent the coupled response across gas supply, generation, transmission, and demand.

But the question has **narrowed**: with device-level failures largely mitigated, the open problem is quantifying residual adequacy risk in a normal (post-EOP-012) winter — **how much firm energy and reserve the system actually deliver when a storm lifts load and suppresses renewables at once.**

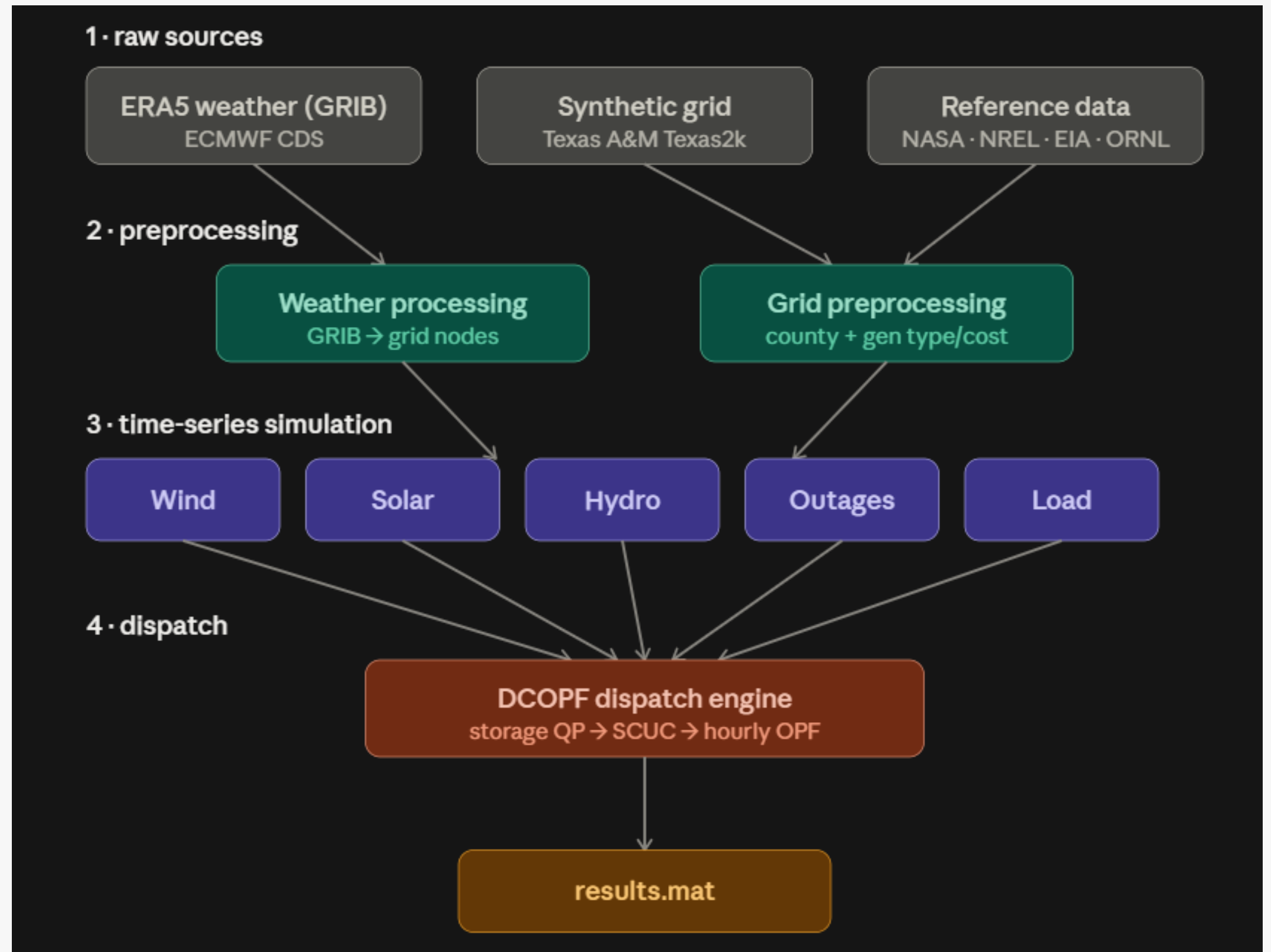
the adequacy questions this framework targets:

- **Partial-system impact** — cold hits only part of the footprint; how does localized weather translate into localized shortfall?
- **Sufficiency** — will committed generation and reserves actually serve load through the event?
- **Fuel & limitations** — how do planned outages, unit limitations, and likely fuel loss enter the assessment?

(Ref: NERC ERO Enterprise CMEP Practice Guide, Jan 2025)

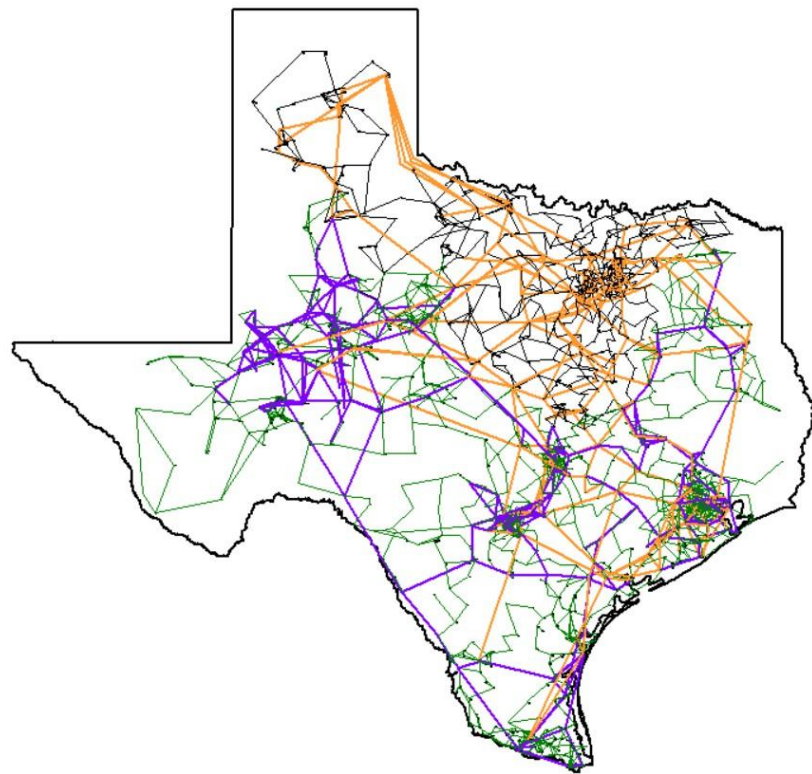
Project Overview

Project Workflow



Synthetic Grid Model Overview

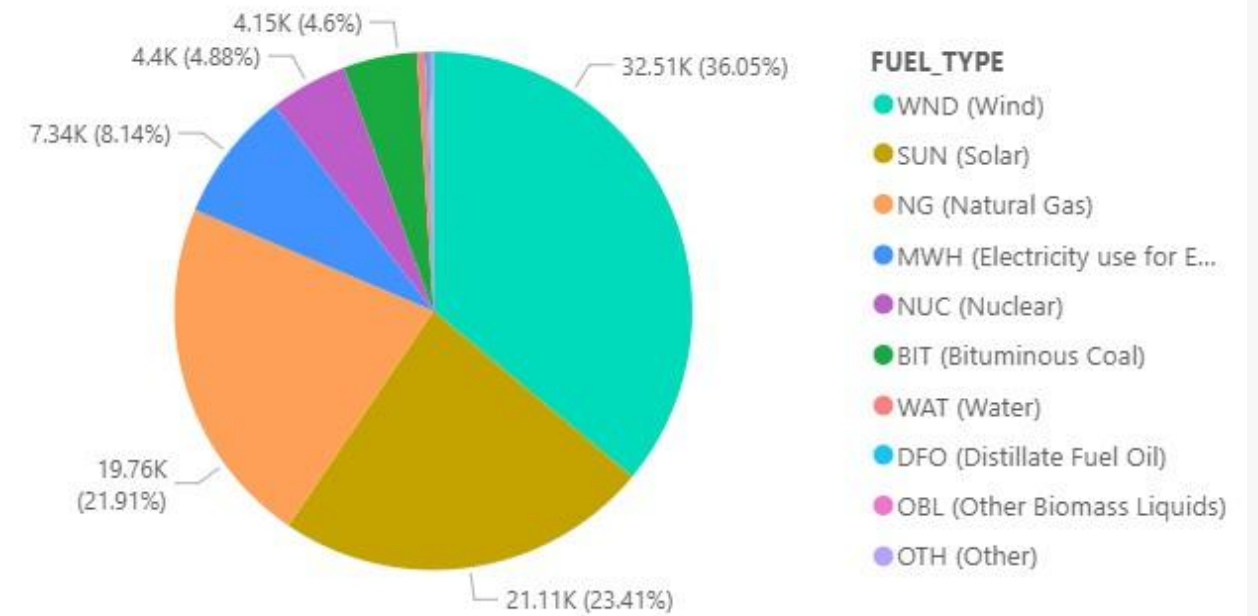
Synthetic grid is a large-scale model of power system that mimics the complexity, topology, and statistical properties of real grids without using confidential data. Each node represents a substation(including data like power generation, demand)



Testing Model: Texas2k Series25
Cases with 2025 level wind, solar,
and batteries

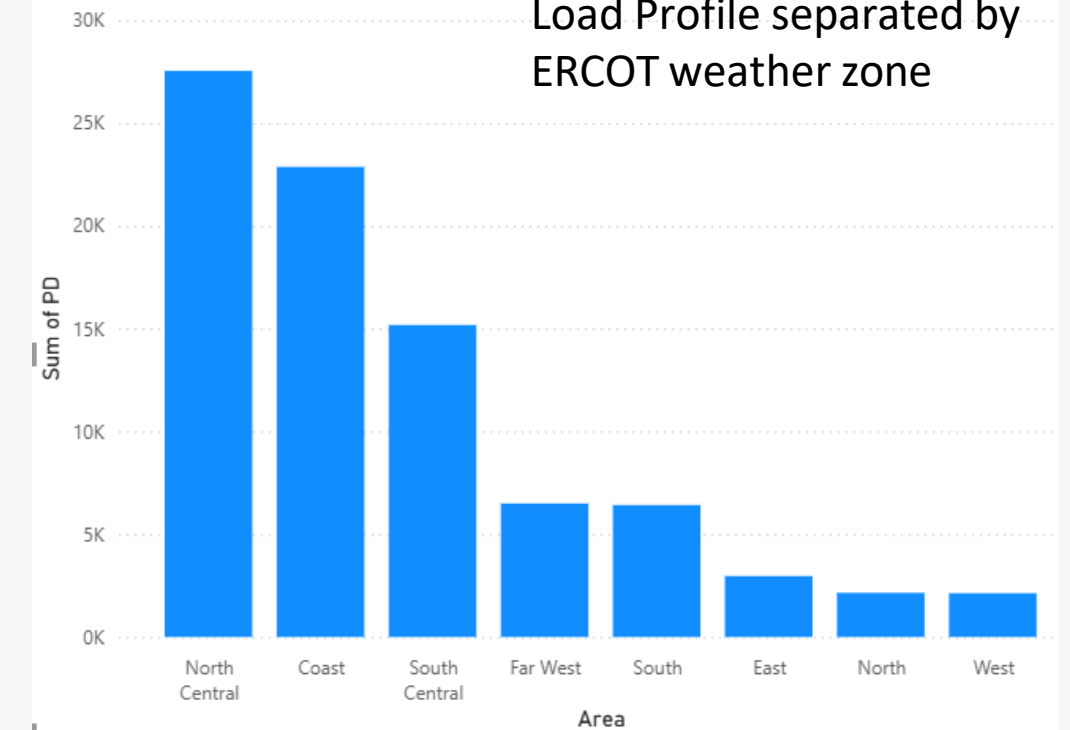
Source: <https://electricgrids.engr.tamu.edu/>

Sum of PG by FUEL_TYPE



Generation Mix

Sum of PD by Area



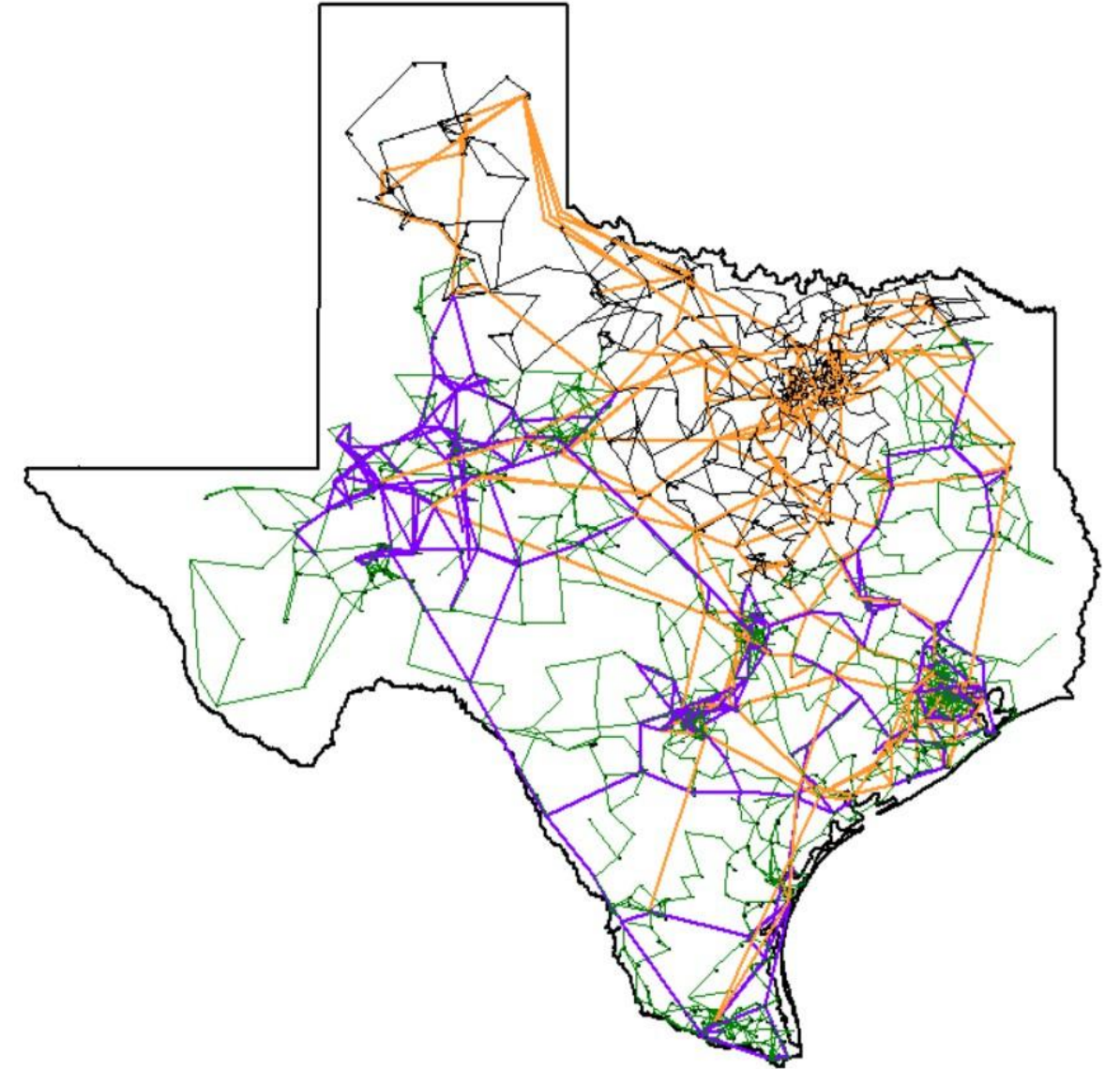
Load Profile separated by
ERCOT weather zone

Project Positioning

Given an extreme cold benchmark temperature event, this framework generates weather-driven, spatiotemporally consistent scenarios across an ERCOT-scale grid — with nodal load, renewable output, battery storage scheduling, and unit commitment, as one coherent hourly time series.

This framework sits at the upstream stage. It produces the weather-stress scenarios that extreme-temperature reliability assessments (such as those under TPL-008-1) need at the start. In these scenarios, the same weather pushes load up and cuts renewable output down at the same time, and this happens only in the areas the weather event actually hits.

The framework builds these scenarios on a synthetic grid for research, and produces active-power scenario inputs only.



Goal & Delivered Capabilities (v1)

Goal:

Build a weather-driven, node-level framework that maps a winter weather time series onto an ERCOT-scale (synthetic) grid and produces a consistent hourly scenario across load, generation, storage, and commitment. Resource- and energy-adequacy metrics fall out of that scenario as one readout.

Delivered & validated (v1) — all against ERCOT 2025 actuals:

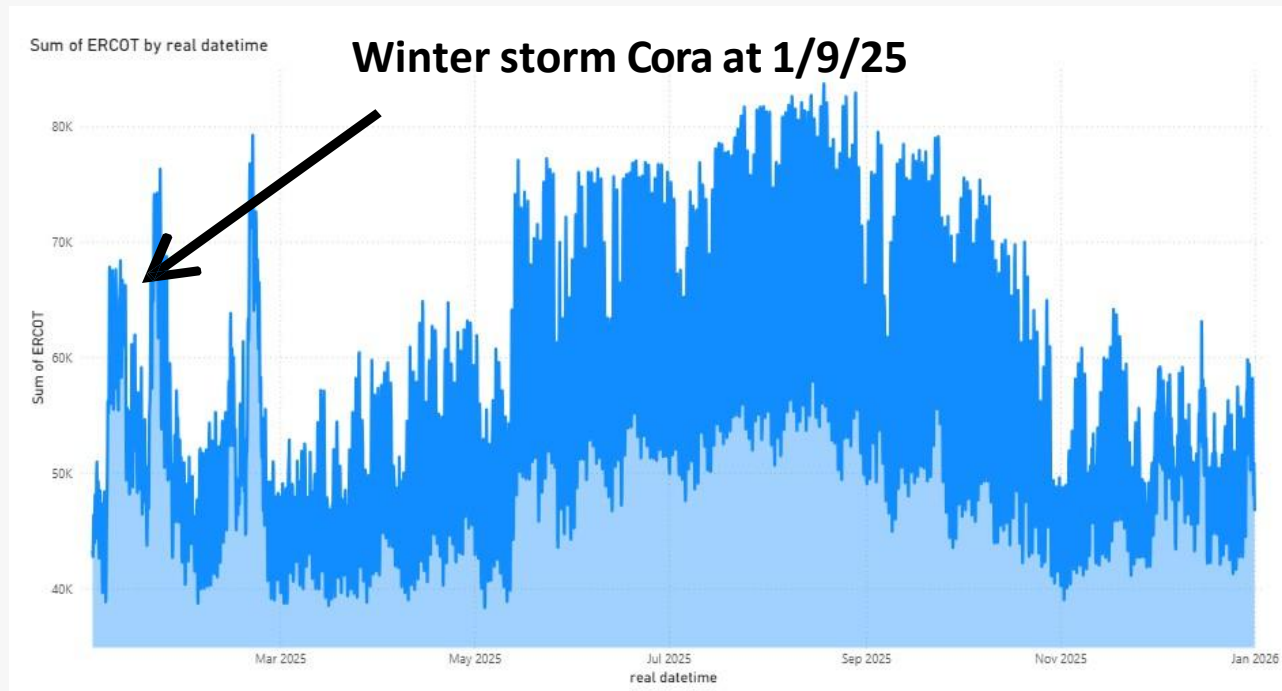
- **wind / solar / hydro generation simulation**
- **Temperature-sensitive bus-level load simulation**
- **generator availability (3-state Markov chain)**
- **Unit commitment modeling (MILP, solved w/ Gurobi)**
- **BESS Modeling (Peak-shaving behavior)**
- **Sequential DCOPF Integration**
- **load shedding mechanism modeling**
- **Ancillary service (Operational Reserves)**

Outputs: a full per-node, per-hour dataset (generation by fuel, unit availability, storage state, nodal LMP, load shed and curtailment by magnitude / timing / location). Adequacy metrics such as LOLH and EUE are derived from it.

Validation window & Scope (v1)

Validation window: 01/01–01/15/2025

This period includes Winter Storm Cora, a record-demand-but-no-emergency storm: the archetype of the post-EOP-012 normal winter this framework characterizes.



ERCOT 2025 Load Profile

Scope:

- This is a resource- and energy-**adequacy screening**, not a security assessment.
- Conducted on a **synthetic network**; results are methodological and are not representative of specific corridors in the actual ERCOT system.
- Produces **active-power scenario inputs only**; does not perform steady-state or transient-stability analysis.

Roadmap — Toward Fuel-Side Coupling

v1 covers the **device-side** residual (independent availability). What remains is the **fuel-side** residual — and that is the roadmap

v1 — synthetic grid benchmark:

- wind / solar / hydro generation
- Temperature-sensitive bus-level load simulation
- generator availability
- Unit commitment
- BESS Modeling
- Sequential DCOPF Integration
- load shedding mechanism
- Ancillary Service (Operational Reserves)

v2 — Fuel-limited thermal:

Add a time-varying fuel-limited energy budget to gas units, split by firm vs interruptible contract, driven by a simple gas-availability proxy, no gas-network model required. A first-order capture of fuel-side scarcity.

V3 — Endogenous gas-vs-heating

competition:

A model of gas demand competition between **gas-fired generation buses and load (heating) buses**, drawn from real gas-distribution operating procedures — reproducing how a cold snap starves generation of fuel. *This is the project's intended core innovation, placed where it belongs: ahead, not done.*

Outputs: LOLH, EUE, nodal Imp, shed magnitude / timing / location, curtailment magnitude / timing / location.

Literature Review

Literature Review: Summary

Existing literature primarily utilizes mathematical optimization and statistical analysis methods to study grid resilience during extreme winter events. Most studies treat winter storms as a purely random variable when building a testing model for resilience analysis. These approaches typically use statistical methods to generate winter storm scenarios and apply meteorological parameters as global constants across the system.

However, this setting ignores the spatiotemporal evolution and its localized inference of weather systems, making it difficult for models to accurately describe the localized impacts that extreme cold exerts on the grid. Even when literature employs spatiotemporal models to capture weather dynamics and localized impacts, they still focus exclusively on the physical damage to transmission lines caused by winter storms.

Beyond this fragmented approach, most literature evaluates single-dimensional responses and overlooks the complex coupled responses and common-mode failures across gas supply, generation, transmission, and load under extreme cold. Most notably, there is a lack of quantitative modeling for fuel limitations.

Therefore, to more accurately assess extreme events like the 2021 Texas Winter Storm Uri, this study argues that the analytical perspective must be expanded from the local component level to the system level. Accurately identifying and integrating the common-mode failure mechanisms triggered by extreme winter storms in the model is a core component for realistically reproducing and deeply analyzing large-scale systemic collapses.

Technical Details

The following modules together produce the per-node, per-hour scenario dataset.

Milestone: Weather Data Processing

Weather Data Processing Logics

This code is used to process ERA5 GRIB data (Texas, 1/1/2025 – 1/15/2025, 00:00-24:00), mapping the raw data to each node of the synthetic grid. The variables extracted from ERA5 include:

Variables	Shortname	Unit
2m temperature	2t	T
2m dewpoint temperature	2d	T
Surface solar radiation downward	ssrd	J/m^2
10m, 100m windspeed components	10u, 10v, 100u, 100v	m/s
Surface pressure	sp	Pa
Snow depth	sd	m
10m 3s wind gust	10fg	m/s
Total precipitation	tp	m
Precipitation type	ptype	XXX

- 1. Data Ingestion and Baseline Alignment:** Parses the underlying multidimensional meteorological files (GRIB), converts the latitude/longitude coordinate systems, and forces alignment against a unified time axis and spatial baseline to eliminate floating-point inaccuracies in the raw data.
- 2. Spatial Mapping and Efficient Extraction (Step 4):** Matches the physical node coordinates of the target synthetic grid with the meteorological grid. It utilizes the Nearest Neighbor method in a vectorized manner to precisely project large-scale gridded data onto specific power grid nodes.
- 3. Data Cleaning and Boundary Repair (Step 5):** Automatically scans for and locates missing values (NaN) caused by geographical boundaries. It achieves seamless data imputation by dynamically expanding the search window outward and calculating the spatial average of valid adjacent grids.
- 4. Feature Derivation and Engineered Output (Step 6):** Converts global time (UTC) to the local time zone, derives necessary physical quantities for engineering applications from base vectors, and finally exports a lightweight, structured data table (CSV).

Milestone:

Wind Generation Modeling

Wind Generation Modeling: Simulation Logics

- 1. **Site Classification:** Extract annual mean windspeed of each wind farm through NASA POWER API to determine their IEC class and assign appropriate normalized power curves.
- 2. **Performance Simulation:** Compute real-time generation time series by mapping the adjusted wind speeds to power curves.

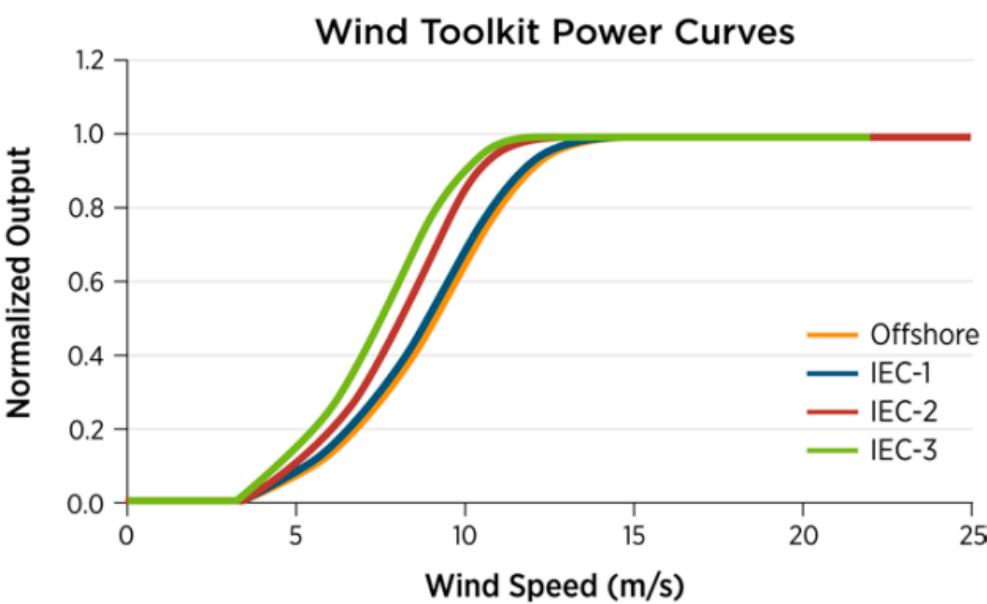


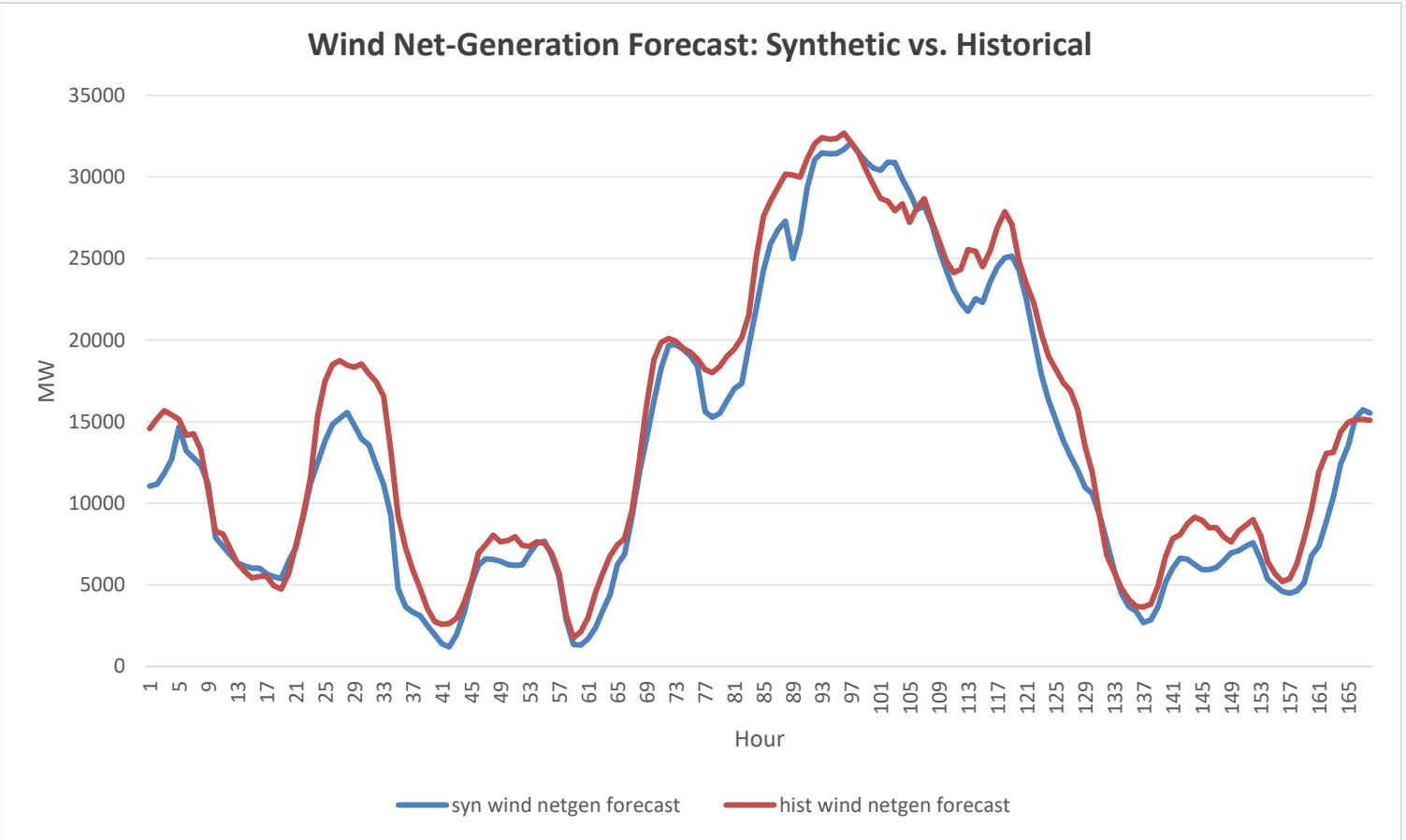
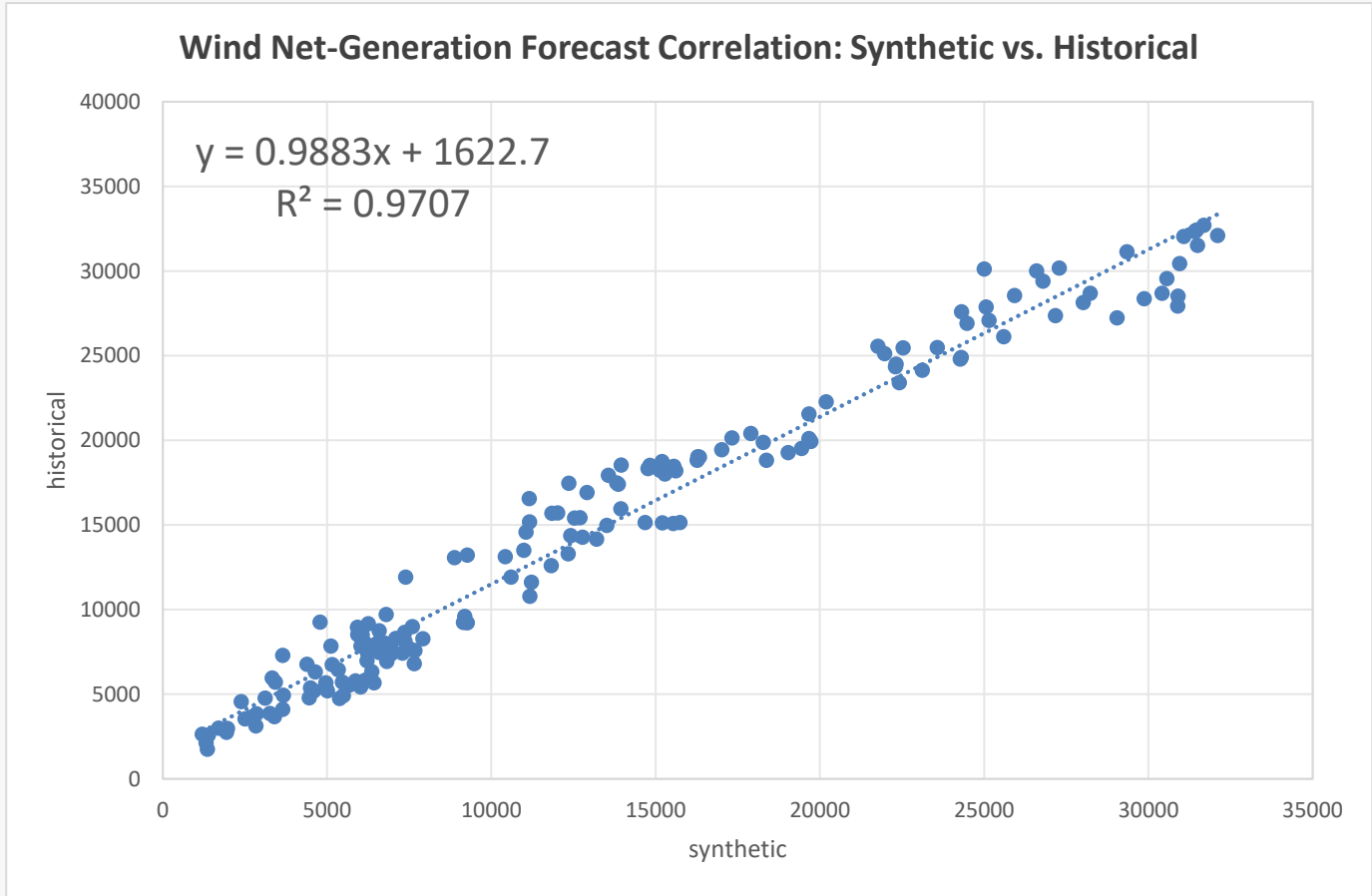
Figure 1. Power curves used to convert modeled wind speeds to power for onshore Class I through Class III locations and offshore locations

Wind turbine class		I	II	III	S
V_{ave} (m/s)		10	8.5	7.5	User defined
V_{ref} (m/s)		50	42.5	37.5	
$V_{50,gust}$ (m/s)		70	59.5	52.5	
I_{ref}	A	0.16			
	B	0.14			
	C	0.12			

In the table (IEC61400-1: 2005):

- 1. Rayleigh distribution is assumed, i.e. $k = 2$.
- 2. V_{ave} is the annual mean wind speed at hub height; V_{ref} is the 50-year extreme wind speed over 10 minutes; $V_{50,gust}$ is the 50-year extreme gust over 3 seconds; I_{ref} is the mean turbulence intensity at 15 m/s.
- 3. A, B and C are the categories of higher, medium and lower turbulence intensity characteristics respectively.

Wind Generation Modeling – Validation: Wind Net Generation

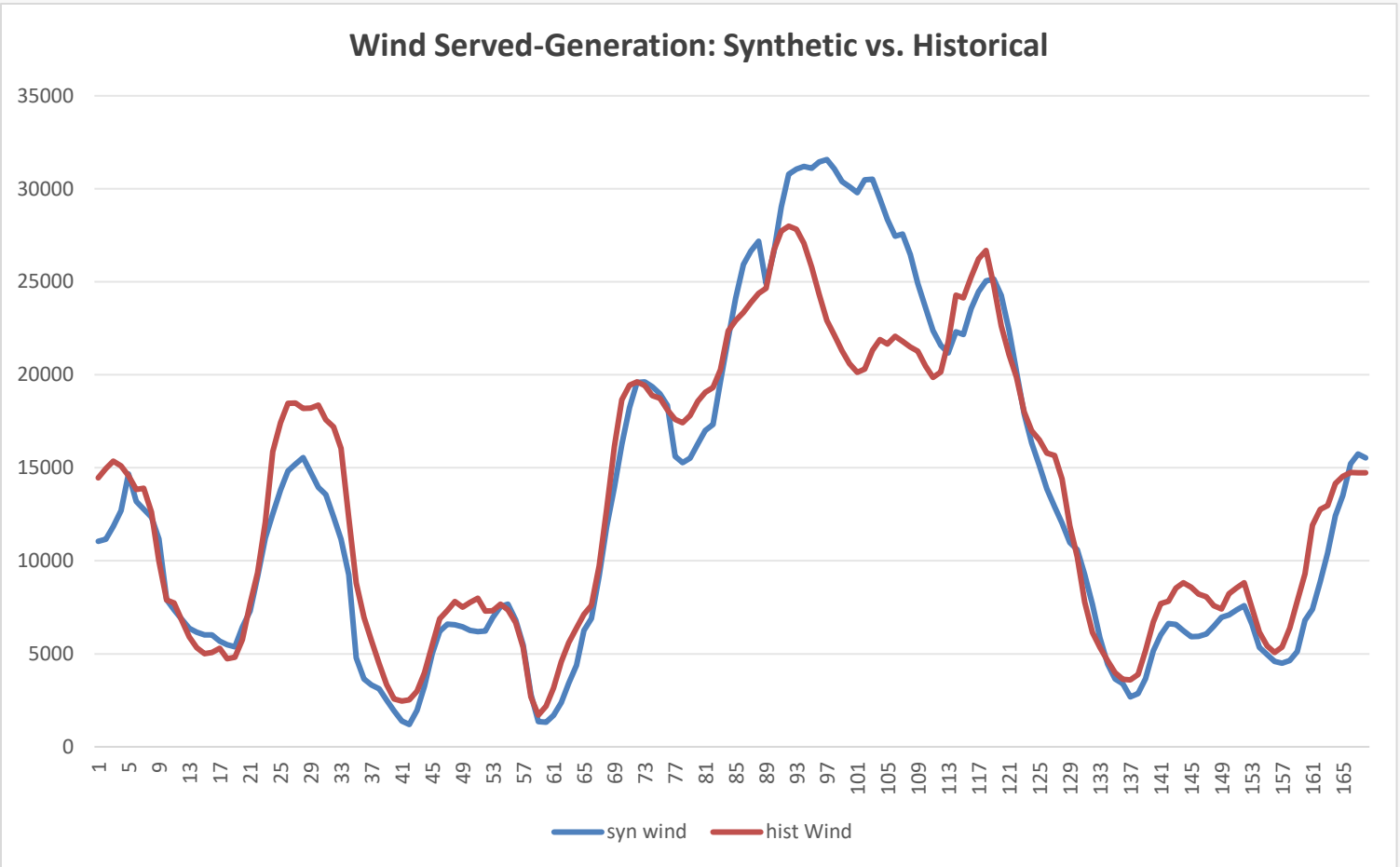
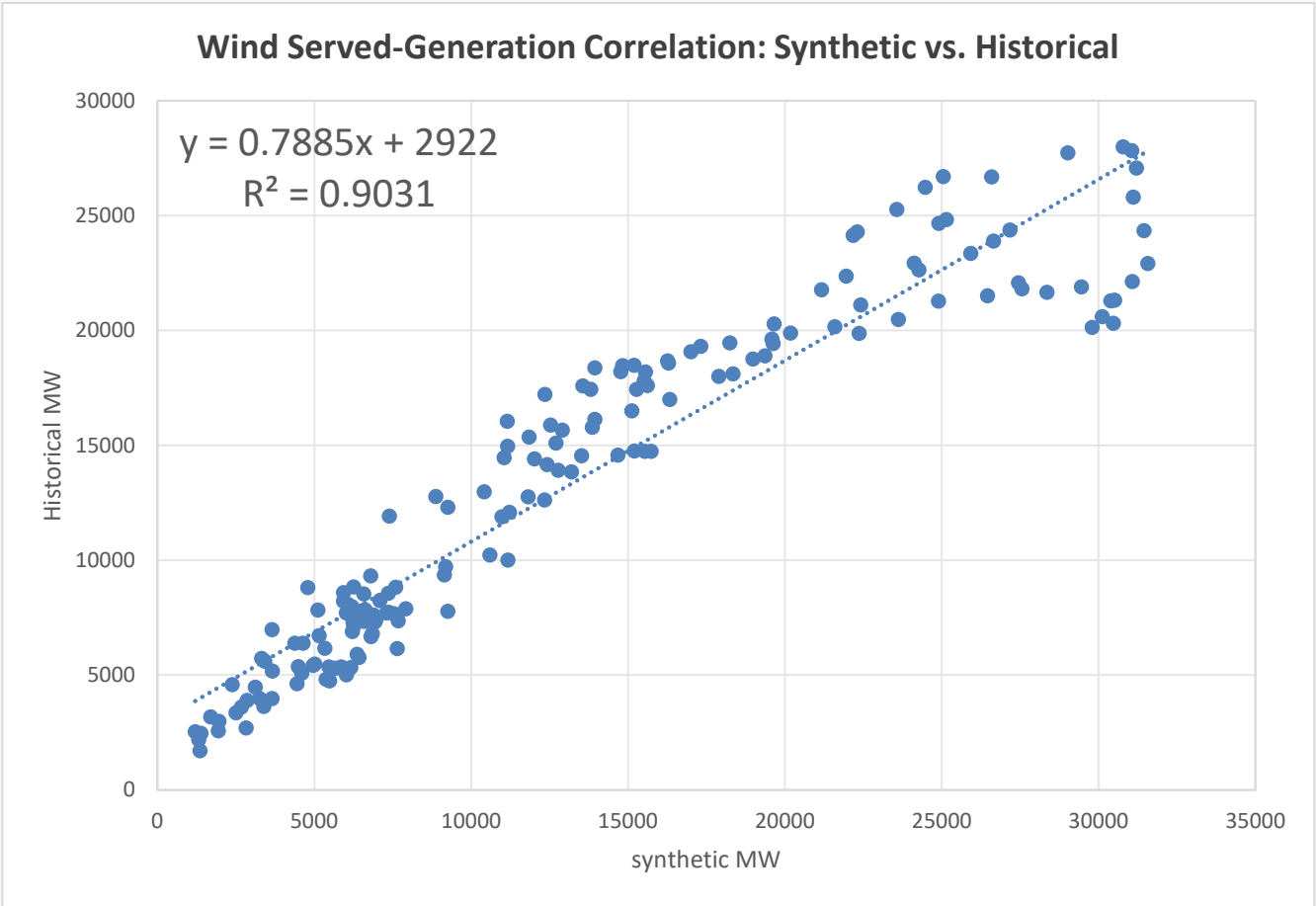


RMSE***

Historical Data Source:

- net gen: ERCOT Wind Power Production - Hourly Averaged Actual and Forecasted Values. Metric: HSL;
- served gen: ERCOT Fuel Mix Report: 2025

Wind Generation Modeling - Validation: Wind Served Generation



RMSE***

Historical Data Source:

- net gen: ERCOT Wind Power Production - Hourly Averaged Actual and Forecasted Values. Metric: HSL;
- served gen: ERCOT Fuel Mix Report: 2025

Milestone: Solar Generation Modeling

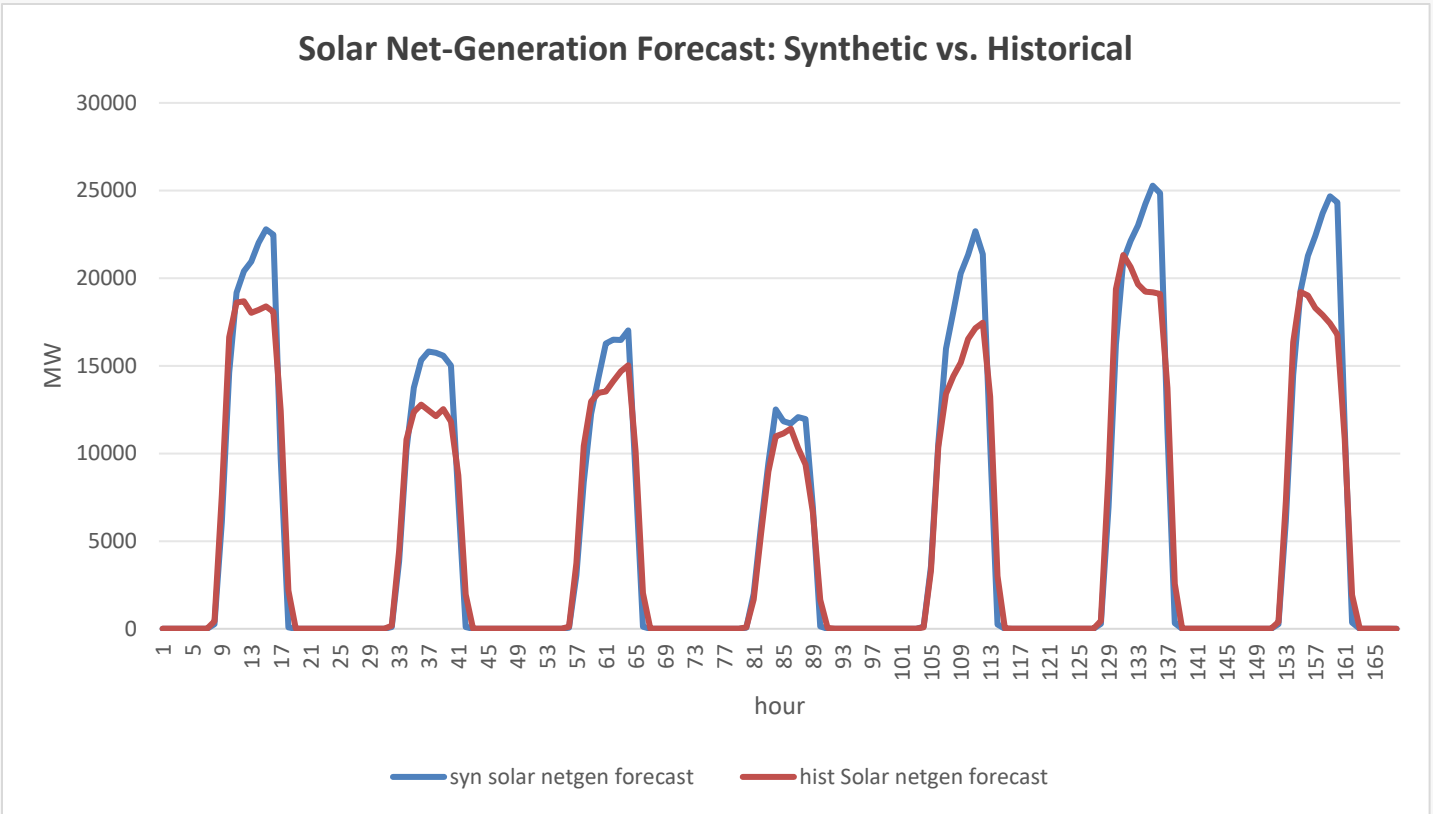
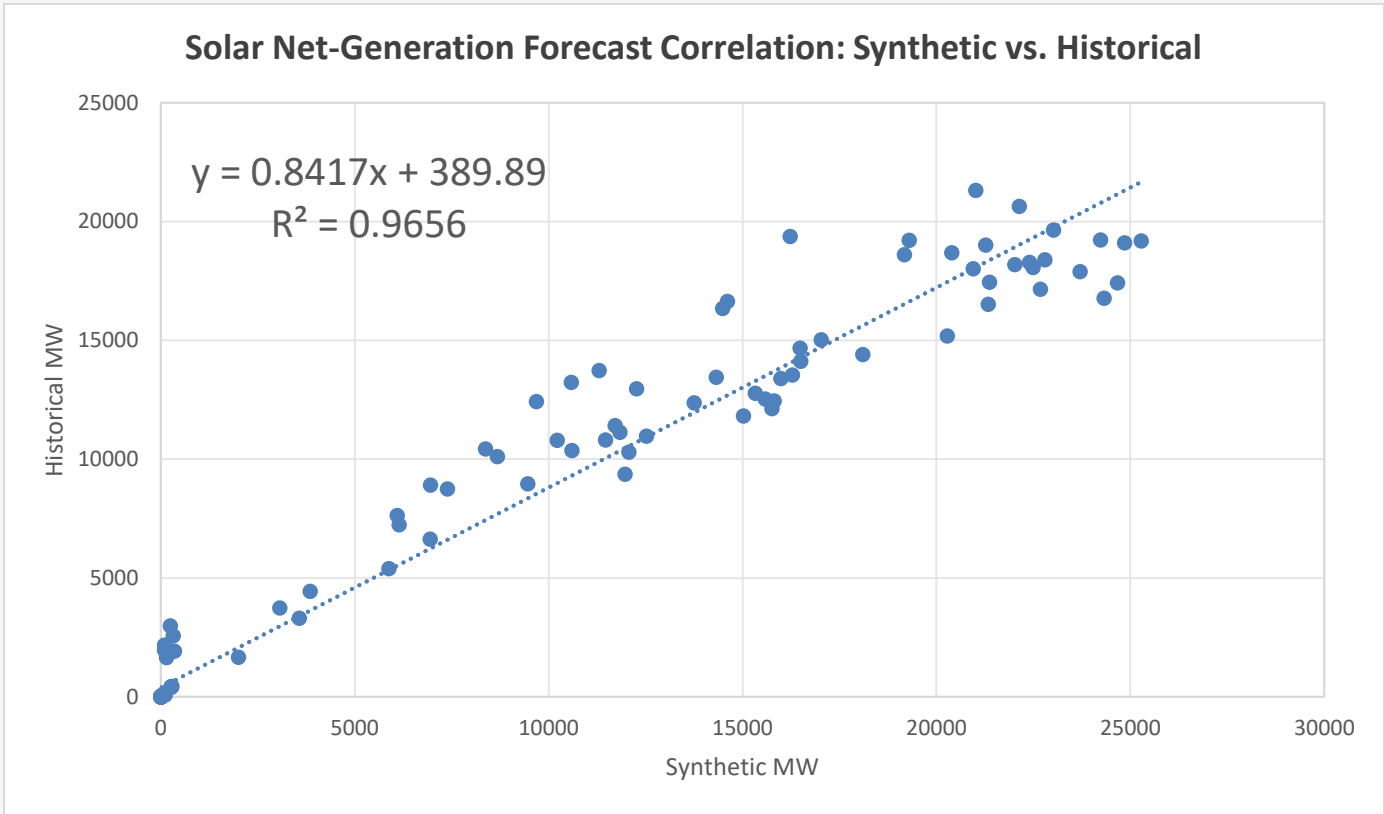
Solar Generation Modeling: Simulation Logics

We apply Pvlb to model solar generation of utility-scale PV solar stations in the synthetic grids.

Since the grid model is based on the ERCOT region. We assume that all solar stations are using single-axis tracking systems [1], Open Rack glass-to-glass module, with GCR = 0.35 [2], and LCR = 1.3 [1]. To consider the power losses due to snow accumulation and elimination, we also apply NREL snow model to correct the model.

1. **Solar Geometry & Decomposing Irradiance:** Calculates apparent zenith and azimuth using pvlib. Applies the ERBS model to decompose Global Horizontal Irradiance (GHI) into Direct Normal Irradiance (DNI) and Diffuse Horizontal Irradiance (DHI).
2. **Tracker Simulation:** Uses DNI and DHI to simulate dynamic single-axis tracking, finding the optimal tilt to calculate Plane of Array (POA) irradiance. (Assumes a Ground Coverage Ratio (GCR) of 0.35 and an OPEN RACK Glass to Glass setup).
3. **Power Conversion & Inverter Clipping:** Uses the SAPM model to estimate operating temperature, and PVWatts to calculate DC power. Applies inverter clipping based on a Load Capacity Ratio (LCR) of 1.3 to determine theoretical AC power.
4. **Losses & Export:** Applies the NREL snow model to calculate derating from panel coverage. Cleans up nighttime missing values and exports the final time-series data as a CSV.

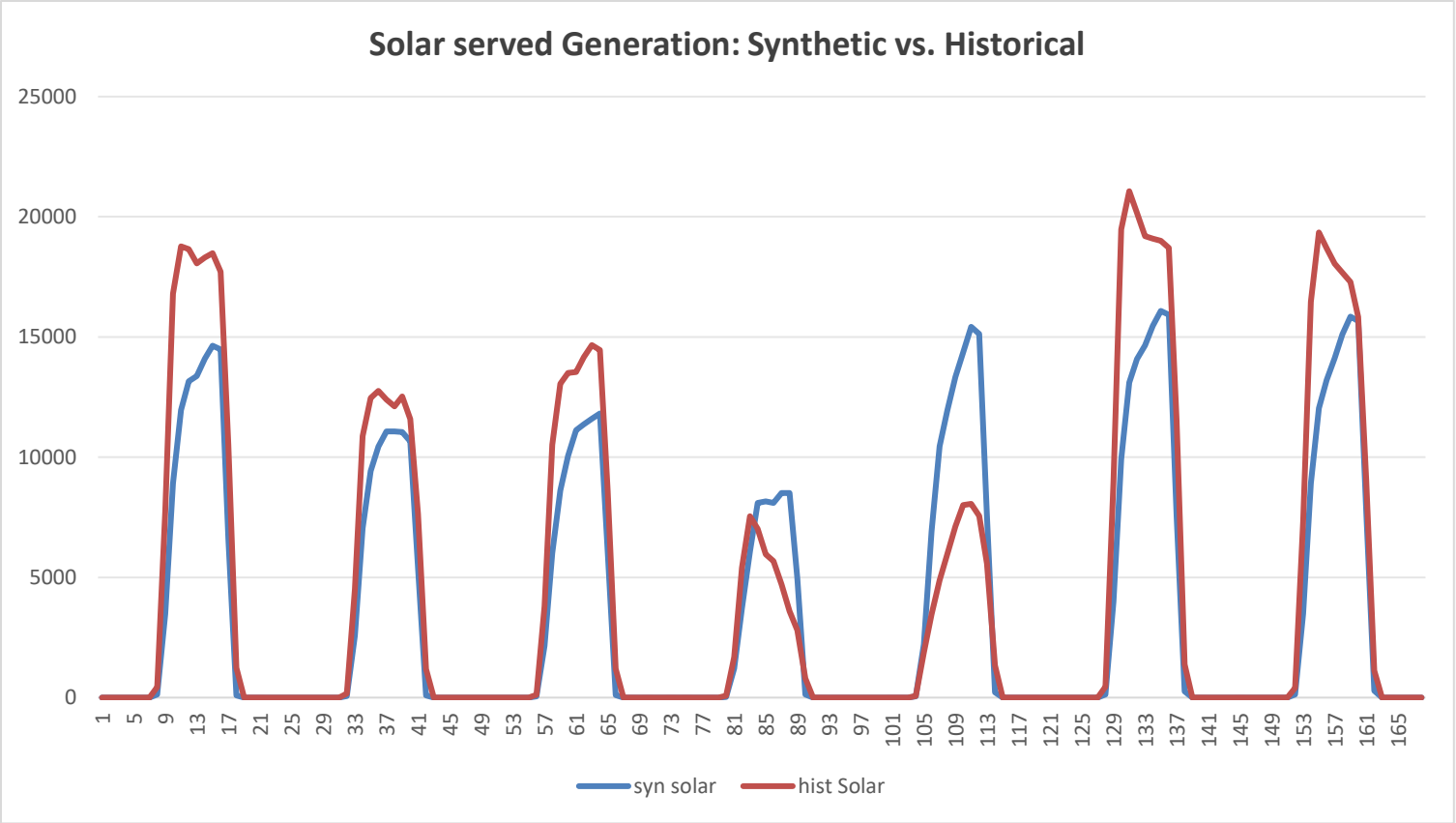
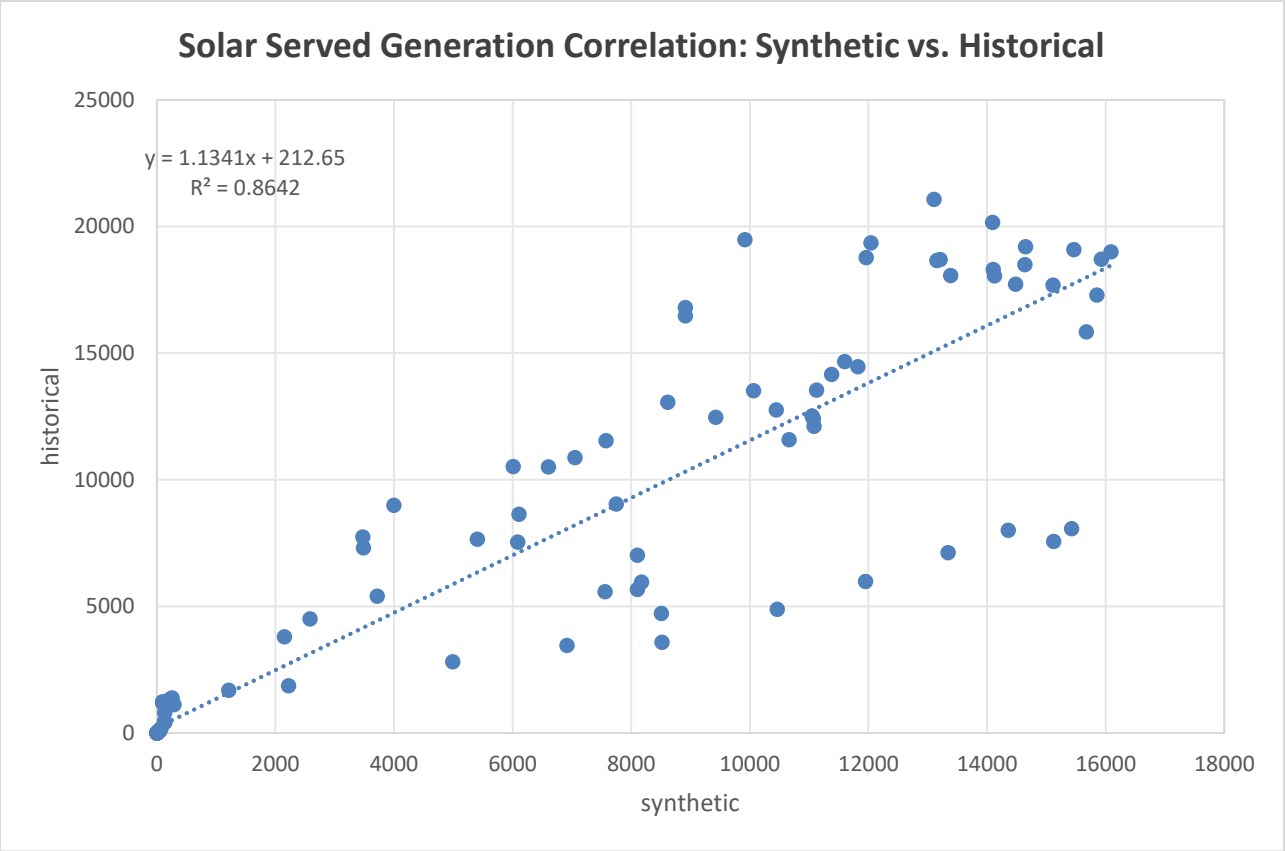
Solar Generation Modeling – Validation: Solar net generation



Historical Data Source:

- net gen: ERCOT Solar Power Production - Hourly Averaged Actual and Forecasted Values. Metric: HSL;
- served gen: ERCOT Fuel Mix Report: 2025

Solar Generation Modeling – Validation: Solar served generation



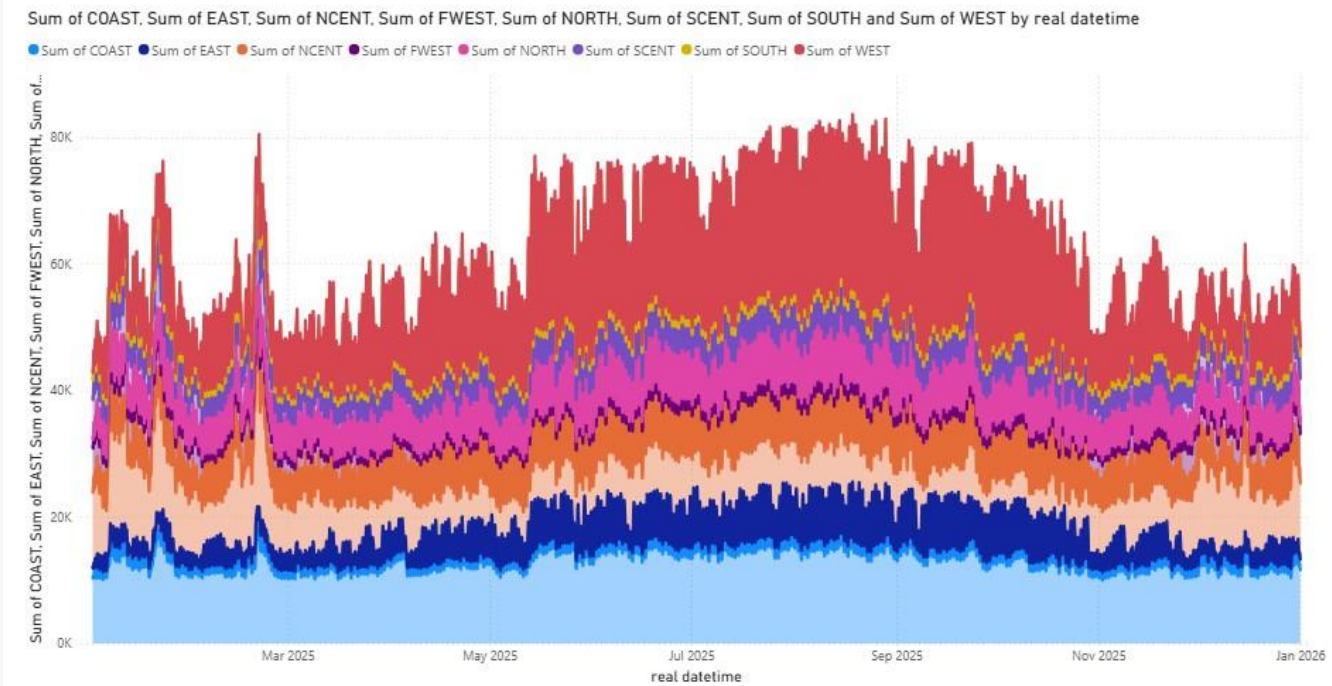
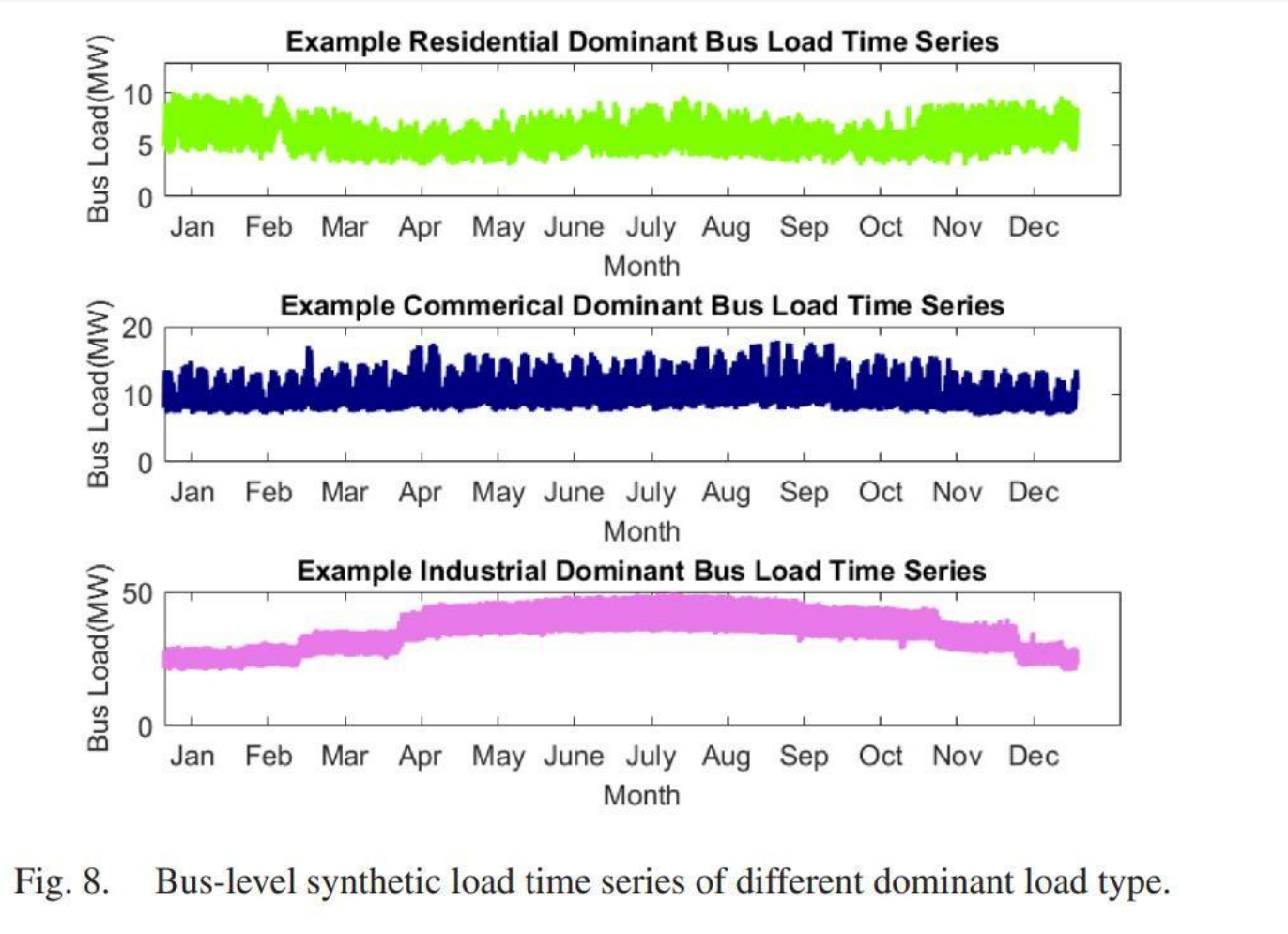
Historical Data Source:

- net gen: ERCOT Solar Power Production - Hourly Averaged Actual and Forecasted Values. Metric: HSL;
- served gen: ERCOT Fuel Mix Report: 2025

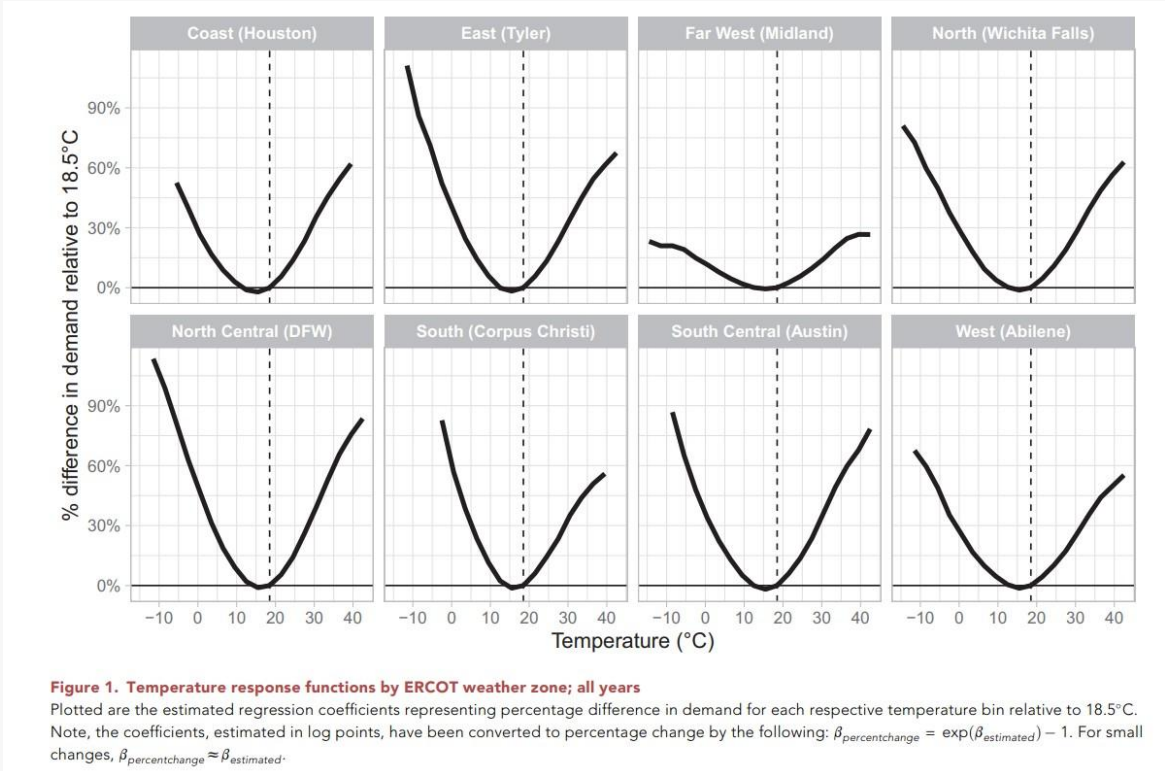
Milestone: Load Modeling

Load modeling: The Core Idea

For each bus on the synthetic grid, 3 load curves (residential, commercial, industrial) with its temperature sensitivity model will be assigned.



2025 ERCOT load profile separated by weather zone



2021 level ERCOT regular winter load temperature sensitivity model separated by ERCOT weather zone

Load Decoupling: Load Ratio in county level

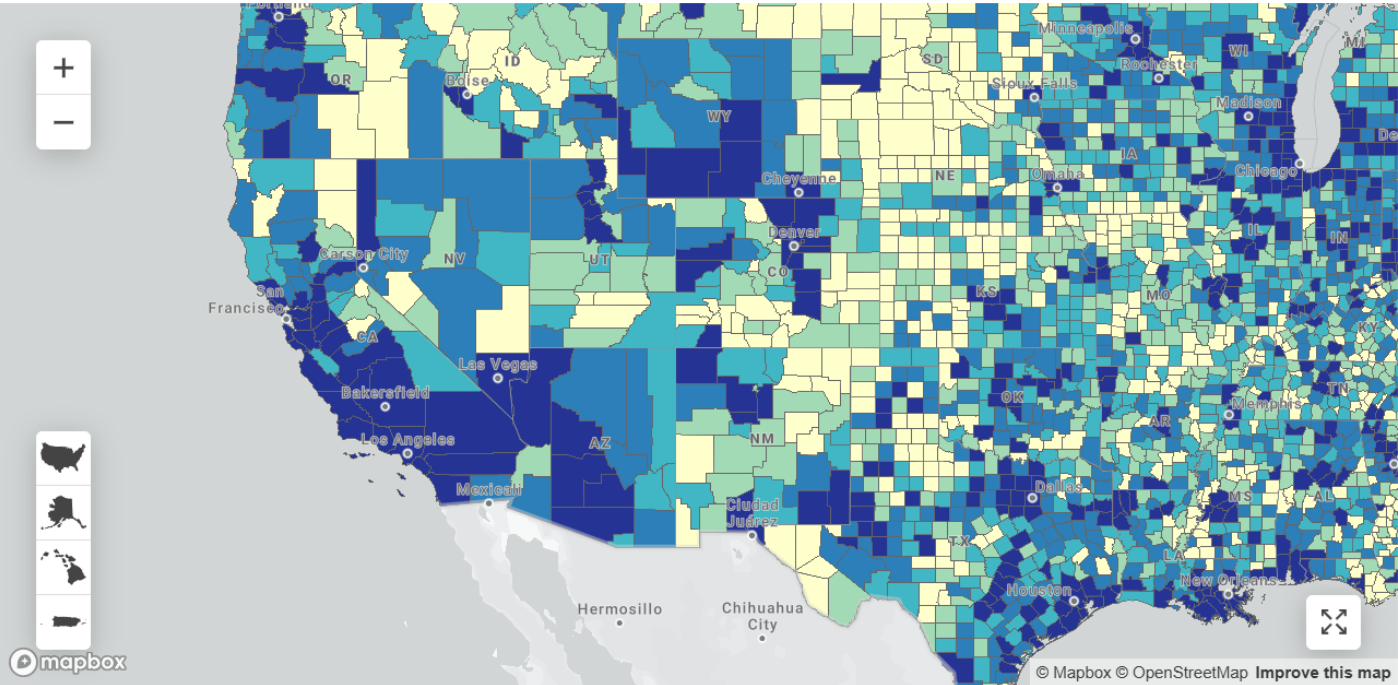
To take the first step on decoupling a bus load, we look into the county and the load ratio of the bus, based on the bus’s coordinates (longitude, latitude).

Data Snapshot:

	A	B	C	D	E	F
1	County Name	State Name	Sector	Year	Source	Consumption MMBtu
3474	Anderson	Texas	commercial	2025	elec	598661.6693
3508	Anderson	Texas	residential	2025	elec	811442.4747
3542	Anderson	Texas	industrial	2025	elec	525741.0723
3576	Andrews	Texas	commercial	2025	elec	188391.6182
3610	Andrews	Texas	residential	2025	elec	267185.9996
3644	Andrews	Texas	industrial	2025	elec	1100175.106
3678	Angelina	Texas	commercial	2025	elec	1225769.238
3712	Angelina	Texas	residential	2025	elec	1476027.126
3746	Angelina	Texas	industrial	2025	elec	1088025.425
3780	Aransas	Texas	commercial	2025	elec	308165.11
3814	Aransas	Texas	residential	2025	elec	502074.0752
3848	Aransas	Texas	industrial	2025	elec	124677.6604
3882	Archer	Texas	commercial	2025	elec	78606.41846
3916	Archer	Texas	residential	2025	elec	178735.7684
3950	Archer	Texas	industrial	2025	elec	568794.5527
3984	Armstrong	Texas	commercial	2025	elec	21737.23182
4018	Armstrong	Texas	residential	2025	elec	29888.04665
4052	Armstrong	Texas	industrial	2025	elec	39485.48561
4086	Atascosa	Texas	commercial	2025	elec	434933.683
4120	Atascosa	Texas	residential	2025	elec	735485.6595
4358	Atascosa	Texas	industrial	2025	elec	260272.5216
5888	Austin	Texas	commercial	2025	elec	343820.6092
5922	Austin	Texas	residential	2025	elec	614518.119
5956	Austin	Texas	industrial	2025	elec	463868.3598
5990	Bailey	Texas	commercial	2025	elec	114651.9404
6024	Bailey	Texas	residential	2025	elec	92457.46247
6058	Bailey	Texas	industrial	2025	elec	217371.1447
6092	Bandera	Texas	commercial	2025	elec	162013.6183
6126	Bandera	Texas	residential	2025	elec	373311.3506
6364	Andrews	Texas	commercial	2025	ng	54944.45752
6398	Bastrop	Texas	commercial	2025	ng	380277.1779
6432	Blanco	Texas	commercial	2025	ng	60197.27129
6466	Calhoun	Texas	commercial	2025	ng	109133.1405

ENERGY CONSUMPTION - NET ELECTRICITY AND NATURAL GAS CONSUMPTION

Aggregate Electricity & Natural Gas Consumption by County

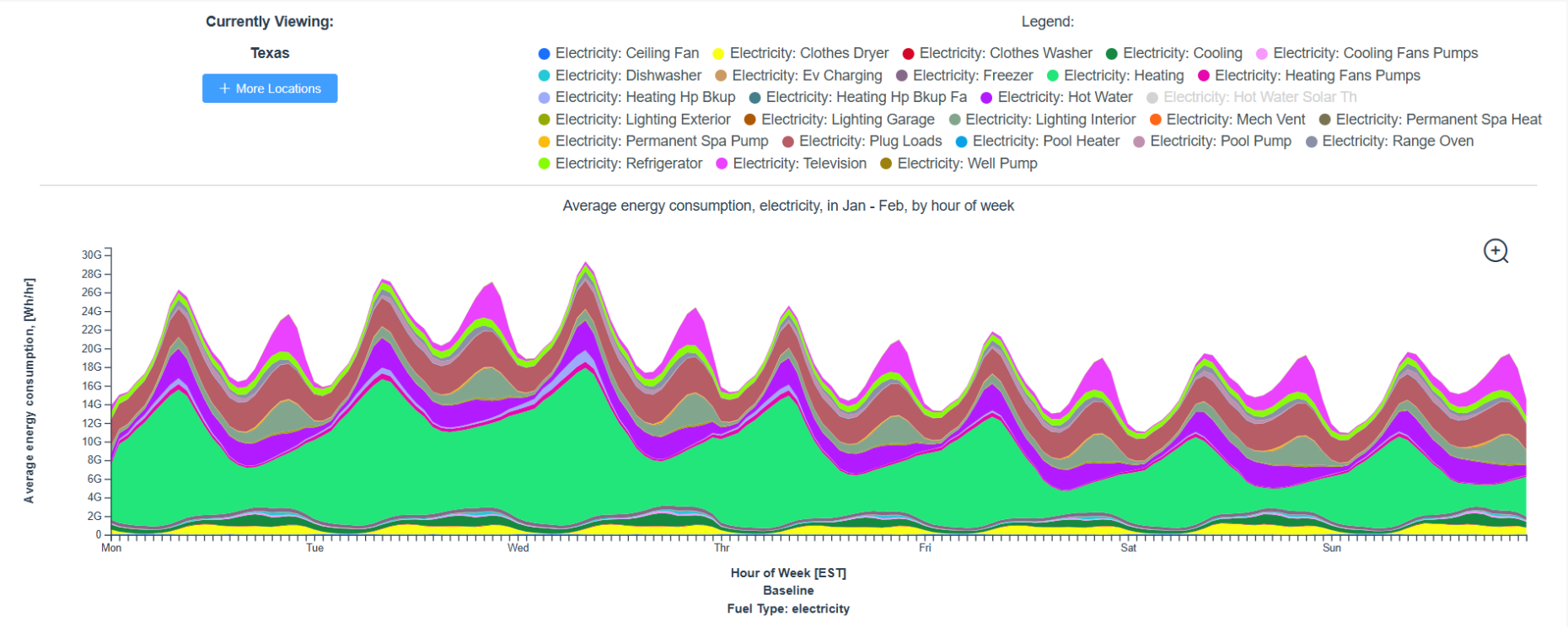


Source: NREL SLOPE: Energy consumption – net electricity and natural gas consumption in county resolution.

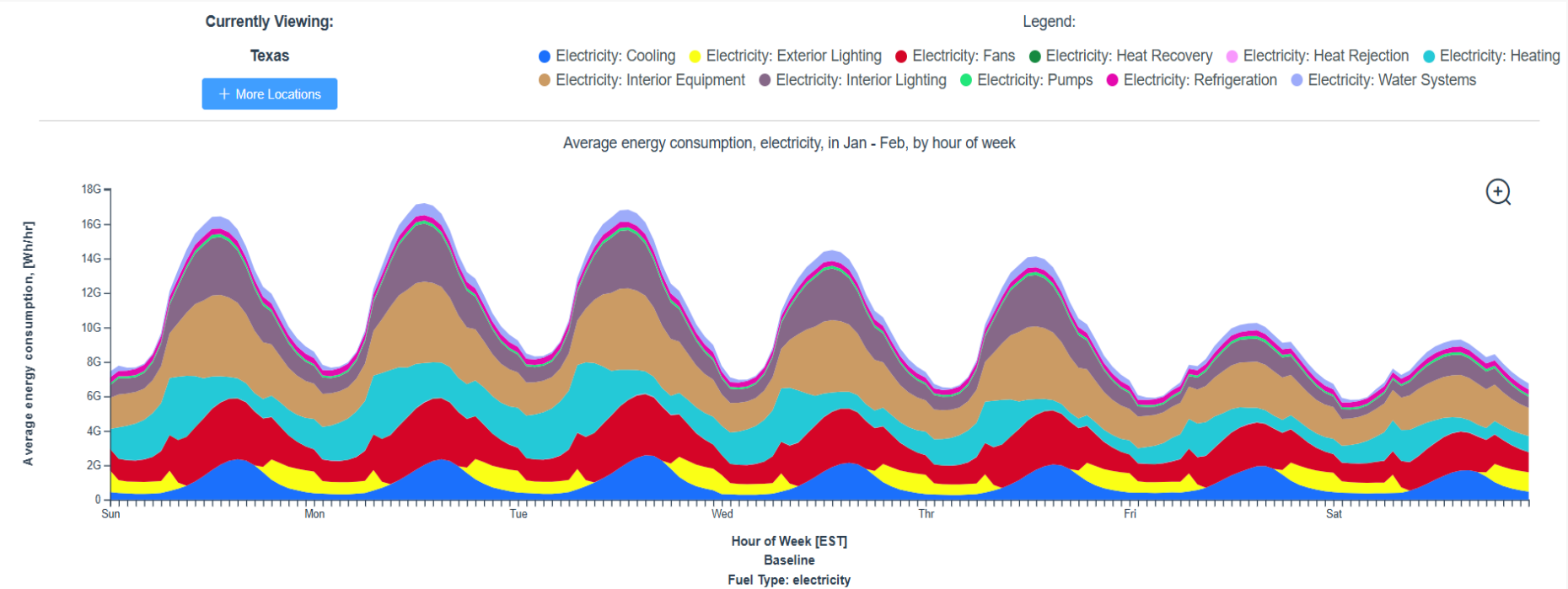
Load Decoupling: Residential Load Curve and Commercial load Curve

Source: NREL ResStock,
ComStock: baseline Texas
Jan – Feb weekly average
energy consumption

ResStock: Texas Residential Load Profile.



ComStock: Texas Commercial Load
Profile.



Load Decoupling: Industrial load Curve

Source: ORNL: Assessment of Industrial Load for Demand Response across U.S. Regions of the Western Interconnect

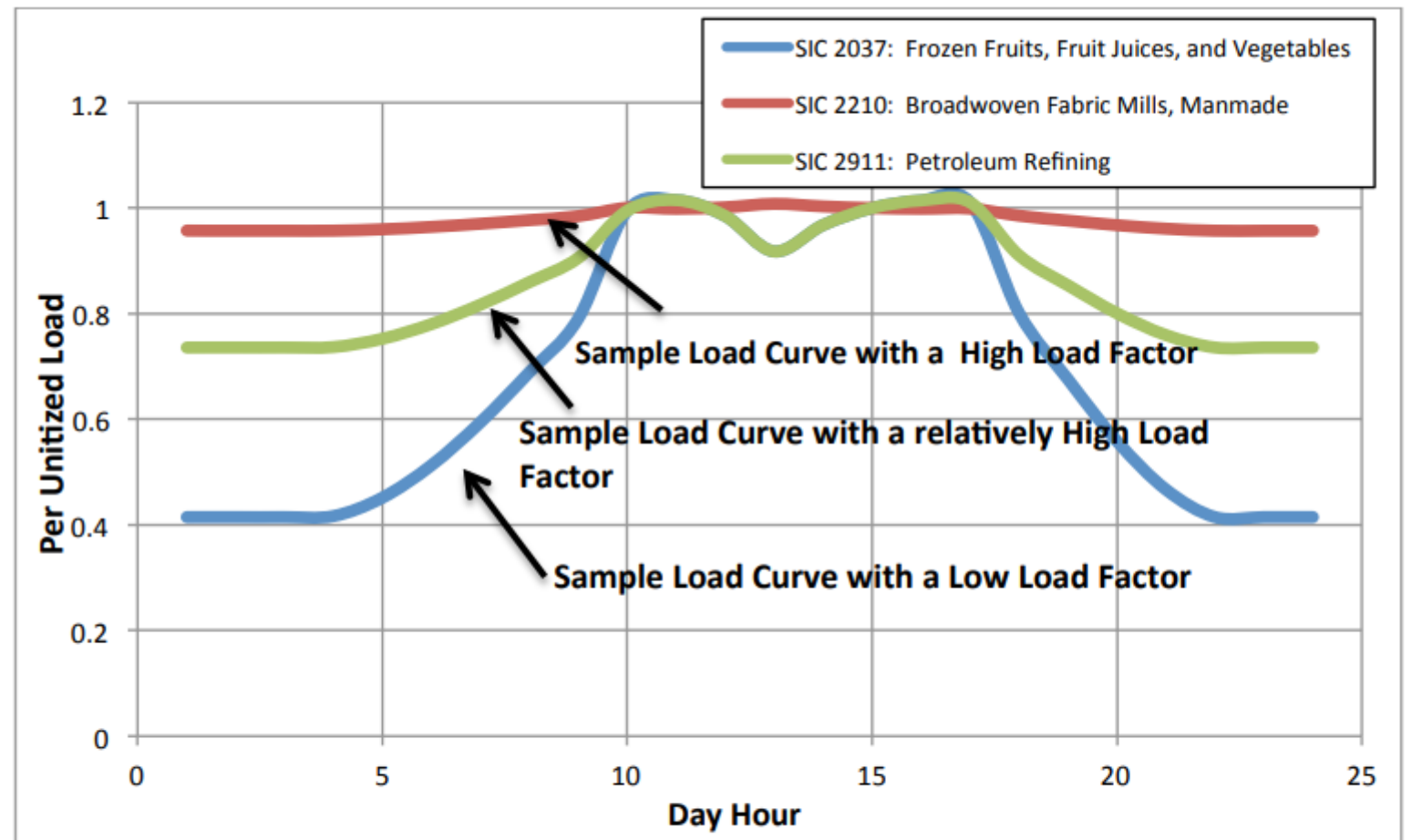


Figure B. 3 Load curves for high load factor and low load factor.

Load Decoupling: Behind-the-meter rooftop solar (BTM DG PV) capacity in county level

Data Snapshot

Source:

EIA861M: Small Scale PV Estimate, Monthly totals

US Census Bureau: County Population Totals 2025

A	B	C	D	E	F	G
FIPS	State	County	County Population	BTM PV Res Cap	BTM PV Comm Cap	
	48001 TX	Anderson	59805	5.483538391	0.6533908	
	48003 TX	Andrews	18914	1.734230334	0.2066421	
	48005 TX	Angelina	88154	8.082866705	0.9631136	
	48007 TX	Aransas	26146	2.397334583	0.2856543	
	48009 TX	Archer	9129	0.837040749	0.0997376	
	48011 TX	Armstrong	1843	0.168985223	0.0201354	
	48013 TX	Atascosa	53590	4.913683176	0.5854897	
	48015 TX	Austin	33625	3.083086337	0.367365	
	48017 TX	Bailey	6963	0.638439559	0.0760732	
	48019 TX	Bandera	22876	2.097507302	0.2499284	
	48021 TX	Bastrop	118908	10.90271019	1.2991119	
	48023 TX	Baylor	3472	0.318348722	0.0379328	
	48025 TX	Bee	32515	2.981310104	0.3552379	
	48027 TX	Bell	402248	36.88223979	4.3947015	
	48029 TX	Bexar	2160088	198.0591167	23.599725	
	48031 TX	Blanco	13581	1.245245964	0.1483772	
	48033 TX	Borden	567	0.0519884	0.0061947	
	48035 TX	Bosque	18918	1.734597095	0.2066858	
	48037 TX	Bowie	92696	8.499324048	1.0127366	
	48039 TX	Brazoria	419080	38.42557092	4.5785971	
	48041 TX	Brazos	249088	22.83895344	2.7213744	
	48043 TX	Brewster	9458	0.867206857	0.103332	
	48045 TX	Briscoe	1474	0.135151502	0.016104	
	48047 TX	Brooks	6579	0.603230484	0.0718779	
	48049 TX	Brown	38711	3.549423203	0.4229314	
	48051 TX	Burleson	20549	1.884143974	0.2245051	
	48053 TX	Burnet	57015	5.227722454	0.622909	
	48055 TX	Caldwell	55150	5.056720044	0.6025332	
	48057 TX	Calhoun	19599	1.797038189	0.214126	
	48059 TX	Callahan	14491	1.328684137	0.1583193	
	48061 TX	Cameron	433946	39.78863892	4.7410134	
	48063 TX	Camp	13380	1.226816214	0.1461812	
	48065 TX	Carson	5805	0.532262191	0.0634217	

	I	J	K	L	M	N	O	P	
EIA 861M Small Scale PV Estimate				Capacity MW					
Year	Month	State	Data Status	Residential	Commercial	Industrial	Total		
2025	12	TX	Preliminary	2907.483	346.441	3.326	3257.25		
County Population data from US Census Bureau									

Load Decoupling: BTM DGPV load curve

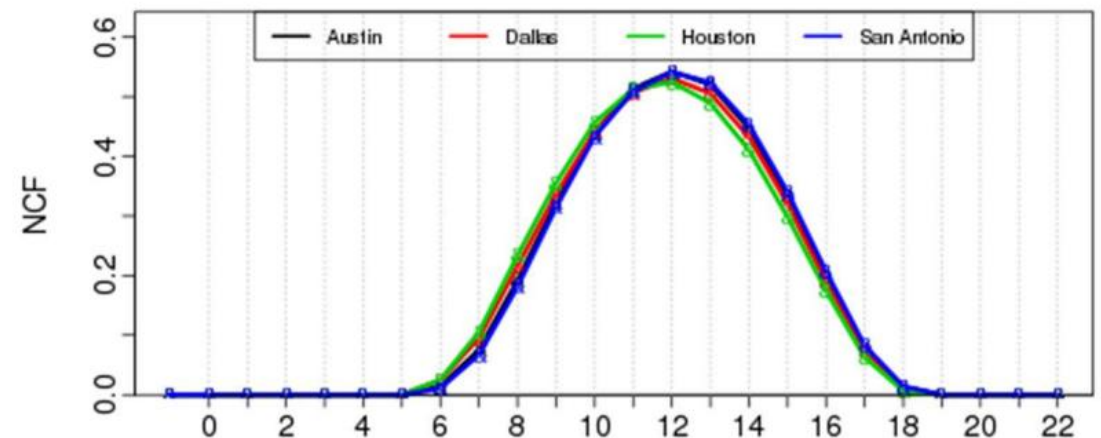


Figure 7.7: Diurnal Net Capacity Factor for DGPV metro areas (local standard time)

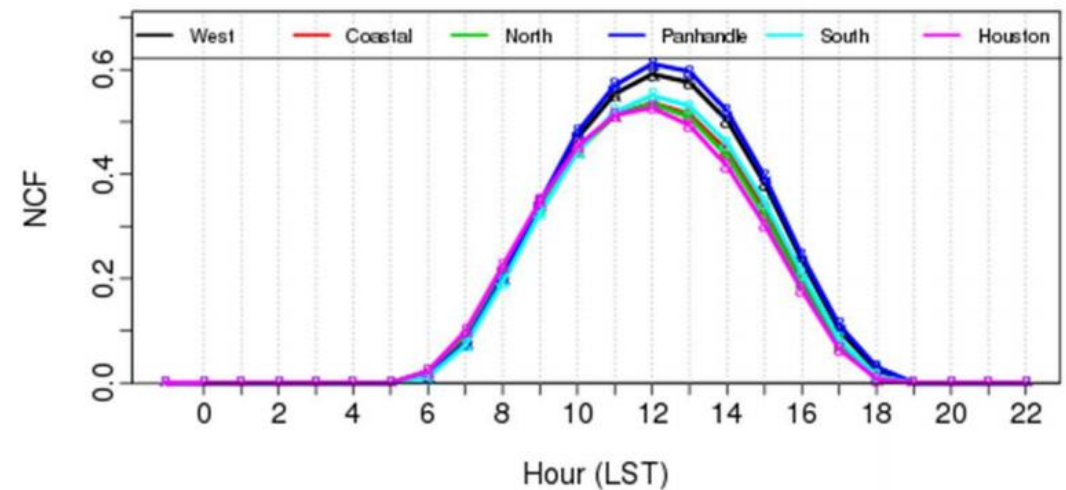


Figure 7.8: Diurnal Net Capacity Factor for DGPV Rural Zones (local standard time)

Source: HOURLY WIND AND SOLAR GENERATION PROFILES (1980-2020) By ERCOT, UL

Load Decoupling: Temperature Sensitivity in residential sector and commercial sector

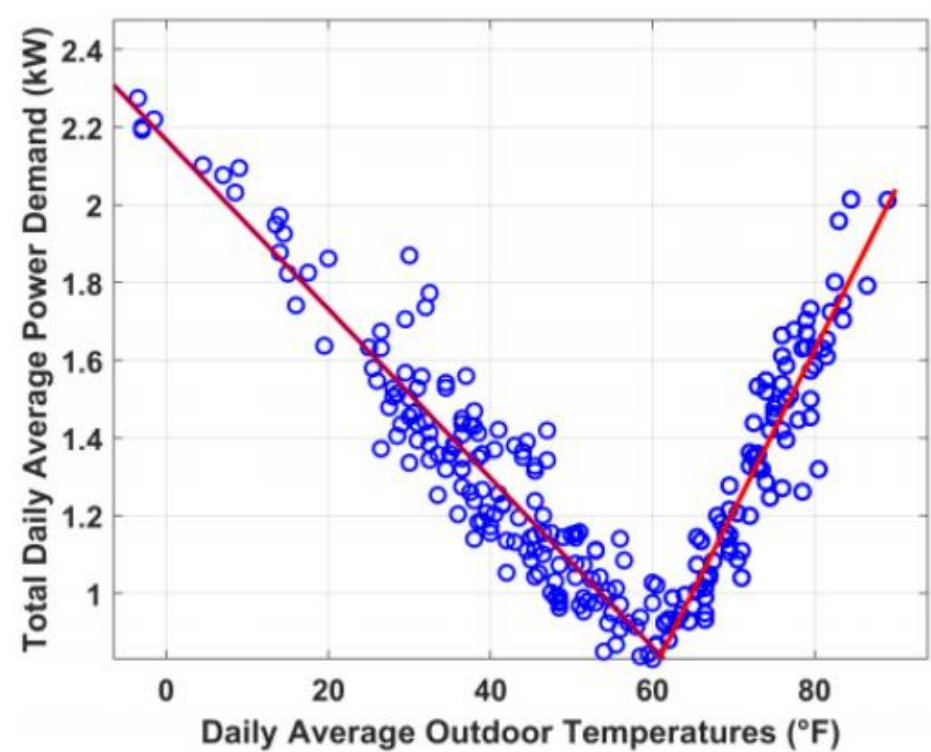


Fig. 5. The household power demand versus the temperature.

Residential Temperature Sensitivity
Source: Distributed Auto-Clustering for Residential Load Profiling Using AMI Data From the U.S. High Plains

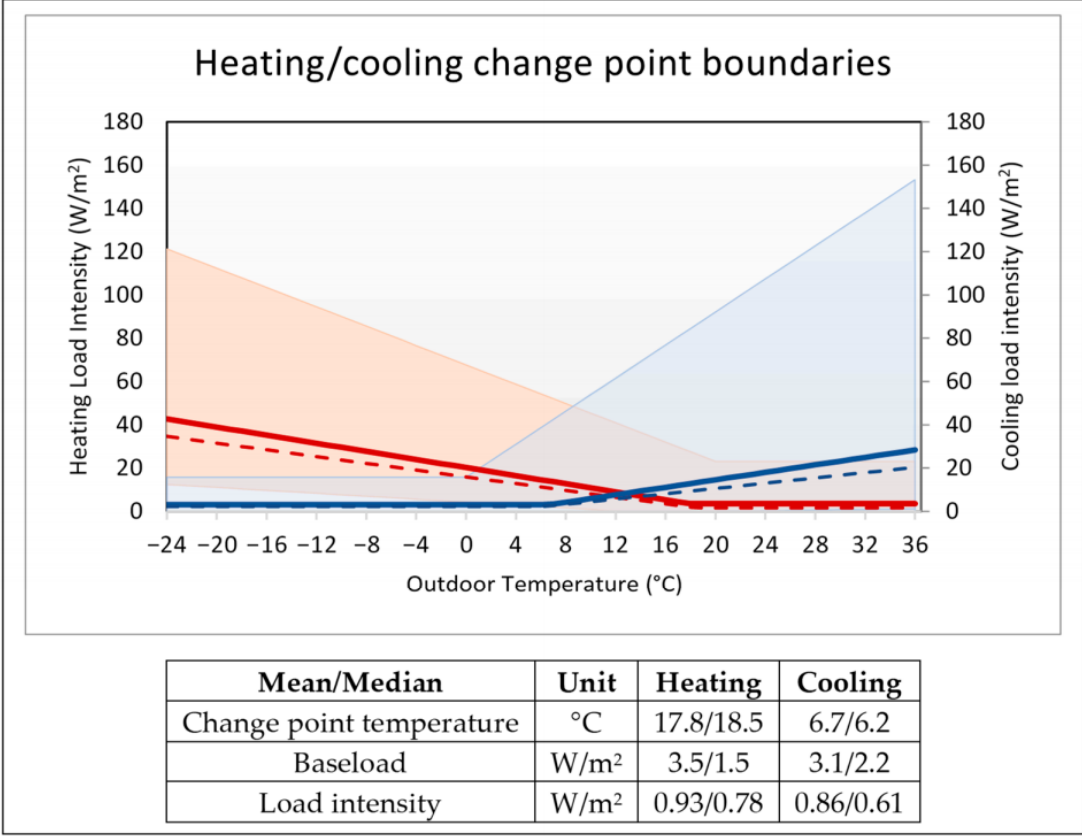


Figure 9. Heating/cooling change point model fit—mean of all study buildings. Solid lines: mean; dashed lines: median.

Commercial Temperature Sensitivity
Source: How Building Energy Use Reacted to Variable Occupancy Pre- and Post- COVID-19 Pandemic - Sensitivity Analysis of 35 Commercial Buildings in Canada

Load Decoupling: A note for Temperature Sensitivity

Load modeling with temperature sensitivity:

The temperature sensitivity of residential and commercial loads primarily stems from heating demand. Due to a lack of comprehensive data, constructing a heating load model using a bottom-up approach presents significant challenges. Specifically, existing public datasets (such as the EIA's RECS survey) lack the required granularity and timeliness for the ERCOT model, both in terms of the penetration rates of various heating technologies and the distribution of fuel types. Without access to first-hand customer data from utilities, the accuracy of a bottom-up modeling approach is difficult to guarantee.

Therefore, adopting a top-down modeling approach driven by temperature sensitivity is currently a more practical choice. The model temporarily utilizes generic parameters from existing literature to approximate multi-regional performance. (Note: The referenced parameters were trained using data from the US High Plains region; applying them directly to other regions may introduce certain modeling biases.)

Residential Space Heating Pathway Assumptions

Pathway	Share of Adoption (%)		Min. Operating Temp (F)	Approx. Seasonal Heating Supplied (%)	Reference HP
	2034	2050			
Air-to-Water	0.2	0.2	-5	100	Taco Comfort System M
Ducted Full	14.5	18.9	5	100	Lennox Central Heat Pump
Ducted Partial	47.2	39.1	5	63-74	
Ductless Full	21.9	26.8	-13	100	Mitsubishi M-Series
Ductless Partial	13.2	10.3	-13	65-76	Daikin DZ6VS
Ground Source Heat Pump (GSHP)	1.7	2.2	-20	100	Energy Star rated models
Packaged Terminal Heat Pump (PTHP)	1.2	2.5	5	81	ICE-AIR HP

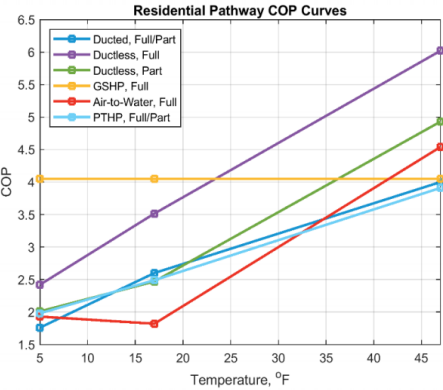


Table HC6.8 Space heating in homes in the South and West regions, 2020

	Number of housing units (million)									
	South Census Region					West Census Region				
	Total U.S. ^a	Total South	South Atlantic	East South Central	West South Central	Mountain				
						Total West	Total Mountain	Mountain North	Mountain South	Pacific
All homes	123.53	46.84	24.84	7.38	14.62	27.72	9.22	4.62	4.60	18.51
Space heating equipment										
Uses space heating equipment	117.74	43.77	22.38	7.33	14.06	25.25	8.90	4.60	4.31	16.35
Has space heating equipment but does not use it	3.92	2.16	1.75	Q	0.37	1.60	0.25	Q	0.23	1.35
Does not have space heating equipment	1.87	0.91	0.71	Q	0.19	0.87	0.06	N	0.06	0.80
Main heating fuel and equipment										
Natural gas	62.71	15.54	6.86	2.55	6.13	16.13	5.77	3.44	2.33	10.36
Central warm-air furnace	53.26	14.36	6.27	2.37	5.72	14.22	5.19	3.11	2.08	9.02
Steam or hot water system	6.51	0.47	0.40	Q	Q	0.50	0.28	0.22	0.06	0.21
Built-in room heater	2.77	0.63	0.16	0.12	0.35	1.35	0.26	0.09	0.17	1.09
Some other equipment	0.18	0.07	Q	Q	Q	0.06	0.03	Q	Q	Q
Electricity	42.57	25.76	14.05	4.28	7.43	7.62	2.55	0.82	1.73	5.08
Central warm-air furnace	13.82	9.01	4.75	1.29	2.98	2.14	0.97	0.35	0.62	1.17
Heat pump	16.13	12.52	7.47	2.35	2.70	1.74	0.84	0.07	0.78	0.89

Source:

- Final 2025 Heat Pump Forecast, ISONE
- 2020 Residential Energy Consumption Survey (RECS) Survey Data, EIA

Load Modeling: Validation - legends

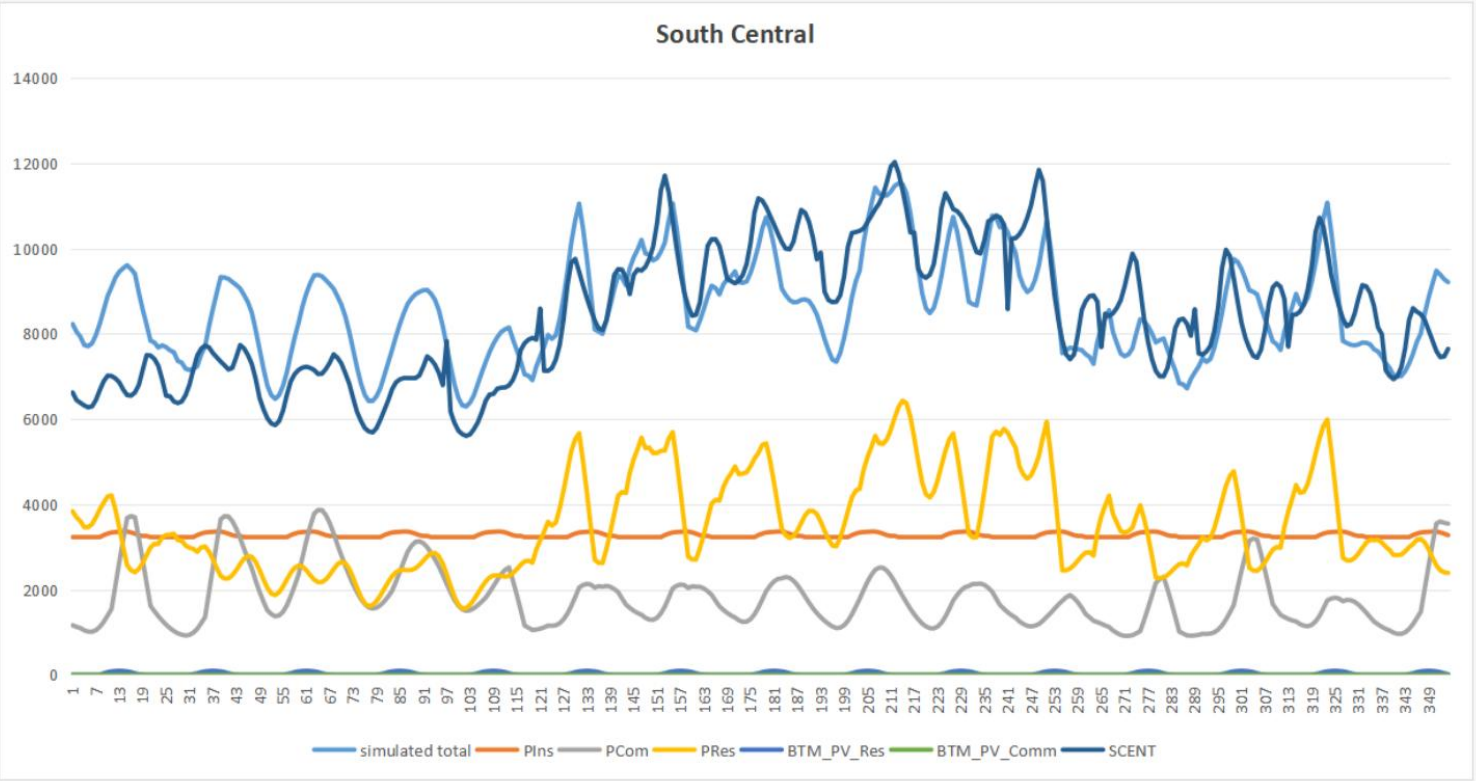
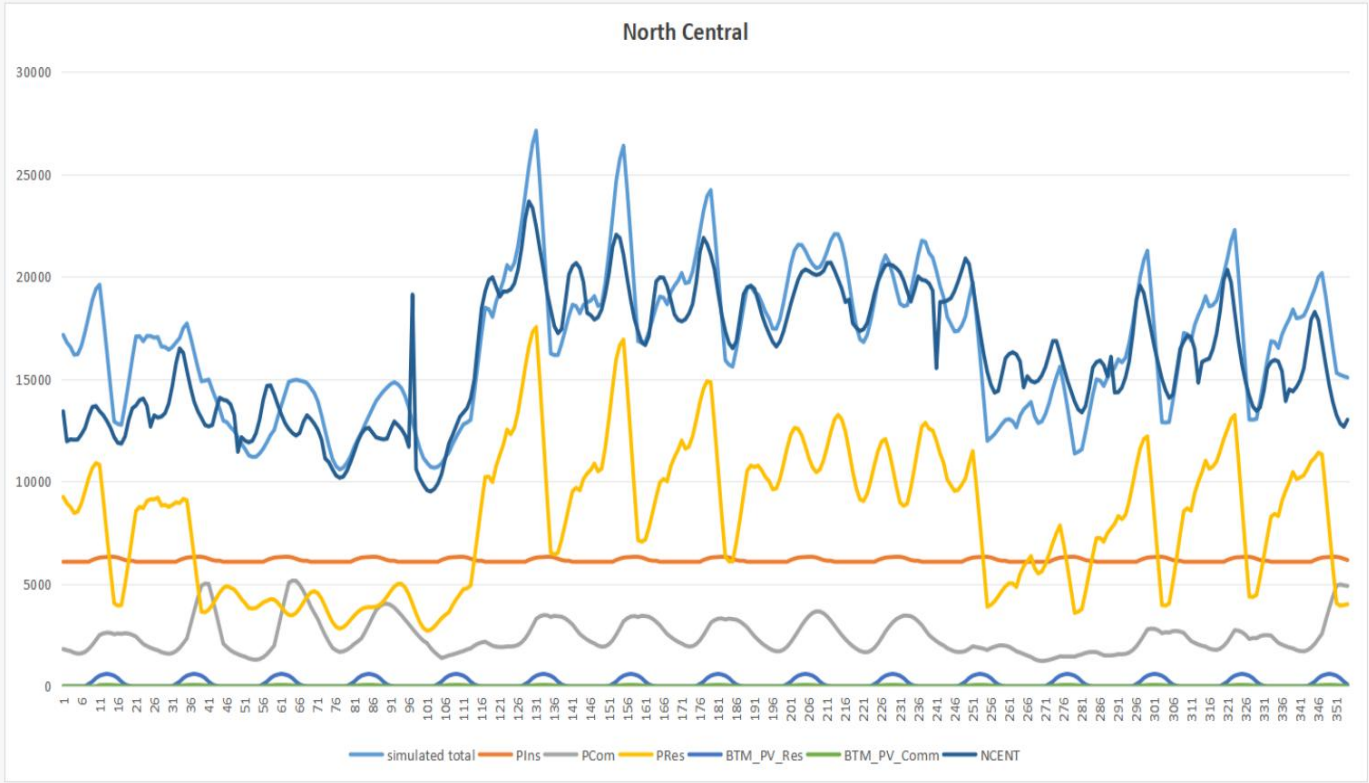
Legends Explain:

- simulated: The total simulated load for the given time step.
- PIns: Active power (P) demand for the Industrial sector.
- PCom: Active power (P) demand for the Commercial sector.
- PRes: Active power (P) demand for the Residential sector.
- BTM_PV: Behind-The-Meter Photovoltaics (rooftop solar)
- avg temp [C]: Average ambient temperature in degrees Celsius.
- avg dewtemp [C]: Average dewpoint temperature, which indicates moisture levels in the air, in degrees Celsius.
- COAST, EAST, FWEST, NCENT, NORTH, SCENT, SOUTH, WEST: historical power demand in ERCOT weather zones.
- ERCOT: The system-wide, aggregate total for the entire grid footprint.

Historical Data Source:

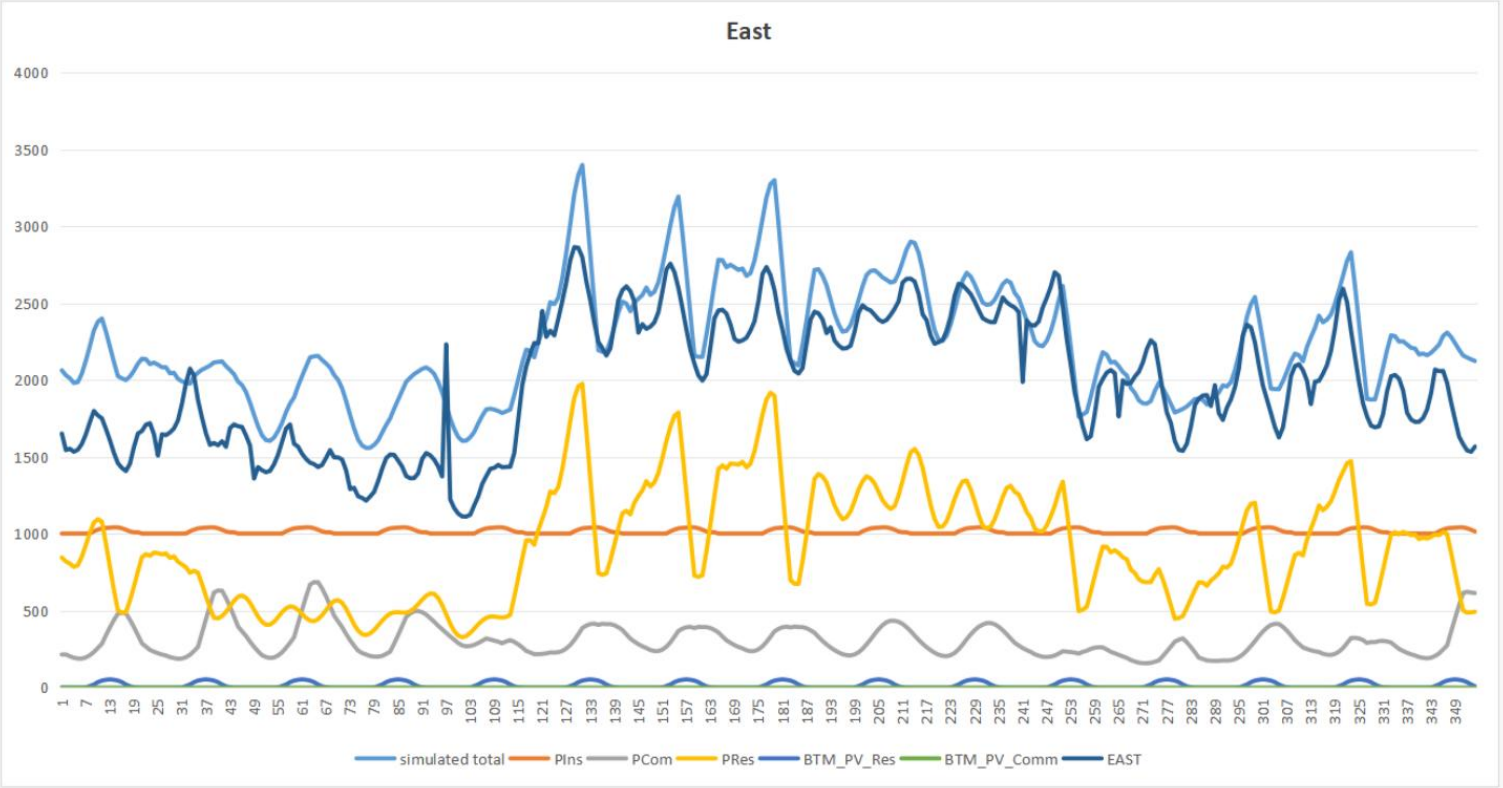
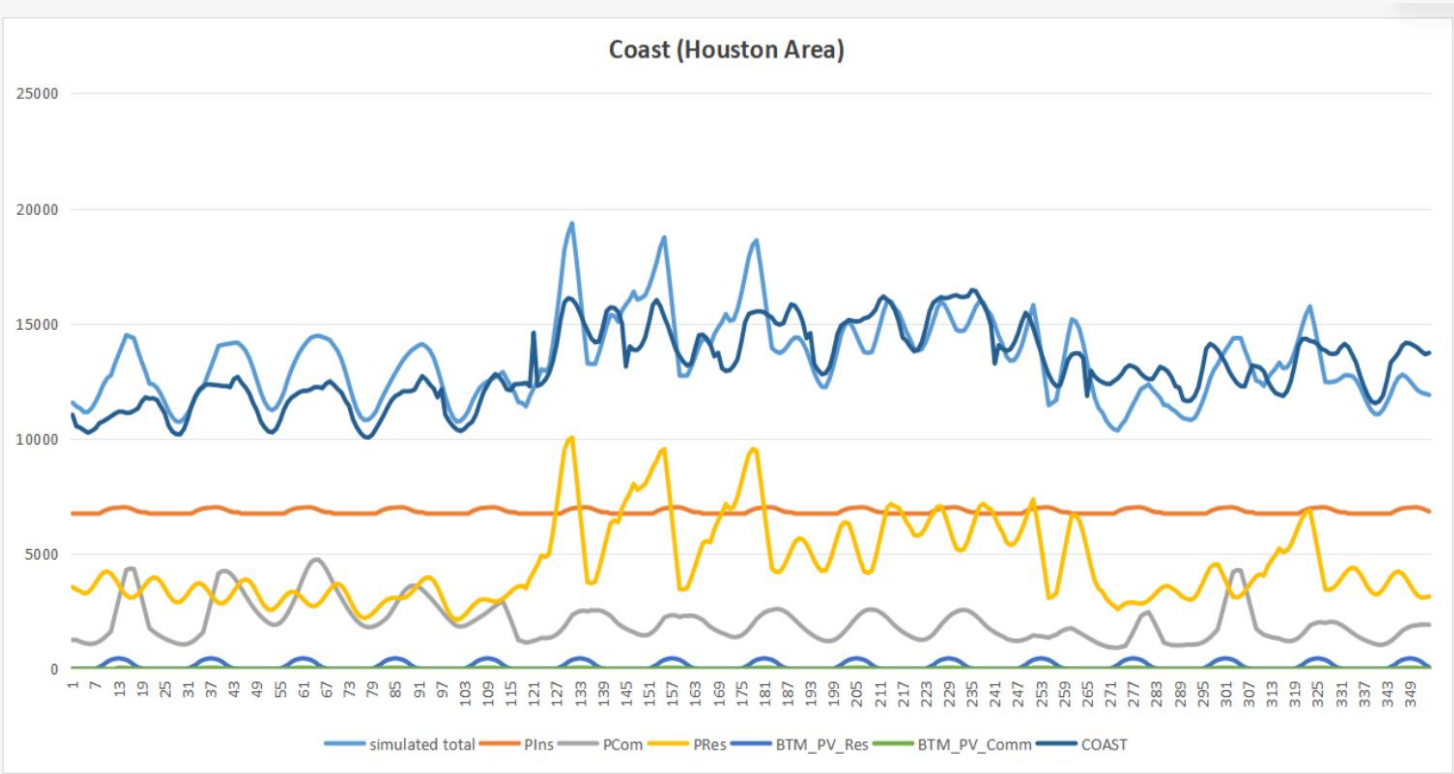
Historical load: ERCOT Hourly Load Data Archives Y2025

Load Modeling: Validation



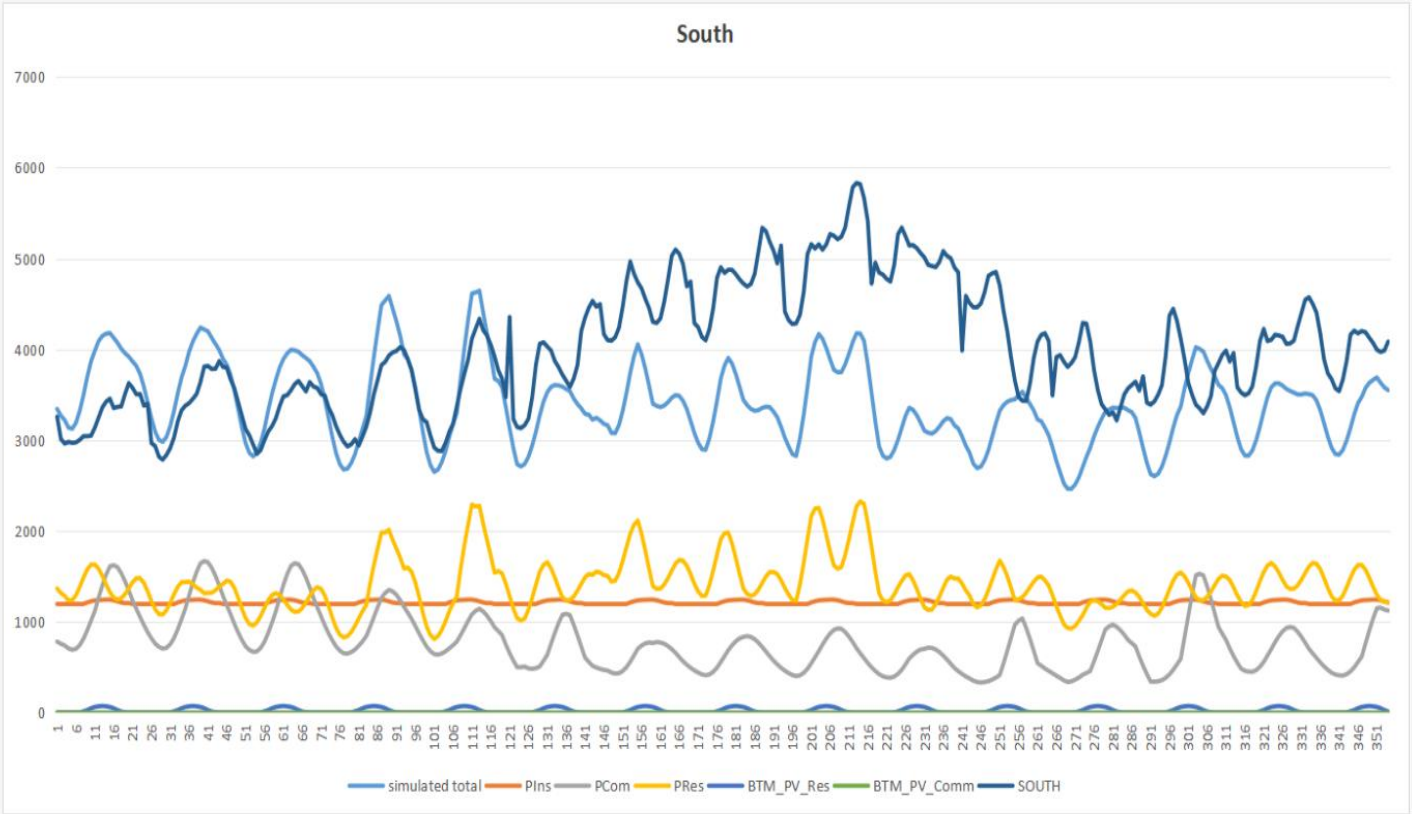
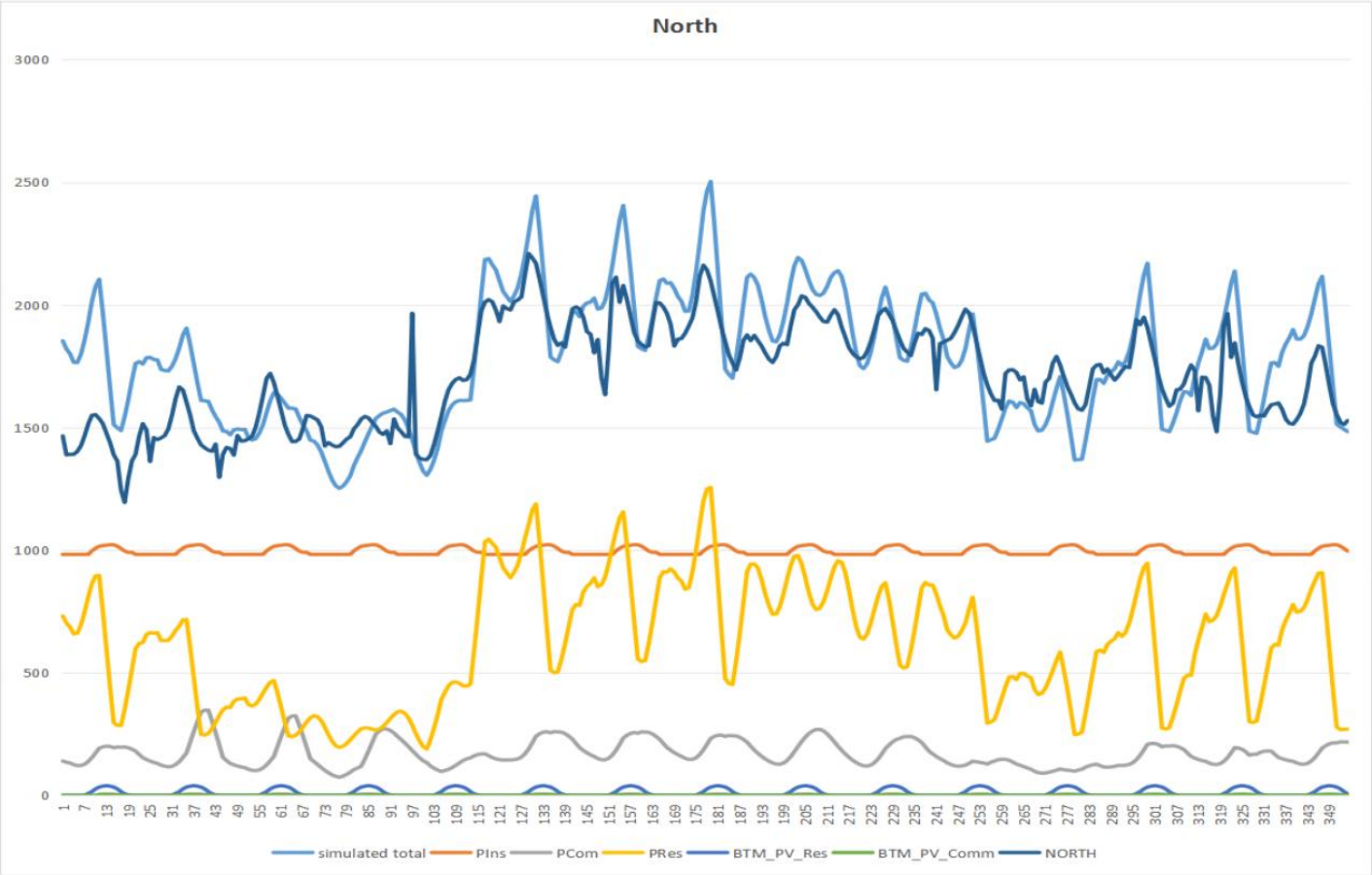
Historical Data Source:
Historical load: ERCOT Hourly Load Data Archives Y2025

Load Modeling: Validation



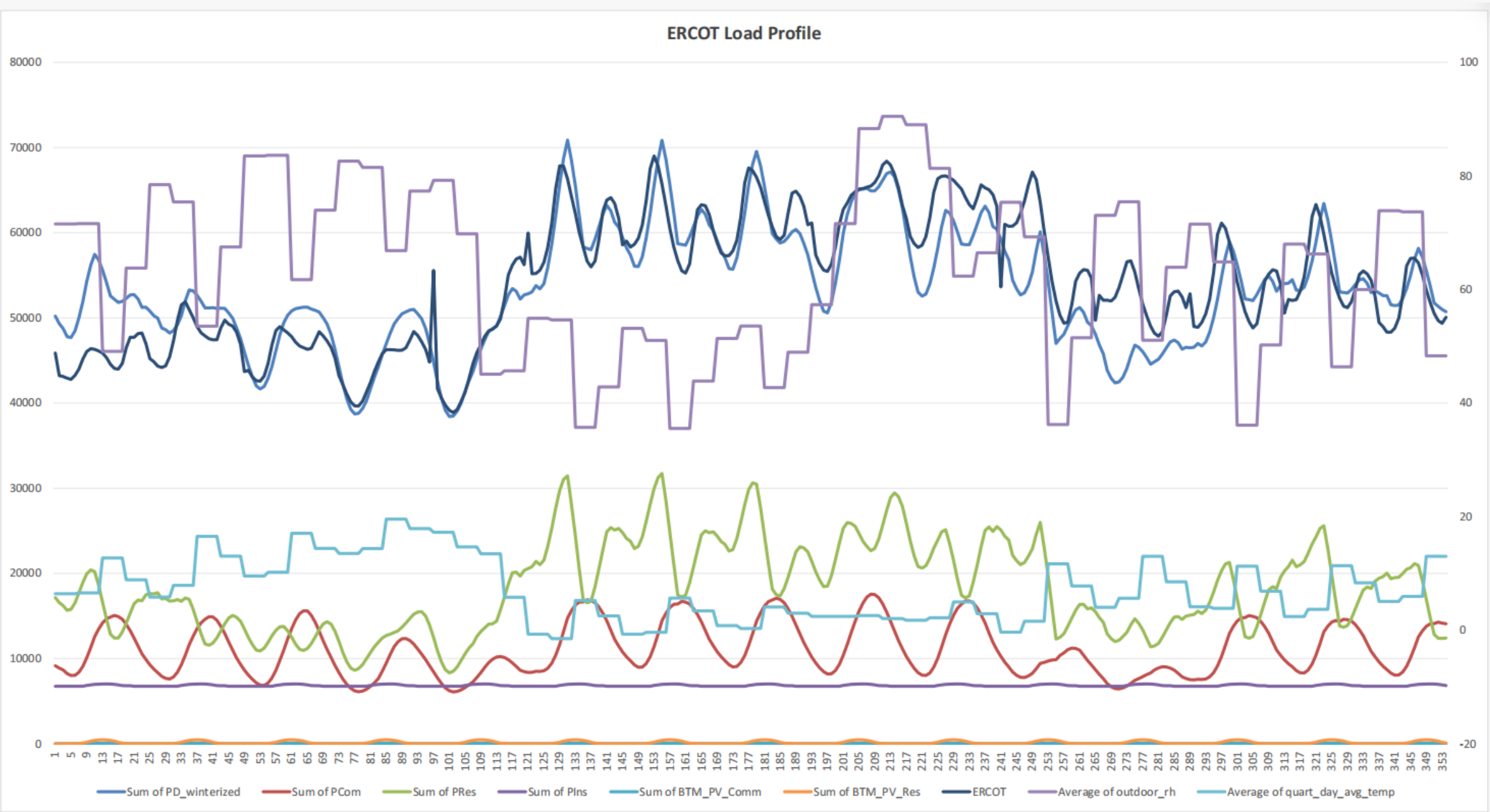
Historical Data Source:
Historical load: ERCOT Hourly Load Data Archives Y2025

Load Modeling: Validation



Historical Data Source:
Historical load: ERCOT Hourly Load Data Archives Y2025

Load Modeling: Validation



Historical Data Source:
Historical load: ERCOT Hourly Load Data Archives Y2025

Milestone:

Load shedding algorithm

Load shedding algorithm Modeling: definition and approach

1. What load shedding is, and when it's needed

DCOPF enforces power balance at every bus: injection must equal outflow. When an hour's combined generation and transmission capacity cannot serve the full load—unplanned outages, transmission congestion, extreme peak—the optimization becomes literally **infeasible**: no solution exists.

Facing the same physical shortfall, a real grid's last resort is to deliberately shed part of the load, sacrificing local supply to save the system. So in the model, load shedding plays a dual role: it is both a **depiction of real "lost load" behavior** and a **feasibility valve** that keeps DCOPF always solvable under physical shortage. The key requirement is that it act **only when truly necessary**, and shed **as little as possible**.

ERCOT 2024 Emergency Operations Overview

2024 Energy Emergency Alert Overview

When electric supply and demand can't be balanced with normal procedures, ERCOT will implement additional actions to avoid the need to enter emergency operations, including starting additional generation, reducing operating reserve levels, and deploying designated demand response programs. If operating reserves continue to drop, ERCOT begins emergency operations using three levels of Energy Emergency Alerts (EEAs). These

EEA levels and actions

 EEA LEVEL 1	 EEA LEVEL 2	 EEA LEVEL 3
If operating reserves drop below 2,500 MW and are not expected to recover within 30 minutes, ERCOT actions include: Bringing any additional available generation online Increasing other generation supplies, including: Importing available generation from neighboring electric grids: up to 1,220 MW Switchable generation that can serve multiple electric grids: up to 1,260 MW Additional information on EEA 1	If operating reserves drop below 2,000 MW and are not expected to recover within 30 minutes, or frequency drops below 59.91 Hz for 15 minutes actions include: Request energy conservation (if not already in effect) from public, including government and municipal agencies: # MW vary Reduce power by deploying remaining demand response programs, including: Deploying operating reserves carried by Load Resources (some large industrial customers who are paid to reduce their power): up to 1,423 MW Load management programs from transmission companies: 372 MW Voltage reduction by transmission companies: 573 MW Additional information on EEA 2	If operating reserves drop below 1,500 MW, or the grid's frequency level drops below 59.8 Hz for any period of time, ERCOT actions include: Instructing transmission and distribution service providers/companies to reduce demand on the electric system, which occurs through the use of controlled outages – these outages impact all customer classes, including residential, commercial, and industrial Additional information on EEA 3

Load shedding algorithm Modeling: definition and approach

2. The Core Technique: A VOLL Virtual Generator

We turn load shedding into a virtual generator the optimizer can dispatch on its own: each load-carrying bus gets a co-located "shed gen." A unit of its positive output is equivalent, in the power balance, to that bus's demand being curtailed by the same amount—the more it injects, the less load is served.

3. The mechanism: setting a price ladder

The mechanism that makes the virtual unit "act only when necessary" lives entirely in its **price**. Each shed gen's marginal cost is set to **VOLL = 5000 \$/MWh** (uniform value of lost load). Because DCOPF minimizes total cost and VOLL sits far above any real unit's marginal cost, the optimizer engages this virtual unit only when no cheaper way to serve load remains. The **uniform** price guarantees that, when shedding is unavoidable, the total MW shed is minimized.

Load shedding algorithm Modeling: Mechanism - the price ladder

Going further, load shedding sits at the very top of a full price ladder, which forms a cheapest-to-most-expensive dispatch order:

1. real generation (cheapest, from 0 to around 20)
2. Non-Spin emergency discharge ($AS_DEPLOY = 800$)
3. storage charge curtailment ($VOLL_CHG = 2500$)
4. firm load shed ($VOLL = 5000$, last resort)

Because $800 < 2500 < 5000$, the optimizer exhausts every cheaper relief measure first—releasing storage emergency capacity, then giving way on storage charging—before touching firm load; **native load is always protected last**. The entire ladder is sorted automatically by merit order within a single DCOPF solve, with no explicit priority-ranking code needed.

Milestone:

Hydroelectric Generation Modeling

Hydroelectric Modeling: Introduction

What hydroelectric is.

Hydropower generates electricity by releasing water through turbines. **Conventional hydro** (run-of-river or reservoir) depends on rainfall and snowmelt—wet years produce more, dry years less. **Pumped storage** works like a giant battery: pump water uphill when electricity is cheap, let it flow back down to generate when prices rise.

What they share: you can install a big turbine, but if there's not enough water, you can't run it at full power all year. **The bottleneck is fuel (water), not equipment**—the same challenge we solved for batteries with SoC tracking.

Why we skip the detailed model in ERCOT.

We *could* model hydro the way we modeled batteries. We don't, because in ERCOT it's simply too small to matter. Conventional hydro totals only a few hundred MW, there is no pumped storage, and Texas hydro's 2024 capacity factor is just **0.09** (vs. a U.S. average of ~ 0.35). That translates to an average output of tens of MW—roughly **0.05% of an 80+ GW system peak**. It's noise.

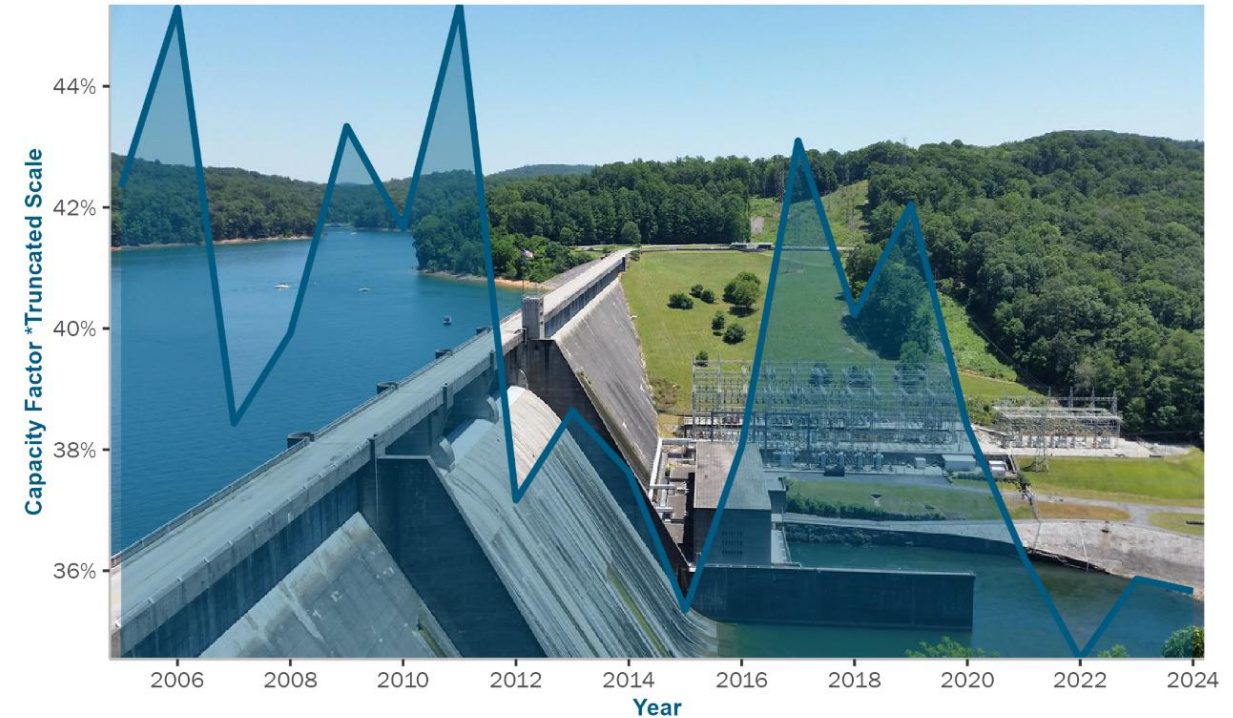
Hydroelectric Modeling: Assumptions and Approach

Simplifying assumptions.

- Maximum output is time-invariant;
- Actual generation is set by grid dispatch.
- That maximum is discounted by the most recent annual capacity factor:

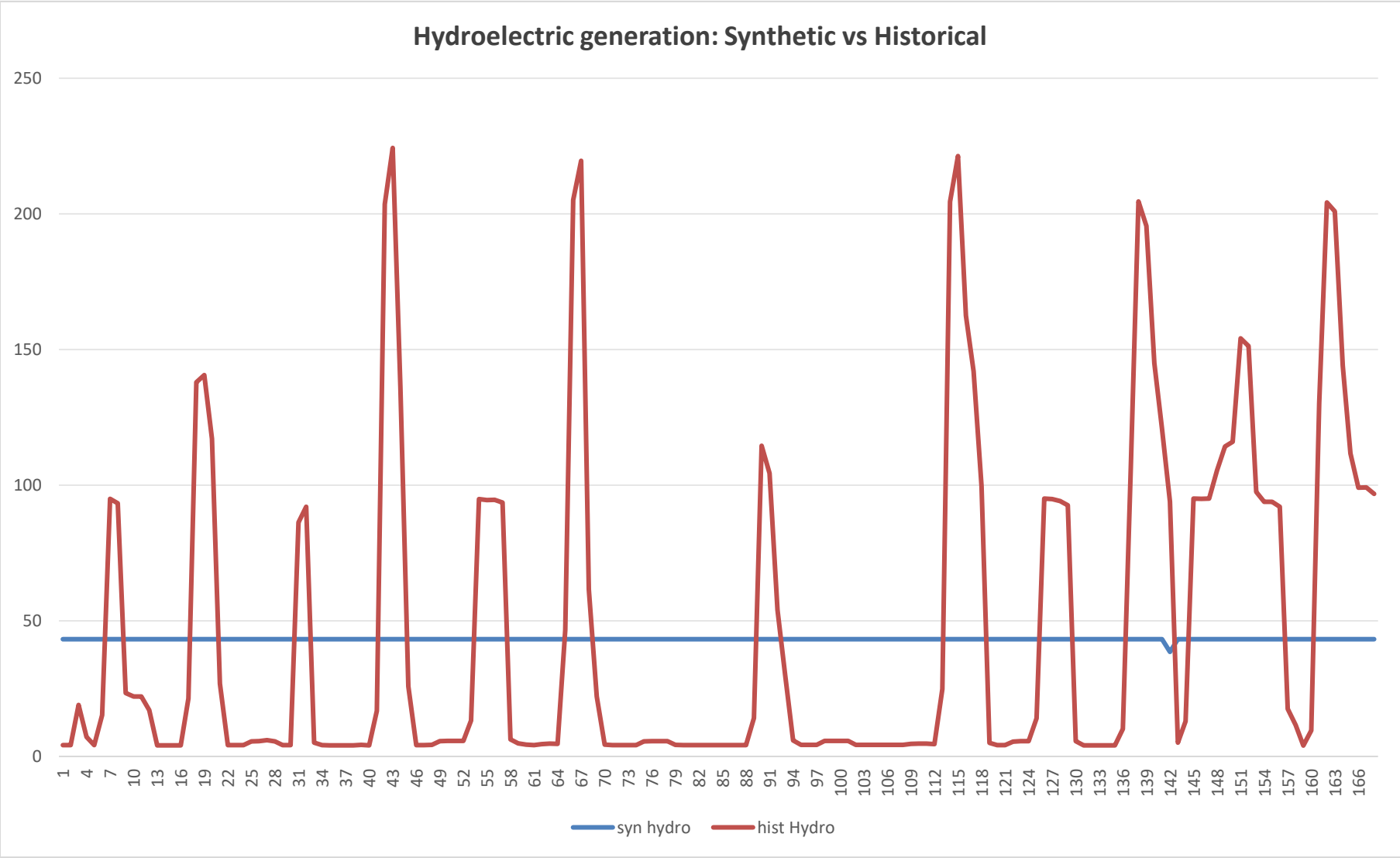
$$P_{\max} = P_{\text{capacity}} \times CF_{\text{TX},2024} \approx P_{\text{capacity}} \times 0.09$$

- Pumped storage is not modeled.



Source: Megan M. Johnson. 2025. Existing Hydropower Assets (EHA) Capacity Factor Plant Database, 2005-2024. HydroSource. Oak Ridge National Laboratory

Hydroelectric Generation Modeling – Validation



Historical Data Source:

- ERCOT Fuel Mix Report: 2025

Milestone:
**Generator Type & Generator Cost
configuration**

generator type configuration: Gas Generator

In power system reliability, different conventional Generator technologies have different failure rates and repair rates. While it is cleared that most coal generators and hydroelectric generators apply to one generator technology, gas generators apply to multiple generator technologies.

Besides, gas generators in this synthetic grid model don't include this technology label, so it must be inferred before the availability model can be applied.

Method: use nameplate capacity as the classifying variable and apply a Gaussian Mixture Model (GMM) to assign each generator to a “steam turbine(NGST)”, “fired combustion(NGCT)” or "combined cycle(NGCC)" cluster.

The following data are the weighted mean capacity, standard deviation, its proportion of generators in each cluster:

Generator Type	Mean [MW]	SD [MW]	Proportion in the US
NGST	232	188	0.1275
NGCC	188	90.5	0.3876
NGCT	65.6	42.5	0.4849

***Cluster parameters derived from EIA-860 y2025 nationwide generator data.**

generator type configuration: Wind Turbine IEC Class Assignment

Why assign an IEC class

A wind turbine's power output at a given wind speed depends on the class it was built for. Turbines at low-wind sites use larger rotors that reach rated power at lower speeds, while turbines at high-wind sites are built more conservatively.

Assigning each wind farm an IEC class lets the model apply the **power curve that matches its site**, instead of one generic curve — so the simulated output reflects where the turbine actually sits.

How the class is assigned

Each wind farm is classified by the **annual mean wind speed** at its location:

IEC class	Annual mean wind speed
Class I (high wind)	> 8.5 m/s
Class II (medium)	7.5 – 8.5 m/s
Class III (low wind)	< 7.5 m/s

The site's long-term mean wind speed is obtained from NASA POWER climatology and extrapolated to hub height. Each class maps to its own **normalized power curve**, which is then scaled by the unit's rated capacity to produce the hourly generation time series.

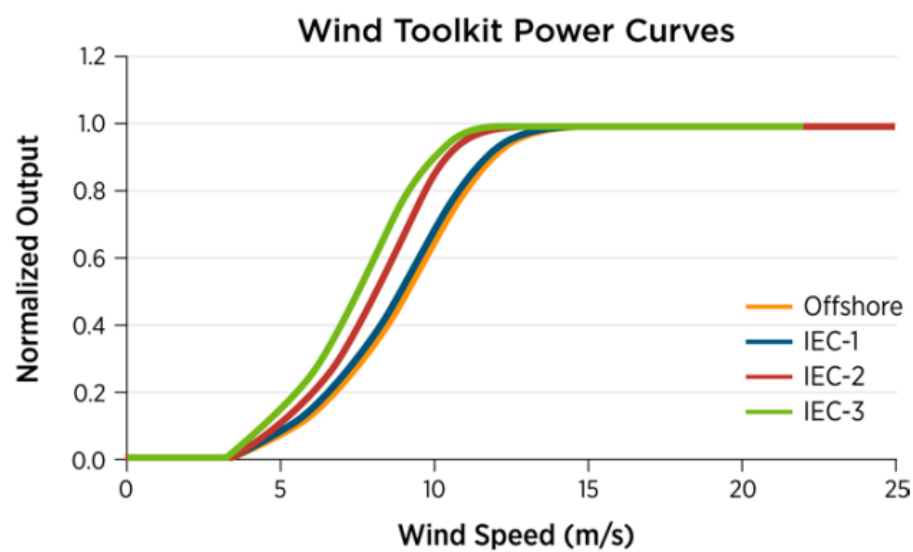


Figure 1. Power curves used to convert modeled wind speeds to power for onshore Class I through Class III locations and offshore locations

Generation Capacity configuration: Battery Storage

In grid resilience analysis, Energy Storage Systems (ESS) play a crucial role in balancing load during extreme events. However, the original synthetic grid dataset only provides the maximum power output (MW) for storage units, lacking the critical energy capacity (MWh) parameter.

To accurately model the discharge capabilities and constraints of battery resources, a realistic storage duration must be inferred for each unit.

Method: Apply an empirical probability distribution to assign a storage duration (ranging from 1 to 6 hours) to each storage unit. The total energy capacity (Storage_MWH) is then calculated by multiplying the assigned duration by the unit's nameplate capacity.

The following data represent the sample counts for the primary duration categories:

Storage Duration	Count
1 hr	313
2 hr	346
3 hr	63
4 hr	370
5 hr	31
6 hr	8

* Empirical distribution derived from EIA-860 (Year 2025) data for operable and proposed Lithium-ion batteries.

Gencost Re-Configuration: All generators

In grid simulation, accurately updating the generator cost curves is essential to realistically model the grid's Unit Commitment (UC) and economic dispatch behaviors. According to [1], The cost function of a generator is defined by the variable – the generation power Pg, and two parameters: fuel cost cf and heat rate b1. see the following formula:

$$\text{Cost}(Pg) = cf * b1 * Pg$$

Note: While thermal generators require specific fuel prices and heat rates to calculate these coefficients, renewable and energy storage resources (e.g., wind, solar, hydro, and batteries) operate with zero generation costs.

[1] Application of Large-Scale Synthetic Power System Models for Energy Economic Studies

*Heat rate distribution parameters derived from EIA-923 Texas plant data; Fuel prices based on EIA EPM March 2025.

Fuel	Fuel price [\$]	Data Reference
Natural Gas	4.18	EIA EPM Table 4.13.A for March 2025
Coal	2.02	EIA EPM Table 4.10.A for March 2025
Distillated Fuel Oil	0.58	EIA EPM Table 4.2
Nuclear	17.13	EIA nuclear data & statistics
Solar, Wind, Hydro, Energy Storage	0	[1]

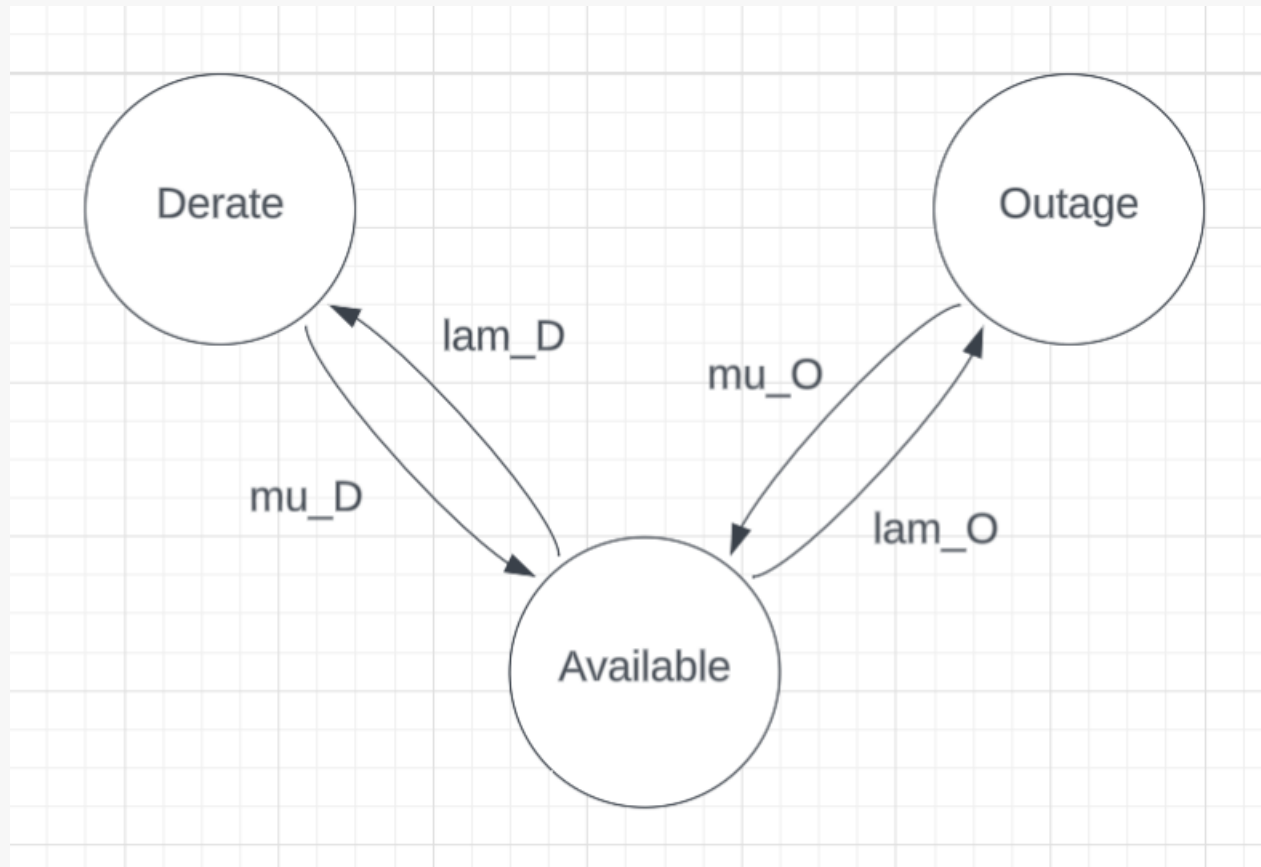
Generator Type	Heat Rate		Ref
	Mean	Std dev	
NGCC	6.891	1.169	EIA923
NGCT	9.867	3.862	EIA923
NGST	11.552	2.569	EIA923
Coal	11.136	0.867	EIA923
Nuclear	10.443	0	EIA table 8.2
Fuel oil	13.083	0	EIA table 8.2

Milestone:
**Generator Availability Modeling for natural gas,
coal, fuel oil, nuclear, hydroelectric generators**

Availability Modeling: Model Assumption

Availability Modeling: Model Assumptions

In practice, natural gas generators experience two distinct levels of abnormal operation under routine failures—**partial deratings** and **full outages**. To build an estimable model on top of limited historical data, we adopt three simplifying assumptions:



- **Assumption 1.** When a failure event occurs, its severity (derate or outage) is fixed at onset and remains unchanged for the entire repair period.
- **Assumption 2.** The model uses a **three-state Markov chain** (Available, Outage, Derate), where the derate and outage states capture different degrees of capacity loss. Failure and repair rates suited to winter conditions are estimated from the **NERC GADS Statistical Brochure and Cause Code Report**, with both time-to-failure (TTF) and time-to-repair (TTR) assumed to follow exponential distributions. For simplicity, no direct transitions are allowed between the outage and derate states.
- **Assumption 3.** Because the synthetic grid units lack technology labels, we infer each unit's type from its capacity using a **Gaussian Mixture Model (GMM)**, then match it to the empirical λ and μ of the corresponding technology in the table above.

Availability Modeling: estimating state transition parameters

Source: NERC GADS Cause Code Report, Generating Unit Statistical Brochure 1. Data collected during year 2024.

Note for data cleaning: Filters are applied for winter season, with all regions covered. Durations which over 90th percentile are truncated

Parameters:

- FOH, FDH, LostMWH, LostMW [cause code report]
- PH, #unit [statistical brochure]

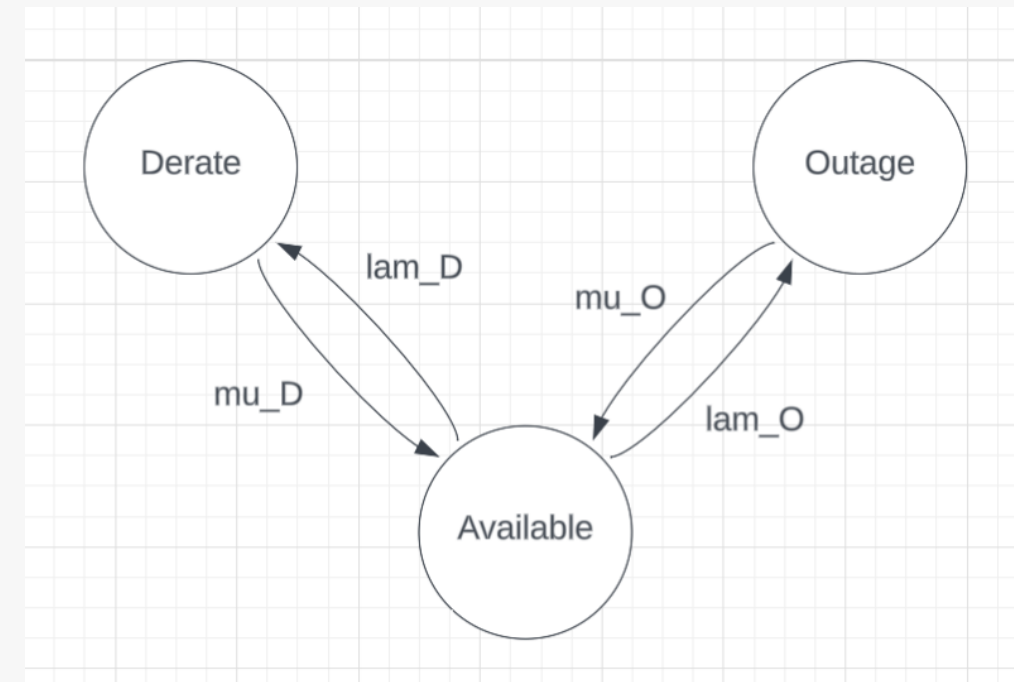
Formulas

$$PH_{\text{winter}} = PH * 0.25$$

$$FDH \text{ or } FOH = \sum duration, duration = round\left(\frac{LostMWH}{LostMW}\right)$$

$$\mu_{\text{outage}} = \frac{\#FOH}{FOH}, \mu_{\text{derate}} = \frac{\#FDH}{FDH}$$

$$\lambda_{\text{outage}} = \frac{\#FOH}{PH_{\text{winter}} * \#unit}, \lambda_{\text{derate}} = \frac{\#FDH}{PH_{\text{winter}} * \#unit}$$



Availability Modeling: estimating state transition parameters

Source: NERC GADS Cause Code Report, Generating Unit Statistical Brochure 1. Data collected during year 2024, filters applied for winter season, all regions covered.

Fuel/Technologies Considered:

- Gas - steam turbine: fossil-steam, fluidized bed, multi-boiler/multi-turbine
- Gas - single cycle: gas turbine/jet engine (Single Cycle Operation)
- Gas - combined cycle: Co-generator Block, Combined Cycle Block
- Coal - steam turbine: fossil-steam, fluidized bed, multi-boiler/multi-turbine
- Nuclear: Nuclear
- Hydroelectric: Pumped Storage/Hydro
- Distilled Fuel Oil: gas turbine/jet engine (Single Cycle Operation)

Note: # unit for fuel oil generator is combined from # unit for diesel generator unit and # unit for jet engine

	coal steam	parameter	MTTF/R		NG CC	parameter	MTTF/R	
	lamO	0.000699684	1429.216292		lamO	0.000474131	2109.119612	
	lamD	0.00407494	245.4024116		lamD	0.000946219	1056.837473	
	muO	0.025695313	38.917603		muO	0.106373223	9.400862069	
	muD	0.091274616	10.95594855		muD	0.124529317	8.030237581	
	hydro	parameter	MTTF/R		NG CT	parameter	MTTF/R	
	lamO	0.000348837	2866.666667		lamO	0.001354743	738.1474623	
	lamD	0.000241383	4142.795699		lamD	0.000293621	3405.756329	
	muO	0.043408049	23.03720238		muO	0.075551871	13.23593964	
	muD	0.024249061	41.23870968		muD	0.071981777	13.89240506	
	Nuclear	parameter	MTTF/R		NG Steam	parameter	MTTF/R	
	lamO	0.000114155	8760		lamO	0.000492591	2030.08	
	lamD	0.00057574	1736.896552		lamD	0.000567465	1762.222222	
	muO	0.0078125	128		muO	0.030109599	33.212	
	muD	0.00511261	195.5948276		muD	0.026282168	38.04861111	
	Fuel Oil		MTTF/R					
	lamO	0.000209744	4767.715232					
	lamD	3.33368E-05	29996.875					
	muO	0.024025457	41.62251656					
	muD	0.002469644	404.9166667					

Availability Modeling: State Model

Building on the assumptions above, we construct a **three-state transition model**. Based on each generator's technology type, the model assigns the empirical transition parameters appropriate to that type and to winter conditions.

When a generator is assigned the **outage** state, its output is 0. When assigned the **derate** state, its derated capacity is determined by the lowest ambient temperature during that derate period and held constant for the entire derate spell.

Source:
Murphy, A time-dependent model of generator failures and recoveries captures correlated events and quantifies temperature dependence
R. Billinton, reliability evaluation of power systems.

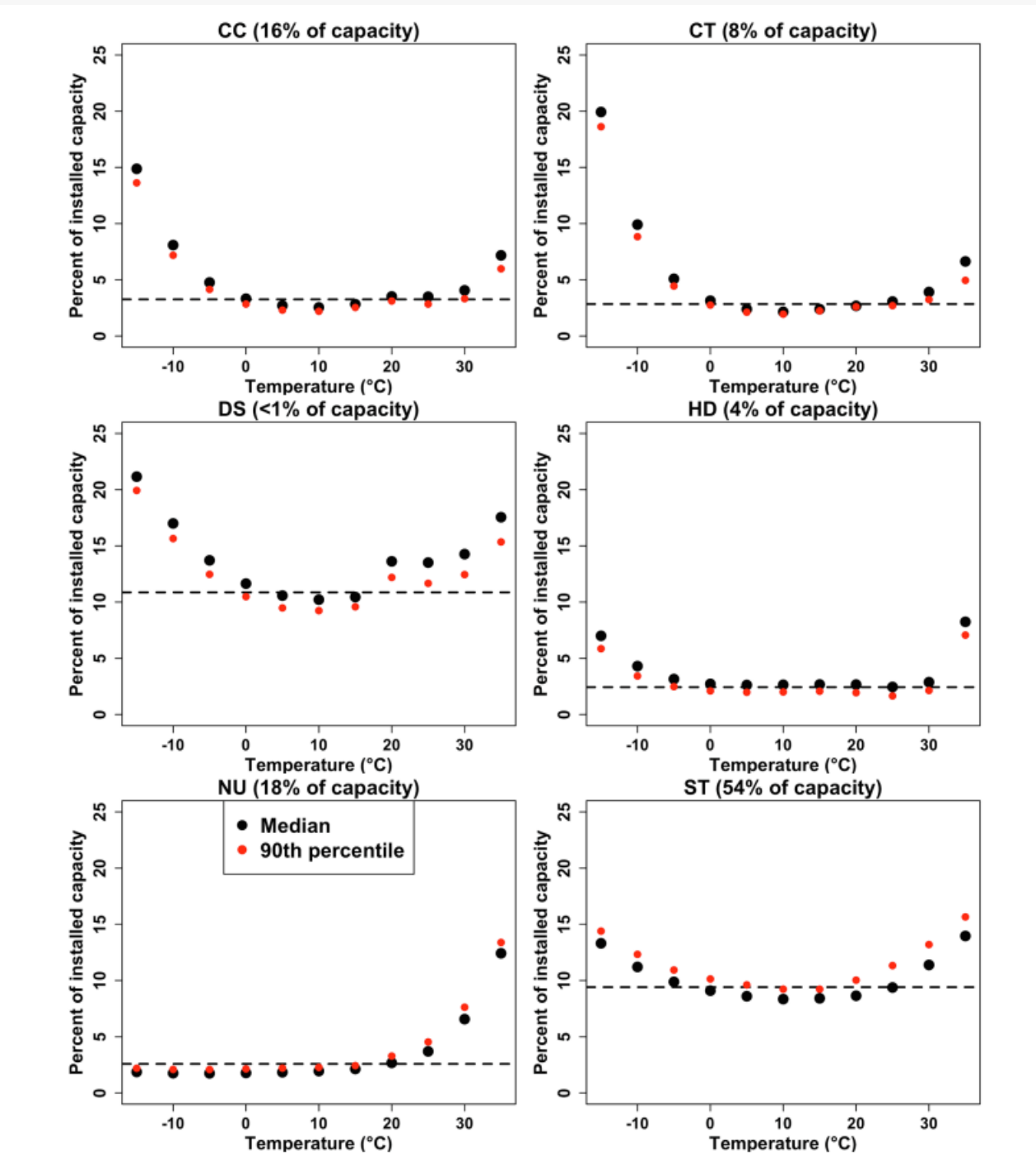


Fig. 6. Expected levels of unavailable capacity as a function of temperature under logistic regression (dots) and current practice (dashed horizontal line) (1995–2018 model fits). Black dots calculated using median load from temperature neighborhood, red dots calculated using 90th percentile load from temperature neighborhood. Temperature neighborhood is defined as ± 10 degrees. Not all generators experience full temperature range; see Fig. A.13 for prevalence of temperatures. CC is combined cycle, CT is simple cycle, DS is diesel, HD is hydroelectric and pumped storage, NU is nuclear, ST is steam turbine.

Conventional Generator Availability Modeling: Results

GEN_ID	FUEL_TYPE	GENERATOR_TYPE	PMAX	Derate	Outage	Duration
398	NG (Natural Gas	fired_combustion	12.3	0	1	4
392	NG (Natural Gas	fired_combustion	13	1	0	2
964	NG (Natural Gas	fired_combustion	16	0	2	18
998	NG (Natural Gas	fired_combustion	16	0	1	15
44	NG (Natural Gas	fired_combustion	17	0	1	7
947	NG (Natural Gas	fired_combustion	18	0	1	6
948	NG (Natural Gas	fired_combustion	18	0	1	2
1046	NG (Natural Gas	fired_combustion	18	0	1	39
130	NG (Natural Gas	fired_combustion	18.3	0	2	23
133	NG (Natural Gas	fired_combustion	18.3	0	1	4
135	NG (Natural Gas	fired_combustion	18.3	0	2	57
136	NG (Natural Gas	fired_combustion	18.3	0	1	10
98	BIT (Bituminous	steam_turbine	391	1	1	131
273	NG (Natural Gas	fired_combustion	18.4	0	1	3
274	NG (Natural Gas	fired_combustion	18.4	0	1	13
275	NG (Natural Gas	fired_combustion	18.4	2	0	15
153	NG (Natural Gas	fired_combustion	18.8	1	0	2
155	NG (Natural Gas	fired_combustion	18.8	0	1	16
156	NG (Natural Gas	fired_combustion	18.8	0	5	71
158	NG (Natural Gas	fired_combustion	18.8	0	1	7
160	NG (Natural Gas	fired_combustion	18.8	0	2	73
161	NG (Natural Gas	fired_combustion	18.8	0	1	7
45	NG (Natural Gas	fired_combustion	19.7	1	0	13
47	NG (Natural Gas	fired_combustion	19.7	0	2	28
985	NG (Natural Gas	fired_combustion	20	0	1	9
165	NG (Natural Gas	fired_combustion	23	1	0	8
245	NG (Natural Gas	fired_combustion	27.6	1	2	41
309	NG (Natural Gas	fired_combustion	28.5	0	1	10
333	NG (Natural Gas	fired_combustion	29	0	2	28
298	NG (Natural Gas	fired_combustion	30	1	1	7
978	NG (Natural Gas	fired_combustion	30	0	2	5
944	NG (Natural Gas	fired combustion	31	0	1	2

Generator availability model: time series

Generator fault report

Milestone:
Unit Commitment Modeling

Unit Commitment Modeling: simulation logics

What is Unit Commitment?

Unit commitment (UC) decides *which* thermal units are online each hour over the week—not *how much* each produces (that is left to dispatch). In short: it sets the on/off schedule, not the power levels.

Why it needs its own layer?

In practice, turning a thermal unit on or off is not free. Starting up costs money, and once a unit switches state it must honor minimum up- and down-times. These are **logical, sequential** constraints that a continuous optimization cannot express—hence a dedicated commitment step.

How we solve it.

Because on/off states are binary while output and costs are continuous—and the two are coupled (a unit can only produce when committed)—UC is a **mixed-integer linear programming (MILP) problem**, solved here with Gurobi.

Unit Commitment Modeling: simulation logics

4. Decision Variables

For each unit i and each hour t , four variables are defined:

$p_{i,t}$ — output (continuous, MW)

$u_{i,t}$ — online status (binary, 1 = on)

$v_{i,t}$ — start-up event (1 = just started this hour)

$w_{i,t}$ — shutdown event (1 = just shut down this hour)

Two types of **slack variables** are also introduced to keep the problem always feasible and to pinpoint stressed periods:

s_t — unserved-demand slack, one per hour

$sr_{z,t}$ — reserve-shortfall slack, one per zone per hour

5. Unit and System Parameters

Unit parameters:

$P_i^{\max}, P_i^{\min}$ — rated output ceiling and minimum technical output

c_{0i} — marginal generation cost (\$/MWh)

c_{1i} — no-load cost (\$/h)

su_i — start-up cost (\$)

$T_i^{\text{up}}, T_i^{\text{dn}}$ — minimum up-time and minimum down-time (hours)

System-level parameters:

R_z — zonal reserve requirement

PEN — penalty price for unserved demand

PENr — penalty price for reserve shortfall

Note: the penalty prices satisfy $PEN \gg PENr$, ensuring the solver prioritizes serving load first and meeting reserves second.

Unit Commitment Modeling: simulation logics

UC parameters are assigned **by fuel/technology type** to reflect differences in physical flexibility across units. See the following:

- Gas Combustion Turbines and oil-fired units are fast-cycling peakers (minimum up/down time of 1 hour);
- gas Steam Turbines and coal units are more "sticky" (minimum up-time of 8-10 hours);
- nuclear is set as **must-run**—it never cycles on economic grounds at the weekly scale, so $u_{i,t} \equiv 1$.

Table 3: Typical Generator Ramp Rates and Start-up/Shut-down Costs

Fuel Type	ramp rate (% of the rated capacity per min)	start-up costs (\$/MW/start)	shut-down costs (\$/MW/shutdown)	starting-up time (hour)	shutting-down time (hour)	minimum on time (hour)	minimum off time (hour)
Coal	0.6-8	100-250	10-25	4-9	2-9	0-12	0-12
Natural Gas	0.8-30	20-150	2-15	2-4	1-4	0-2	0-1
Nuclear	0-5	1000	1000	24	24	days	days
Hydro	15-25	0-5	0-0.5	0-1	0-1	0-1	0-1
Wind	Non-dispatchable						

Ref: Application of Large-Scale Synthetic Power System Models for Energy Economic Studies

Unit Commitment Modeling: simulation logics

6. The Complete Problem Formulation

Objective — minimize total weekly cost (generation + no-load + start-up) plus the two slack penalties:

$$\min_{\{x\}} \sum_t \sum_i \left(c_i^{(1)} p_{i,t} + c_i^{(0)} u_{i,t} + s u_i v_{i,t} \right) + \text{PEN} \sum_t s_t + \text{PEN}_r \sum_{z,t} s r_{z,t}$$

Subject to:

(A) Power balance: $\sum_i p_{i,t} + s_t \geq D_t$

(B) Zonal reserve: $\sum_{i \in Z} P_i^{\max} u_{i,t} - \sum_{i \in Z} p_{i,t} + s r_{z,t} \geq R_z$

(C) Output ceiling: $p_{i,t} \leq P_i^{\max} u_{i,t}$

(D) Minimum output: $p_{i,t} \geq P_i^{\min} u_{i,t}$

(E) State transition: $u_{i,t} - u_{i,t-1} = v_{i,t} - w_{i,t}$

(F) Minimum up-time

(G) minimum down-time

7. Solution Method: Rolling Horizon

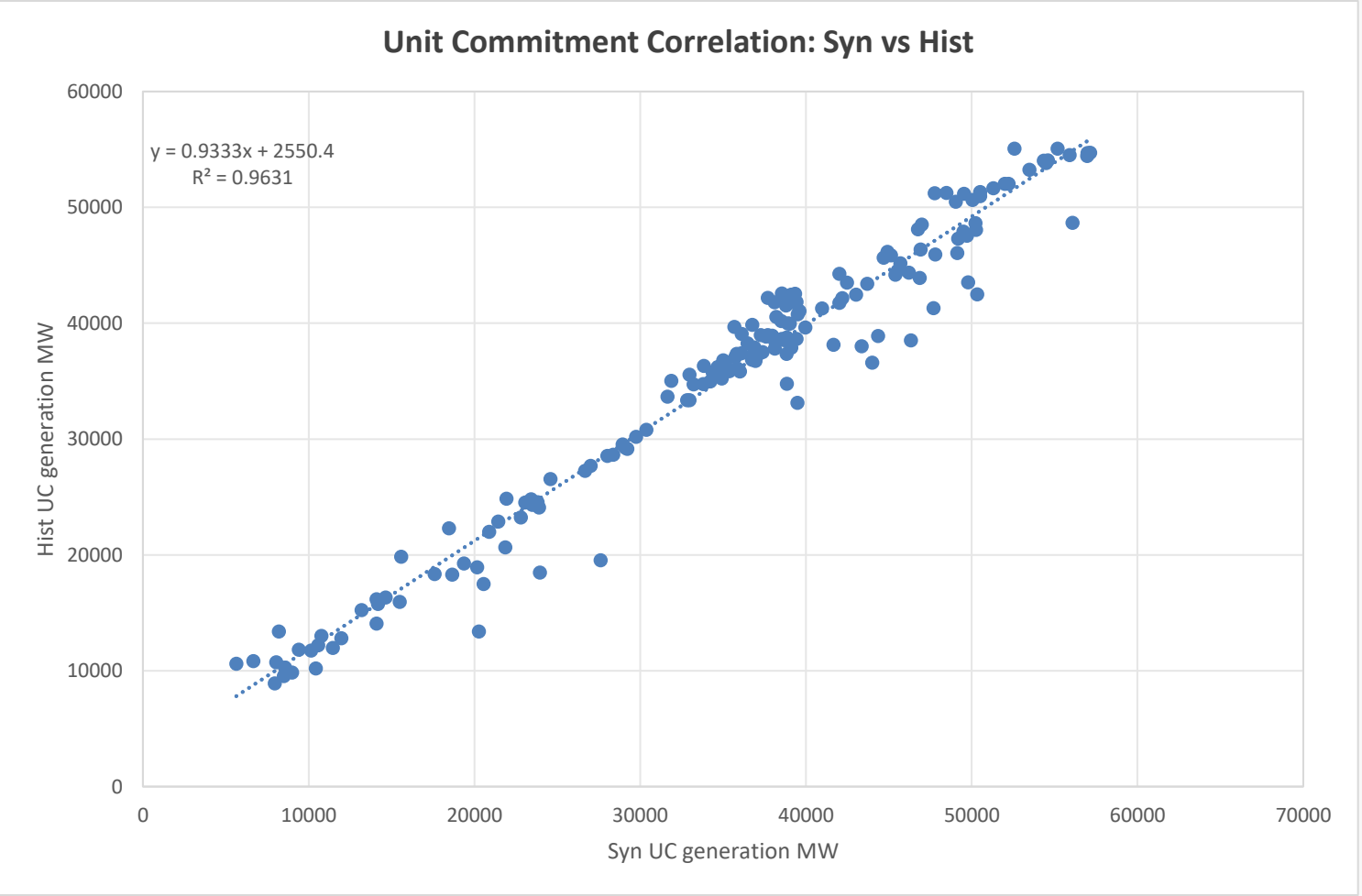
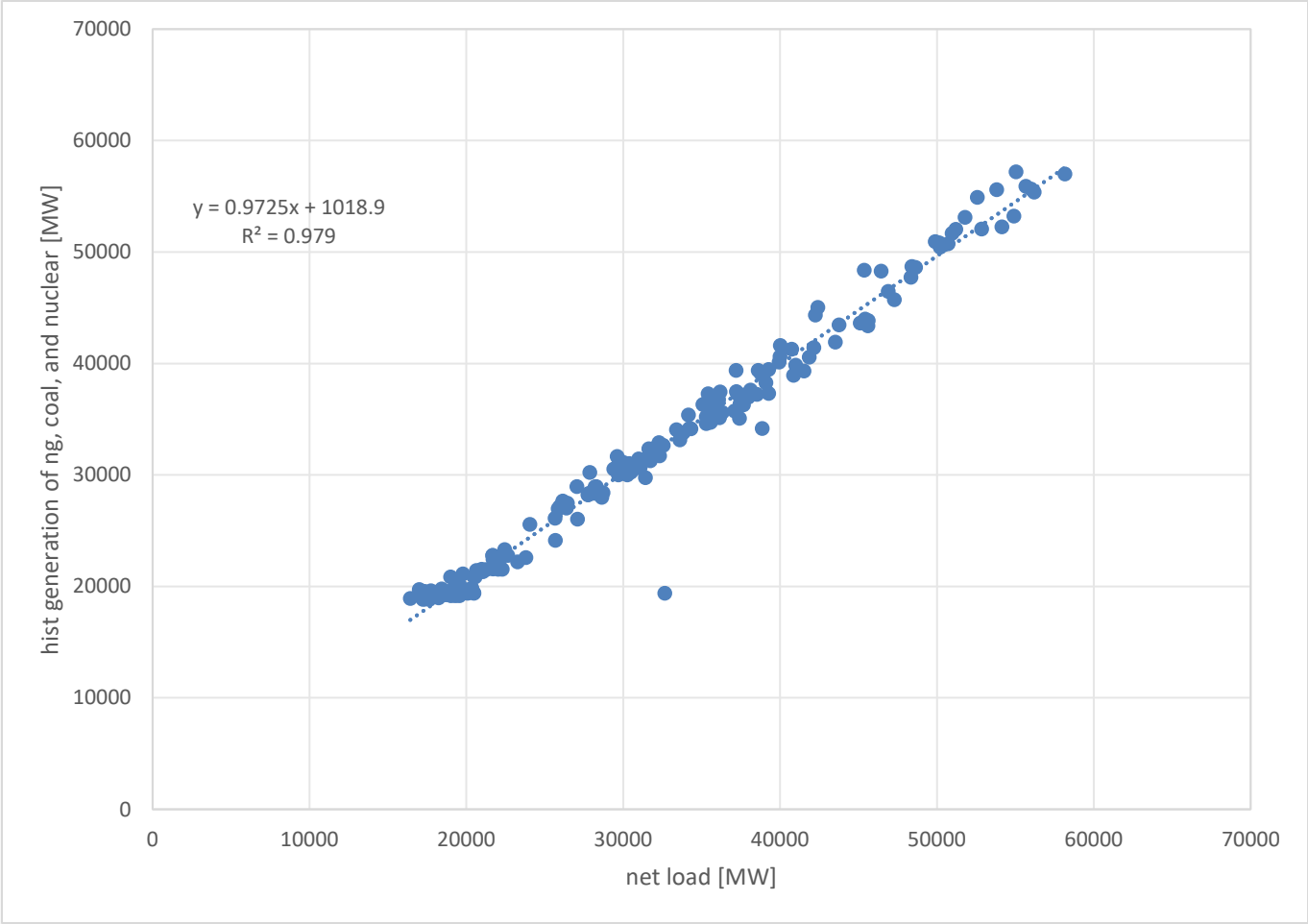
The full 168-hour weekly MILP is large, so we solve it with a **rolling-horizon** scheme: each solve covers one block, of which only the first 24 hours of decisions are committed, while looking 12 hours further ahead as a foresight window (to avoid myopic decisions at the block boundary).

Between blocks, the state of each unit is carried forward, so the minimum up-/down-time constraints are stitched together correctly across blocks. If a block's MILP fails to solve, the model falls back to a greedy commitment ordered by marginal cost as a safety net.

Reference:

- A rolling-horizon unit commitment framework with flexible periodicity
- A Tight MIP Formulation of the Unit Commitment Problem with Start-up and Shut-down Constraints

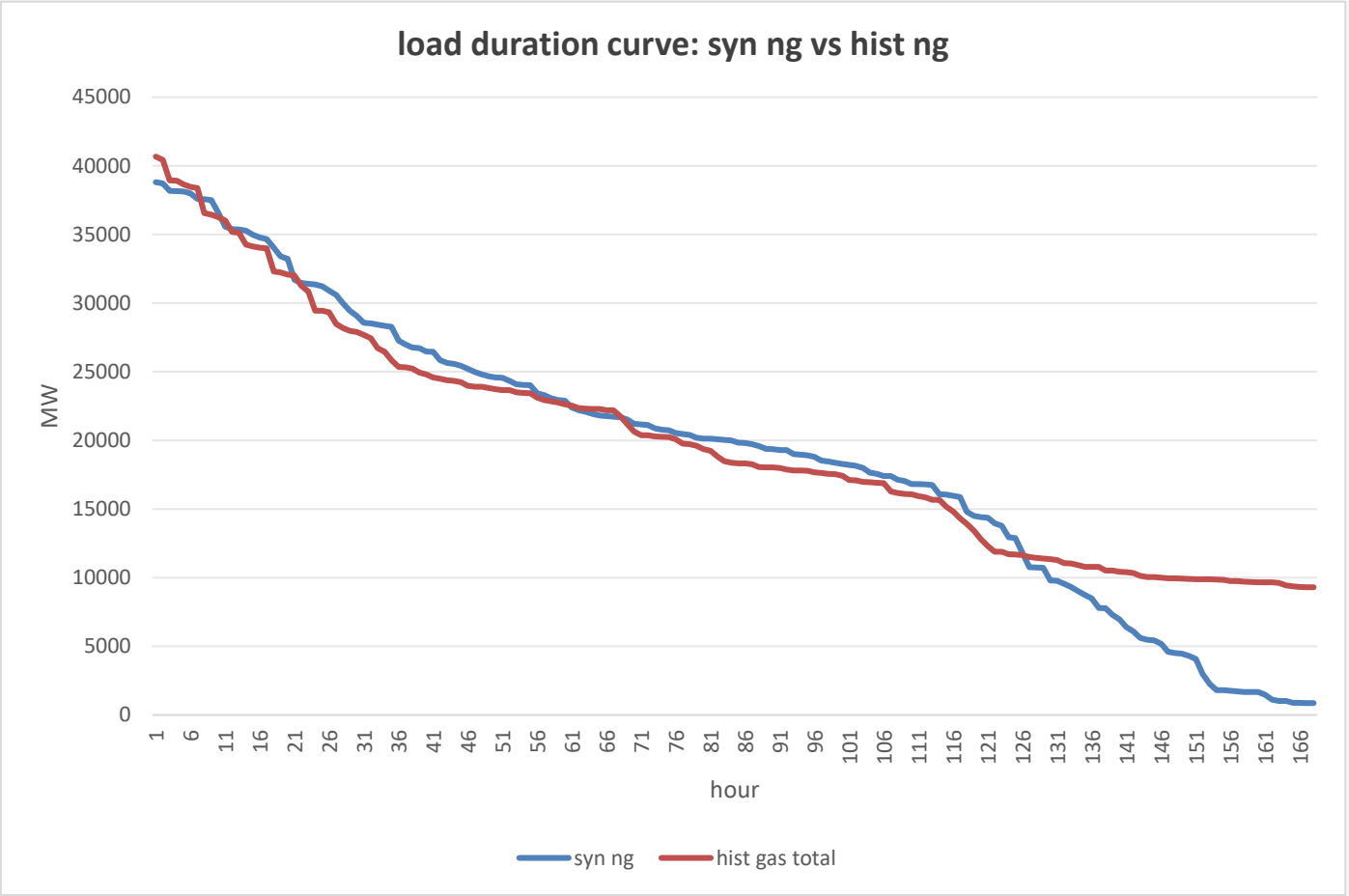
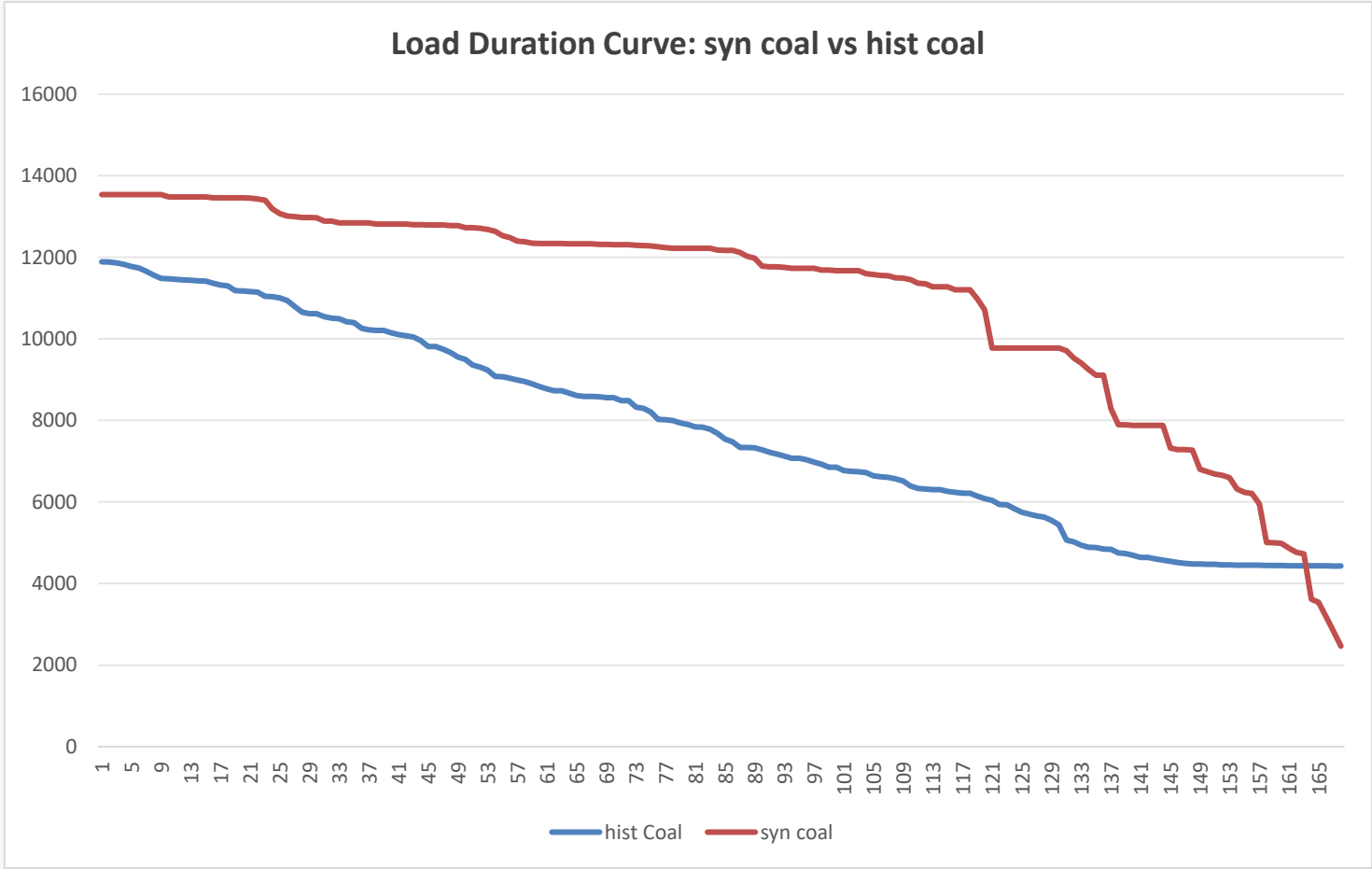
Unit Commitment Modeling – Validation: system design



Historical Data Source:

- ERCOT Fuel Mix Report: 2025

Unit Commitment Modeling – Component Validation: Coal & Natural gas

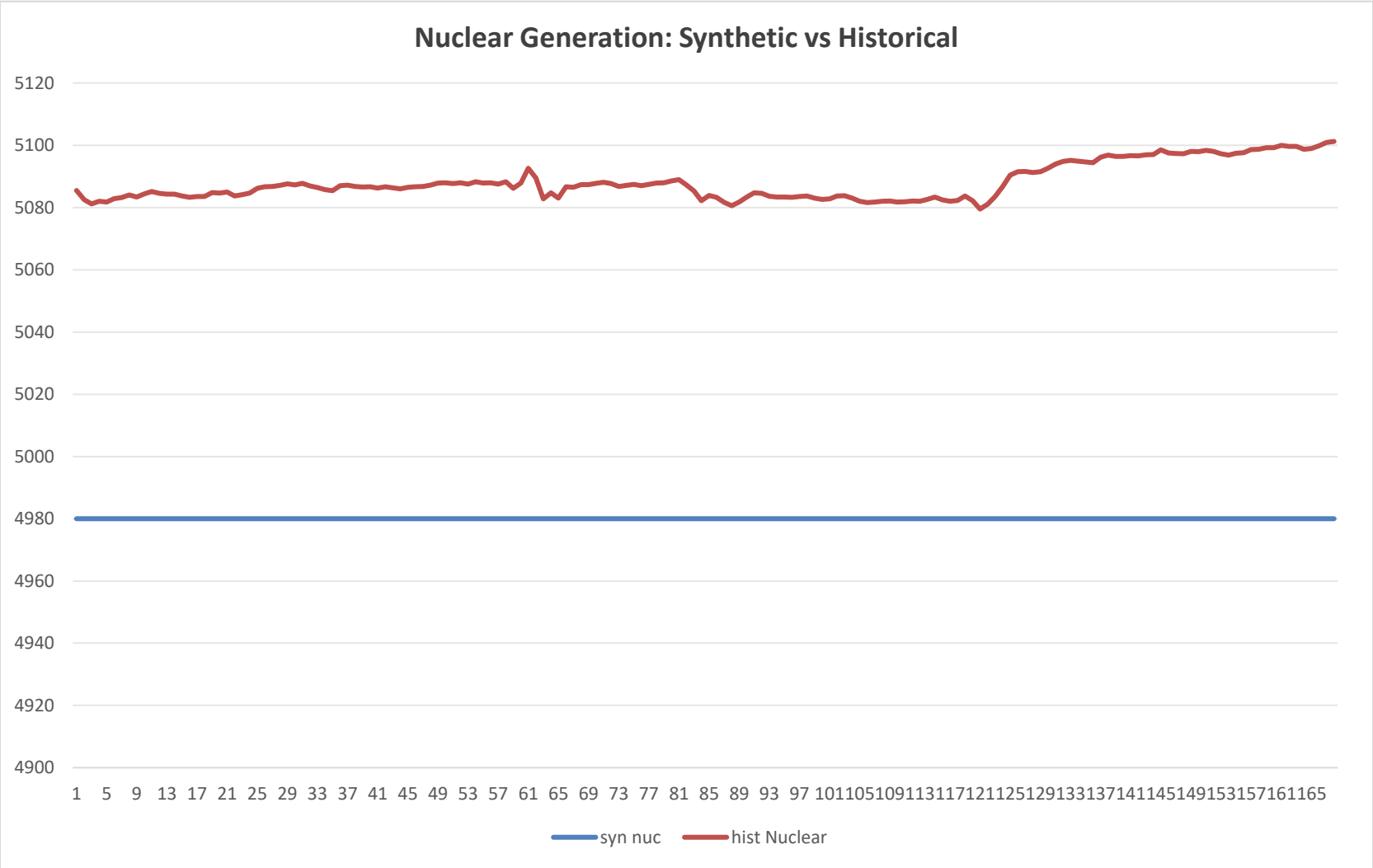


Historical Data Source:

- ERCOT Fuel Mix Report: 2025

Unit Commitment Modeling – Validation: Nuclear

average syn nuc	4980
average hist nuc	5088.076281
mean bias error	-108.0762811



Historical Data Source:

- ERCOT Fuel Mix Report: 2025

Milestone:

Battery Storages Modeling

Battery Storages Modeling: simulation logics

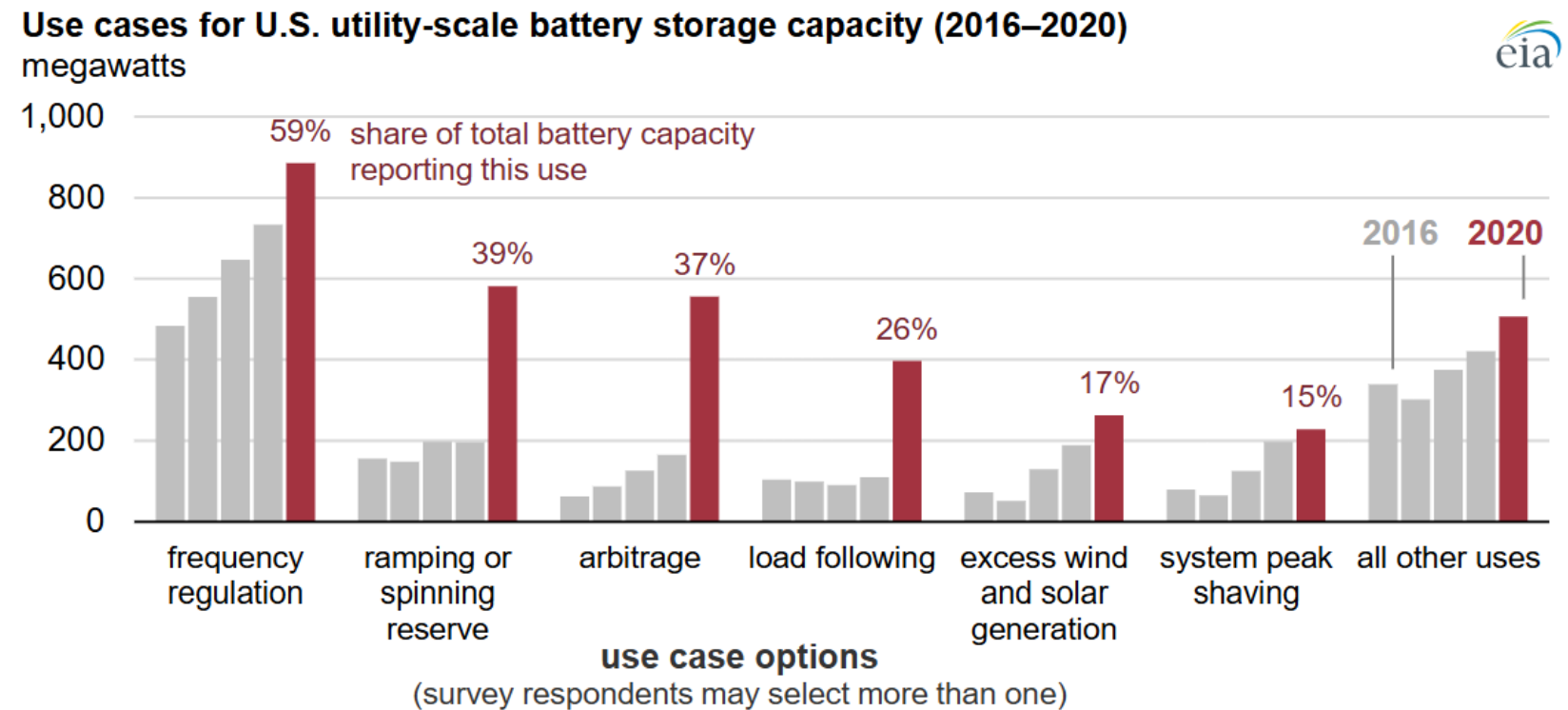
What a battery does — two jobs.

A battery is an *energy-limited* device that can both draw power (charge) and inject power (discharge) on the grid. In an electricity market it earns its keep in two ways, and we model both:

- **Peak shaving** — energy arbitrage: charge when net load is low, discharge when it peaks, flattening the curve.
- **Operational reserves** — hold standby capacity that the grid can call on for reliability.

Where storage sits in our model pipeline.

The DCOPF simulation is a one-way pipeline, and storage goes first: we schedule charge/discharge, *then* commit thermal units, *then* run the hourly power flow. Once scheduled, the storage profile is frozen and passed downstream as a known quantity—mirroring ERCOT's actual dispatch order.



Source: U.S. Energy Information Administration, [Annual Electric Generator Report](#)

Battery Storages Modeling: simulation logics

Step 1 — Split each battery's power and energy by market role.

Every battery's rated power (and its energy, by the same ratio) is divided between the two jobs: a share for peak shaving and a share for reserves. The reserve share is **AS_FRAC = 0.60**, the midpoint of the 55–70% range ERCOT batteries actually devote to ancillary services.

Within the reserve share we carve out a Non-Spin slice (**NONSPIN_FRAC = 0.40**)—the *only* reserve product that actually discharges meaningful energy at the hourly scale. The rest (RRS / ECRS / regulation) occupy reserve **capacity** but release essentially no net energy, since over an hour their charging and discharging roughly cancel.

Ref:

AS_FRAC: EIA860 Energy Storage

NONSPIN_FRAC: ERCOT Projected Ancillary Service Requirements, for Y2027 winter

Battery Storages Modeling: simulation logics

Step 2 — Peak-shaving schedule: a parameter-free QP, solved per weather zone.

Charge/discharge timing comes from a quadratic program whose goal is to flatten net load:

$$\min_{c,d} \sum_t (NL_t + c_t - d_t - \bar{N}L)^2$$

subject to SoC dynamics, power limits, energy limits, and a closed charge/discharge cycle. The magnitude of charging and discharging is set entirely by the battery's own power and energy ceilings—there are no tuning parameters

Note:

$$NL_t = \text{Load}_t - P_{\text{wind},t}^{\text{net}} - P_{\text{solar},t}^{\text{net}}$$

$\bar{N}L$: average net load

c_t : charging power; d_t : discharging power

Step 3 — Inject the fixed storage profile into DCOPF, split by direction.

To avoid the dispatchable-load ambiguity in MATPOWER, charge and discharge are handled separately: **discharge** enters as a generator (capped by real-time SoC), while **charge** enters as a bus load, served by the power flow from the cheapest local source.

Ref: H. Wang et al., *Battery Energy Storage Scheduling for Optimal Load Variance Minimization*

Battery Storages Modeling: problem formation

core idea: aggregate all batteries in a zone into one "equivalent battery," solve a least-squares peak-shaving QP that flattens system net load NL toward its mean, then distribute the result back to individual units in proportion to their power ratings.

Decision variables

For the aggregated battery, over horizon $t = 1, \dots, T$ (here $T = T_h$):

- c_t — charge power (MW)
- d_t — discharge power (MW)
- e_t — state of charge, SoC (MWh)

These are stacked into $x = [c_{1:T}; d_{1:T}; e_{1:T}]$, giving $n = 3T$ variables in total.

Ref: H. Wang et al., *Battery Energy Storage Scheduling for Optimal Load Variance Minimization*

Objective

Let $a_t = NL_t - \bar{NL}$ (the deviation of net load from its mean). The goal is to minimize the squared deviation of *adjusted* net load from the mean:

$$\min_{c,d,e} \sum_{t=1}^T (NL_t + c_t - d_t - \bar{NL})^2 = \sum_t (c_t - d_t + a_t)^2$$

In standard QP form $\frac{1}{2}x^T Hx + f^T x$:

- $H: H_{cc} = H_{dd} =, H_{cd} = H_{dc} = -2$
- $f: f_c = 2a_t, f_d = -2a_t$

The quadratic term is $(c_t - d_t)^2$ and the linear term is $2a_t(c_t - d_t)$. Together, $(c - d)^2 + 2a(c - d)$, plus the constant a^2 , gives $(c - d + a)^2$

This is a **convex QP** (H positive semi-definite, penalizing only net output $c - d$). Charging raises net load and discharging lowers it, so the optimal solution is naturally "charge in the valley, discharge at the peak."

Battery Storages Modeling: problem formation

Constraints

1) SoC dynamics (equality constraints, T rows):

$$e_t = e_{t-1} + \eta_c dt c_t - \frac{dt}{\eta_d} d_t, t = 1, \dots, T$$

At $t = 1$, the given initial value e_0 enters as the right-hand-side term. Here $\eta_c = \eta_d = 0.898$ (round-trip efficiency RTE ≈ 0.807).

2) Power / energy bounds (box constraints):

$$0 \leq c_t \leq P_{\text{agg}}, 0 \leq d_t \leq P_{\text{agg}}, 0 \leq e_t \leq E_{\text{agg}}$$

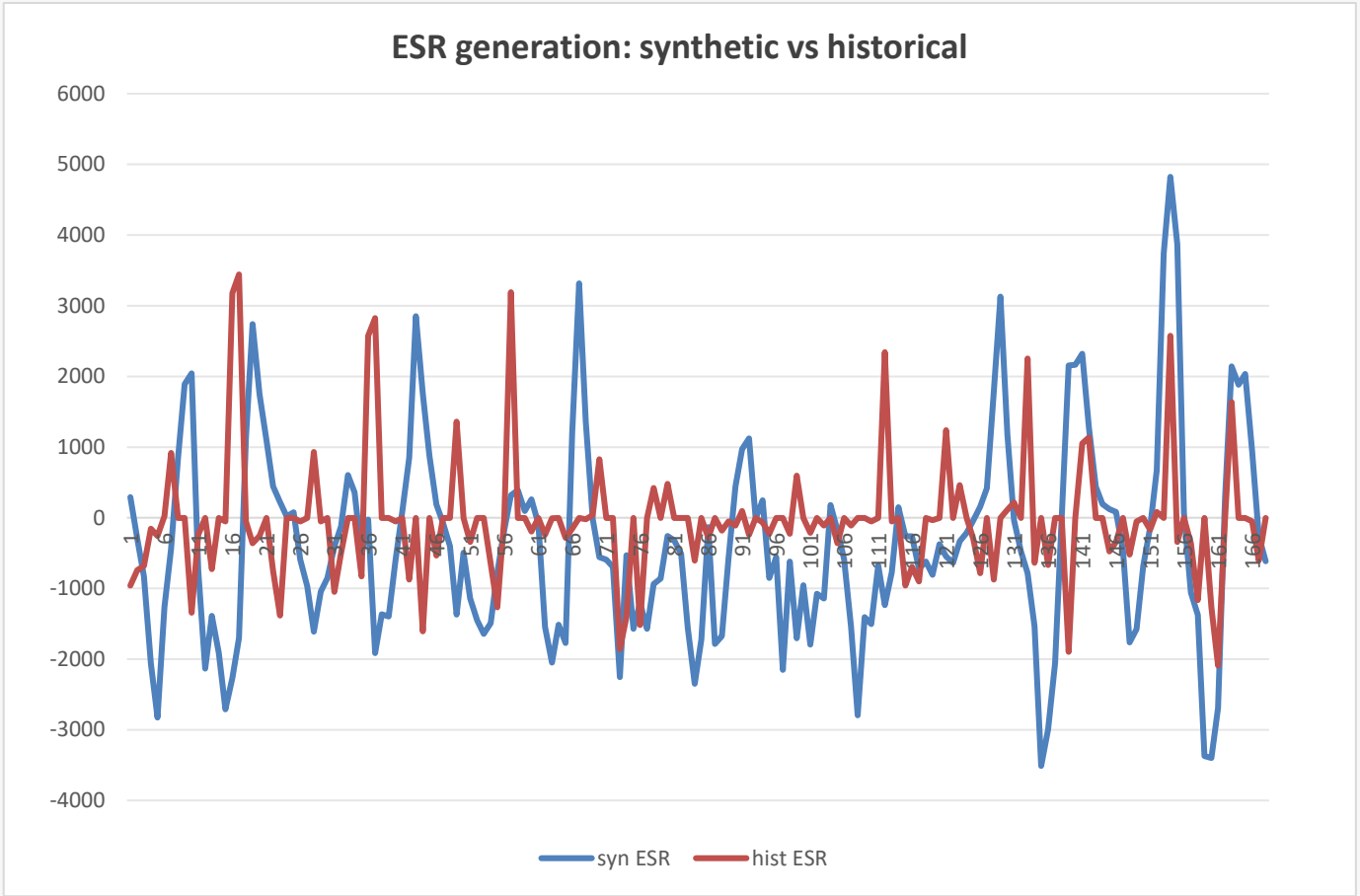
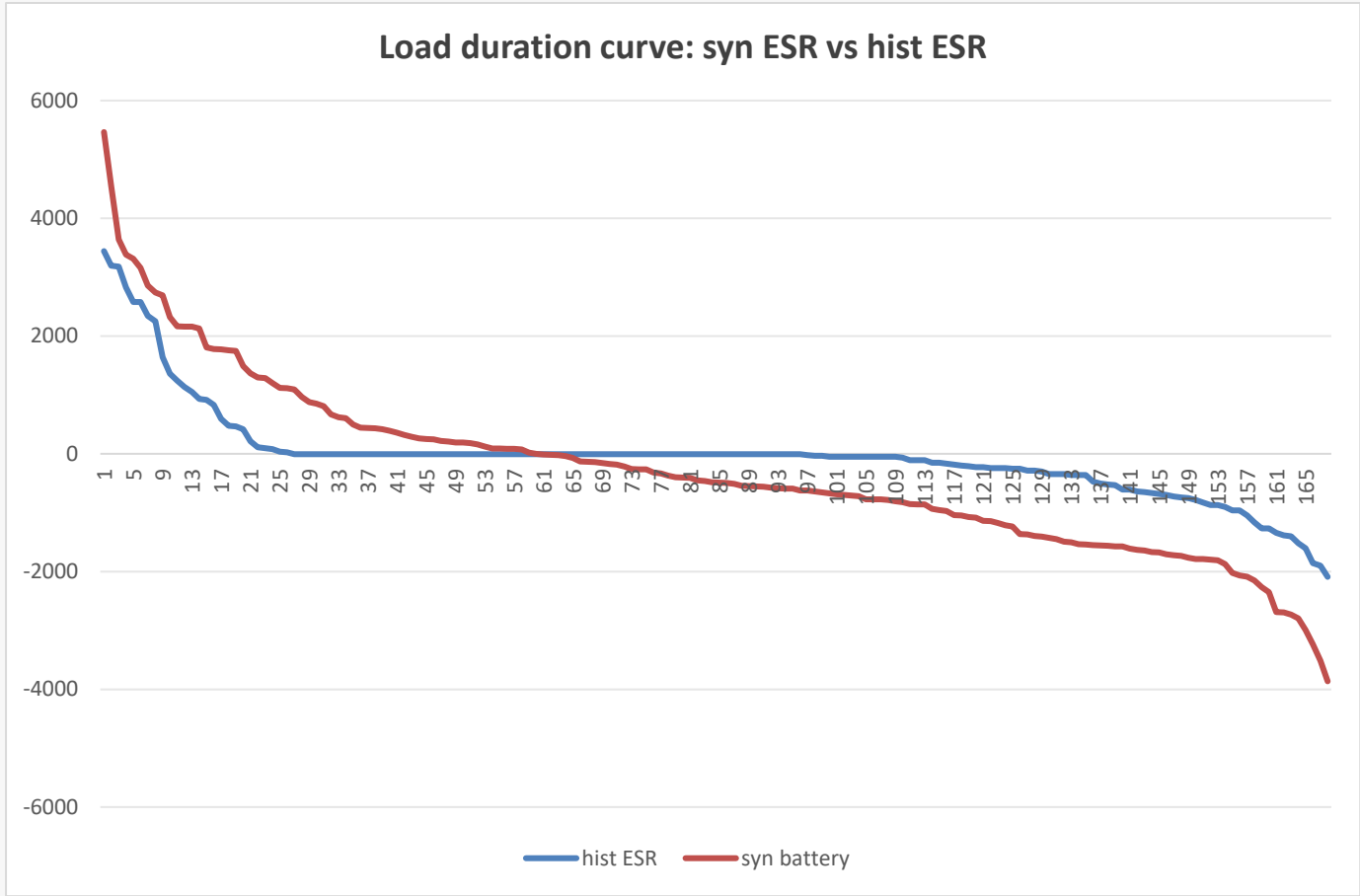
where $P_{\text{agg}} = \sum P_{\text{rated}}$ and $E_{\text{agg}} = \sum E_{\text{rated}}$ are the zone's aggregate power and energy ceilings.

3) Cyclic boundary:

$$e_T = e_0$$

This constraint locks the terminal SoC back to its initial value, guaranteeing the storage is energy-neutral over the window.

Battery Storages – Validation: system design



Historical Data Source:

- Ancillary Service Component Capacities: Projected Ancillary Service Requirements
- Historical charging & discharging data: ESR Integration Report 01-08-25

Milestone:
Ancillary Service (Operational Reserves)

Ancillary Service Modeling: Definition

What is an Ancillary Service (operating reserve)?

Beyond the **energy** needed to serve real-time load, a power system must also hold an operating reserves to absorb sudden shocks: a unit tripping offline, load or renewable forecast errors, frequency swings.

This standby capacity is the **operating reserve**, or **ancillary service (AS)**. The key thing to understand about AS is that it is **capacity, not energy**. Under normal operation these products deliver near-zero net energy; they are a safety margin, called upon to actually generate only when the system tightens or a contingency hits.

In ERCOT this margin is split into several products, differing mainly in *how fast* they must respond:

- **Reg-Up / Reg-Down** — regulation (continuous, second-by-second frequency control)
- **RRS** — responsive reserve (fast, frequency-triggered)
- **ECRS** — contingency reserve
- **Non-Spin** — non-spinning reserve (the slowest, and the one that actually sustains energy delivery)

Ancillary Service Modeling: Definition

Ancillary Service		Purpose	Response type	Operational range	Response expected within	Theoretical max. response duration	Typical procurement range (approx.)
Regulation Service Up and Regulation Service Down		Continuously correct minor frequency deviations	Pre-fault	Frequency outside 60 Hz	5 seconds	1 hour	190 – 880 MW
Responsive Reserve Service	Primary Frequency Response	Arrest frequency deviations, and provide additional dispatchable capacity	Pre-fault/ manual	Frequency < 59.983 Hz	1 minute	1 hour	2,300 – 3,200 MW (PFR to be min. 1,150 MW in 2024)
	Under Frequency Response	Arrest frequency deviations	Post-fault	Frequency < 59.7 Hz	0.25 seconds	1 hour	
	Fast Frequency Response	Arrest frequency deviations	Post-fault	Frequency < 59.85 Hz	1 minute	15 minutes	
ERCOT Contingency Reserve Service		Arrest significant frequency deviations, and provide additional dispatchable capacity	Post-fault/ manual	Frequency < 59.91 Hz or Forecasted 10-minute net load ramp > available 10-minute ramping capability	10 minutes	2 hours	900 – 3,000 MW
Non-Spinning Reserve Service		Provide additional dispatchable capacity when real-time reserves are low	Manual	Physical Responsive Capability < 3,200 MW or Forecasted 30-minute net load ramp > available 30-minute ramping capability	30 minutes	4 hours	1,900 – 3,400 MW

Ancillary Service Modeling: architecture

Overall architecture

Operating reserve is supplied jointly by two resource types: **thermal headroom** (unused capacity on already-committed units) and an **AS share carved out of storage**. These play distinct roles across two solve stages.

What we reproduce

AS is held as *capacity* at the commitment stage and delivers no energy in normal hours. In reality, when real-time tightens, ERCOT releases reserved AS into SCED *before* shedding load. We reproduce this conditional release — opened **only** to the Non-Spin slice.

Capacity layer (UC stage)

Every AS product enters unit commitment as a **zonal reserve requirement**. Storage carries its share first *RzStor*; thermal only commits for the leftover *RzThermal*.

Per zone z :

$$R_z = R_{zStor} + R_{zThermal},$$

$$R_{zThermal} = \max(R_z - R_{zStor}, 0)$$

This is the mechanism that stops UC from over-committing coal/gas just to hold headroom. **No energy is dispatched here.**

Energy layer (DCOPF stage)

Only the **Non-Spin** slice may release as real energy, and only under a trigger modeled after ERCOT's Physical Responsive Capability (PRC).

Ancillary Service Modeling: The storage carve-out

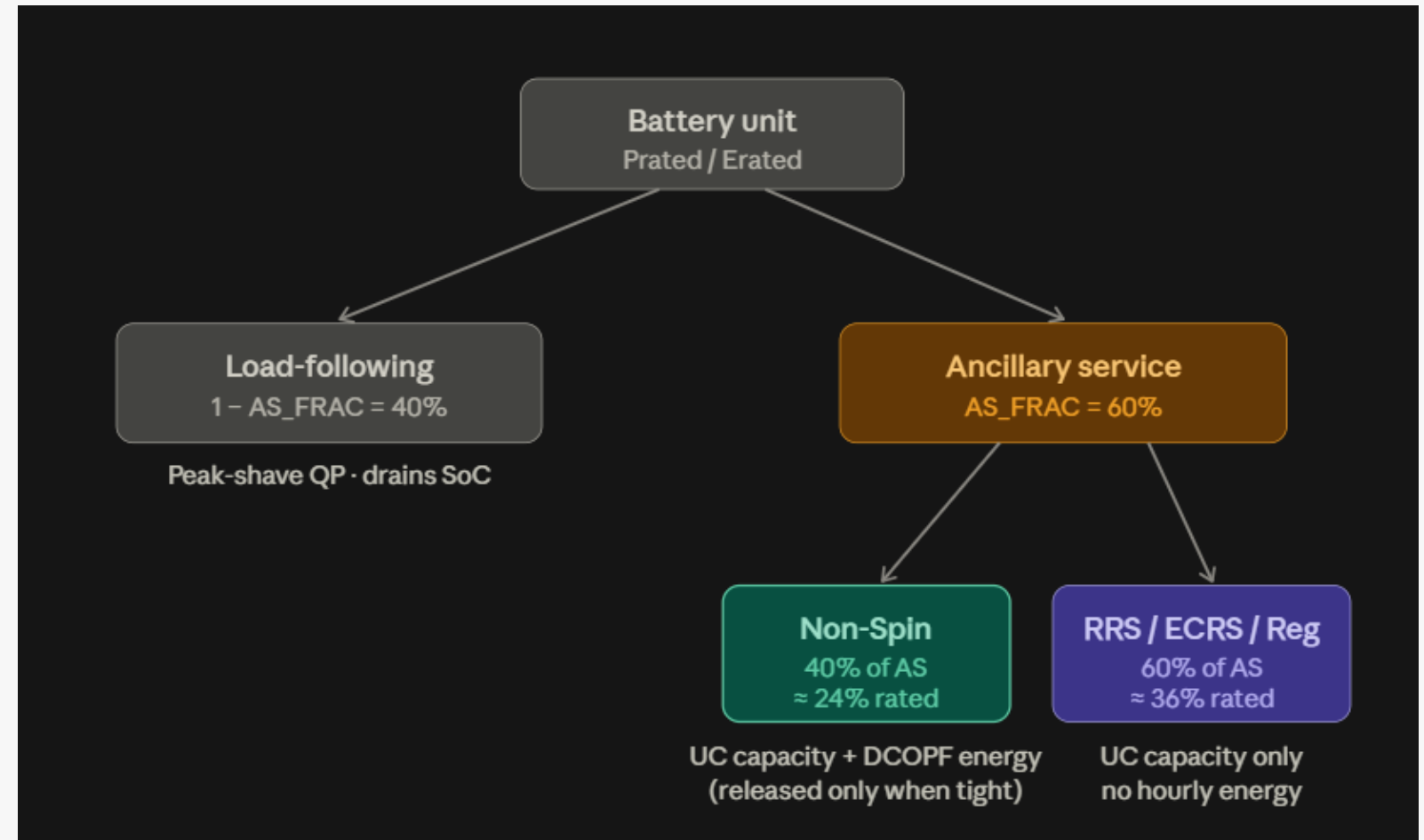
Each battery's rated power (and energy, carved in the same proportion) is split in two levels:

Level 1 — load-following vs. reserve

- **Load-following / Peak shaving:** drives the peak-shave QP, drains real SoC
- **Ancillary service:** pledged to reserve, held out of load-following

Level 2 — inside the AS share

- **Non-Spin:** the only DCOPF-deployable energy
- **RRS / ECRS / Reg:** capacity only, never energy



Ancillary Service Modeling: Assumptions & Limitations

A1 — Only Non-Spin delivers real energy.

Among AS products, only Non-Spin sustains meaningful energy; RRS / ECRS / Reg net ≈ 0 energy per hour. So the latter stay *capacity-only* (they force thermal headroom at the UC layer) and never enter DCOPF as an energy source.

A2 — The Non-Spin share is firm.

Its release is treated as sufficient and firm, so it does **not** draw down real SoC — only load-following discharge moves socReal. This keeps the DCOPF SoC ledger consistent with the UC assumption that each battery holds its AS reservation every hour.

Known limitation of A2.

With no energy budget, a long stretch of consecutive armed hours is under-shed / LMP under-priced, since the firm slice never depletes. The arming gate bounds this to genuinely tight hours; a future fix clips emgCap by socReal and drains it. (*A self-check prints the max consecutive armed run to flag this.*)

Ancillary Service Modeling: Implementation

How it releases.

Rather than detecting a shortfall and patching it afterward, we let the market do it in a single pass. Each battery is given an emergency-discharge option priced *above* normal generation cost but *below* the value of lost load.

That pricing does two things at once: it stays dormant when the system is comfortable (too expensive to bother), and it always undercuts shedding firm load (far cheaper than a blackout). So the dispatch decides on its own whether to tap it, and how much — the same way ERCOT releases reserved capacity into the market before load shedding.

When it releases.

The slice is armed only when the system is genuinely tight — when the reserve margin left after normal generation falls below a set threshold. Above that margin it stays fully locked; below it, the battery's Non-Spin energy becomes available for the dispatch to draw on. This mirrors how an ERCOT operator calls on Non-Spin only once physical reserves run low.

Why it behaves like ERCOT

- The tight-system trigger plays the role of ERCOT's Physical Responsive Capability falling low.
- The above-cost / below-VOLL price reproduces the reserve-before-shed ordering (ORDC/ASDC).
- Because the trigger is judged *before* the dispatch runs, it reflects reserves that are actually online, not a hindsight number.

Milestone: DCOPF Integrator

DCOPF Integrator modeling: Overall Design

The whole codebase is a single forward pipeline: three solve stages hand off in turn, with each upstream solution becoming a fixed input downstream. This layering has two justifications:

- **Solvability.**

Forcing 168 hours of integer commitment, network constraints, and storage temporal coupling into one monolithic MILP would be too large to solve. Split apart, each subproblem sits comfortably within its solver's reach.

- **Fidelity to reality.**

This sequence maps directly onto ERCOT's market operation:

DAM → RUC → SCED

the ISO first fixes commitment day-ahead, then clears network-constrained economic dispatch in real time, interval by interval. The model's layering is itself a depiction of how the market actually runs.

DCOPF Integrator modeling: The Three-Stage Relay

Three solve stages run in sequence — each hands a *fixed* result to the next, like a baton. Nothing downstream can revise what an upstream stage decided.

Stage 1 — Storage planning first.

Storage is scheduled before anything else. It produces a fixed weekly charge/discharge plan, plus how much reserve each zone's batteries can pledge. It goes first because batteries *reshape the net load* — shaving peaks, filling troughs — that every downstream stage must serve. Only once the curve is flattened does it make sense to commit thermal against what's left.

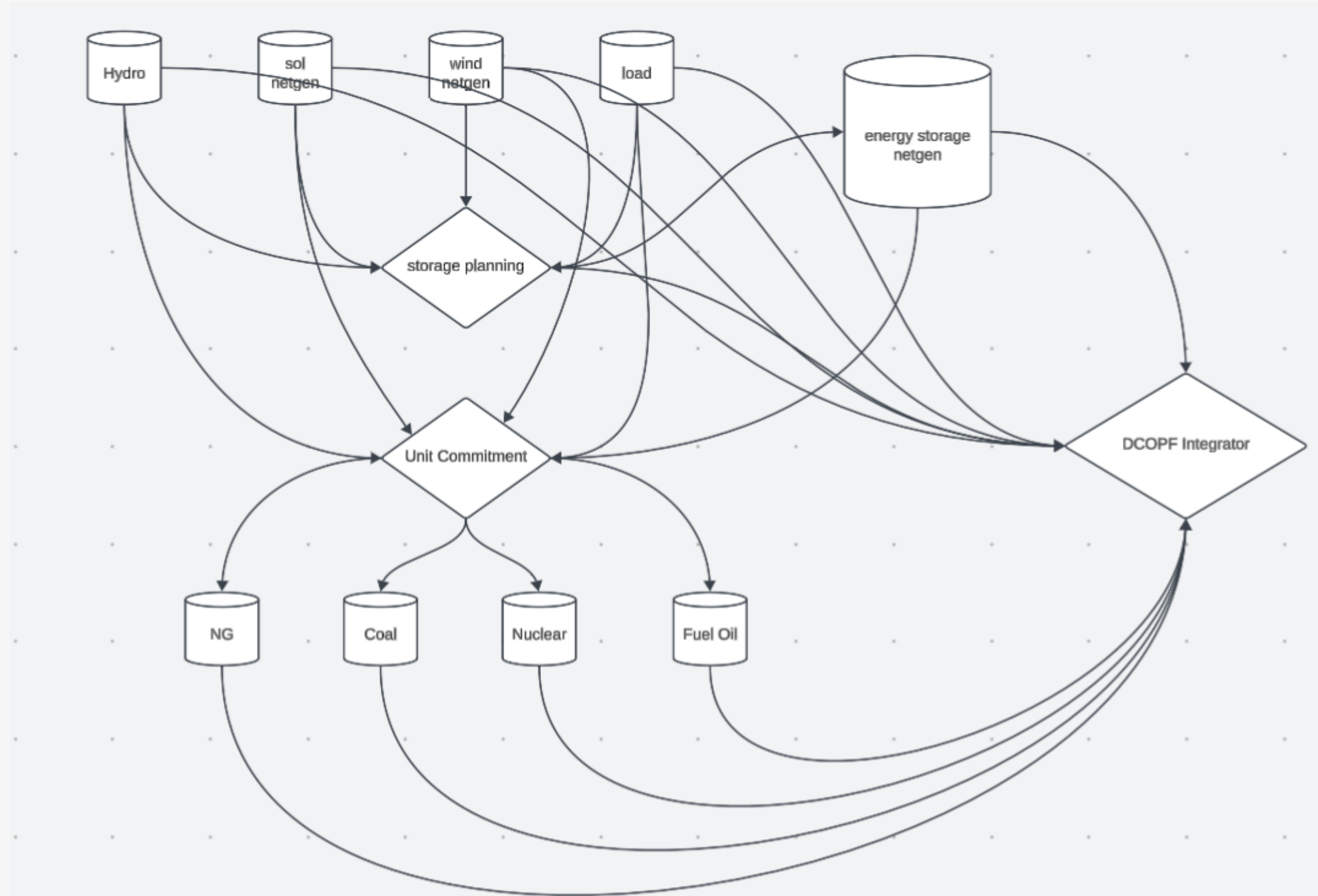
Stage 2 — Unit commitment.

Working on the flattened residual (load minus wind, solar, hydro, and the storage plan), this stage decides which thermal units are on or off, and forces each zone to hold its operating-reserve margin — batteries cover their share first, thermal fills the rest. It reasons on each unit's rated capacity and ignores the transmission network (copper-plate). Its output is a fixed day-ahead schedule handed to Stage 3.

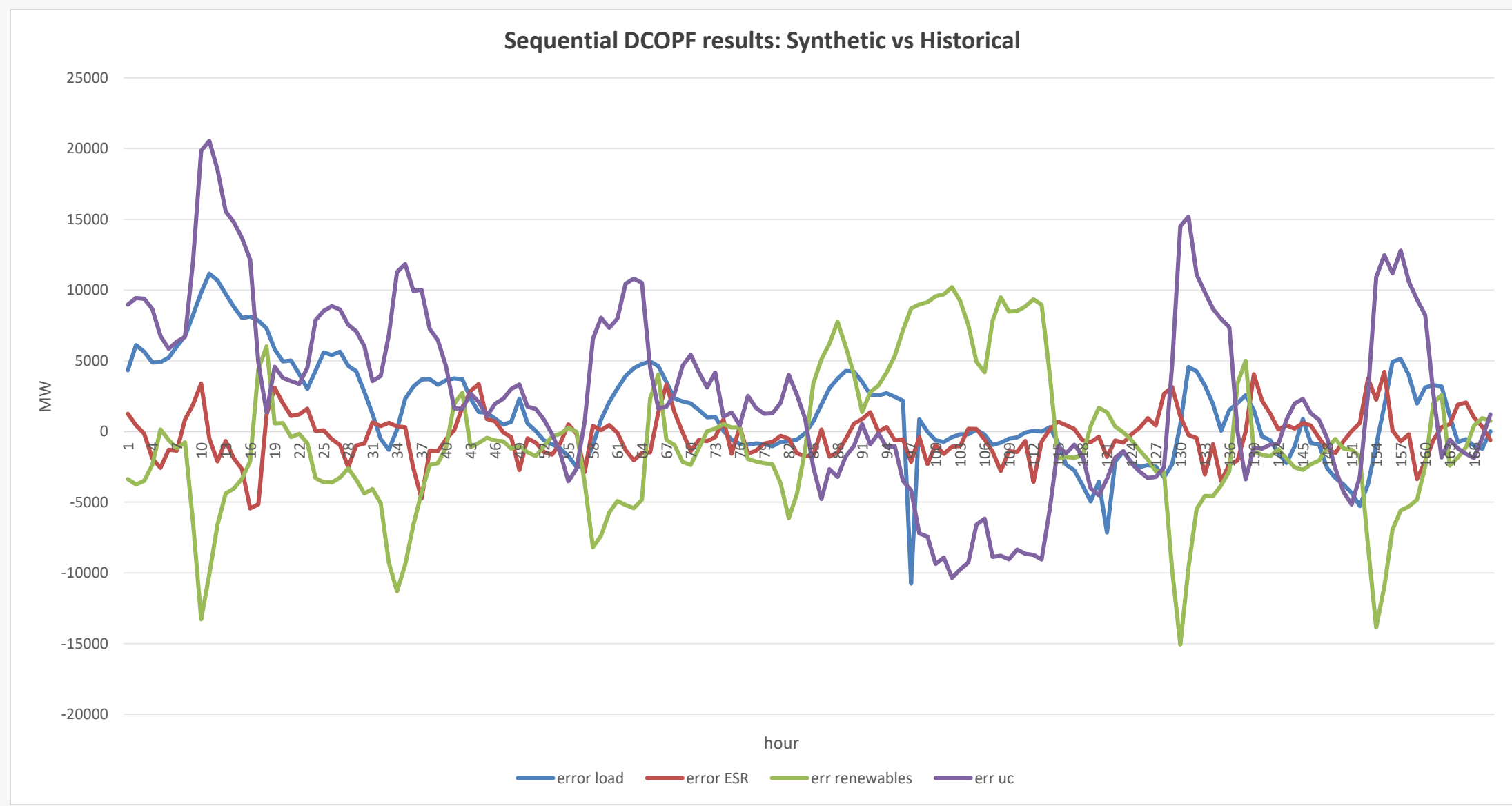
Stage 3 — Hourly DCOPF integration.

The end of the pipeline, where every component converges. Each hour is assembled into one network-constrained case: renewables can be dispatched down, thermal is capped by its committed-and-available capacity, storage injects or charges on its fixed plan, the Non-Spin reserve is released only when the system is tight, and load-shedding sits on standby as a last-resort valve priced at the value of lost load. A single solve then clears both the physical dispatch at every node and the nodal prices (LMPs).

DCOPF Integrator modeling: Overall Design – Data flow diagram



DCOPF Integrator modeling: validation



Note:

Error = synthetic data – historical data

Err_uc = error_coal + error_ng + error_nuc

Err_renewables = error_solar + error_wind + error_hydro

Current Milestones: Summary

Current Milestones: Summary

	Planning	Programming	Validation	
			Net gen	Served
Weather Data Processing	Done	Done	Done	
Gas Availability	Done	Done	Done	Done
Coal Availability	Done	Done	Done	Done
Hydropower Generation	Done	Done	Done	Done
Wind Generation	Done	Done	Done	Done
Solar Generation	Done	Done	Done	Done
Nuclear Availability	Done	Done	Done	Done
Distillated Fuel Oil Availability	Done	Done	Done	Done
Energy Storage Modeling	Done	Done	Done	Done
Unit Commitment: NG, Coal, Oil, Nuclear	Done	Done	Done	Done
Operation Reserves (Ancillary Service)	Done	Done	Done	Done
Gas supply competition Modeling	Will be updated at Ver2			
Generation Cost Reconfiguration	Done	Done	Done	Done
Load Splitting	Done	Done	Done	Done
Load Shedding VOLL	Done	Done	Done	Done
DC Link?	Pending			
DCOPF integration	Done	Done	Done	Done